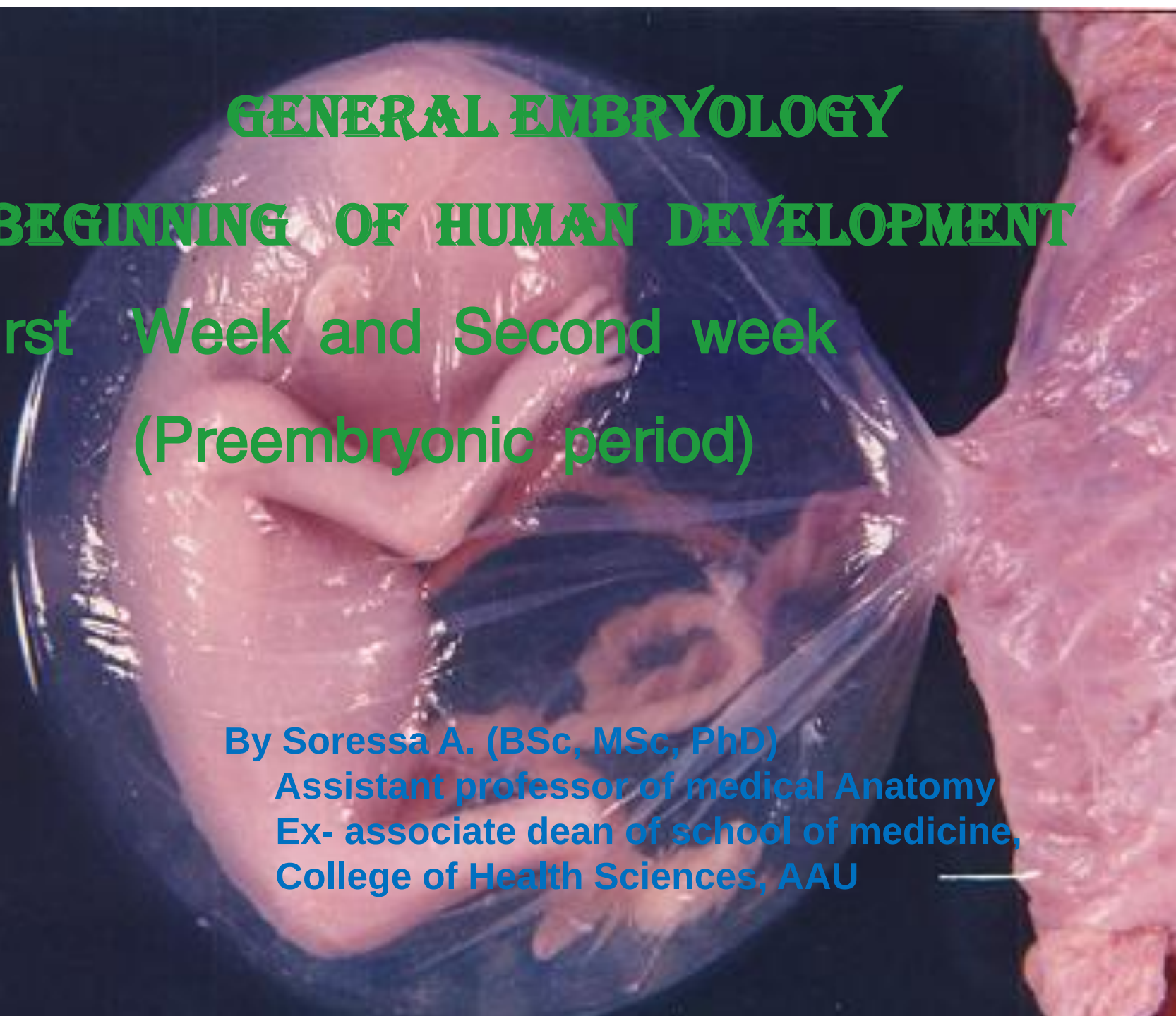


GENERAL EMBRYOLOGY

THE BEGINNING OF HUMAN DEVELOPMENT

**First Week and Second week
(Preembryonic period)**

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Objectives

- At the end of this lecture student will be able to:
- Define fertilization
- Describe necessary events during fertilization
- Define cleavage and blastocyst formation
- Evaluate developmental events of 1st week and the 2nd week
- Identify site of implantation
- Explain the process of implantation

Introduction to general embryology

Embryology :

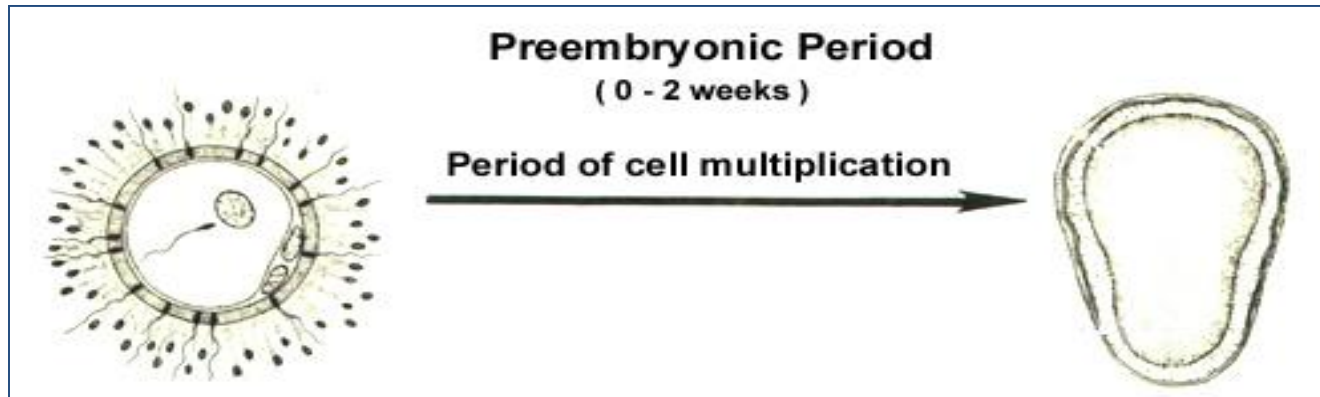
- is a science that studies the **normal development** as well as **birth defects** of a human being in the maternal **uterus**

Normal development

38 weeks from fertilization to parturition

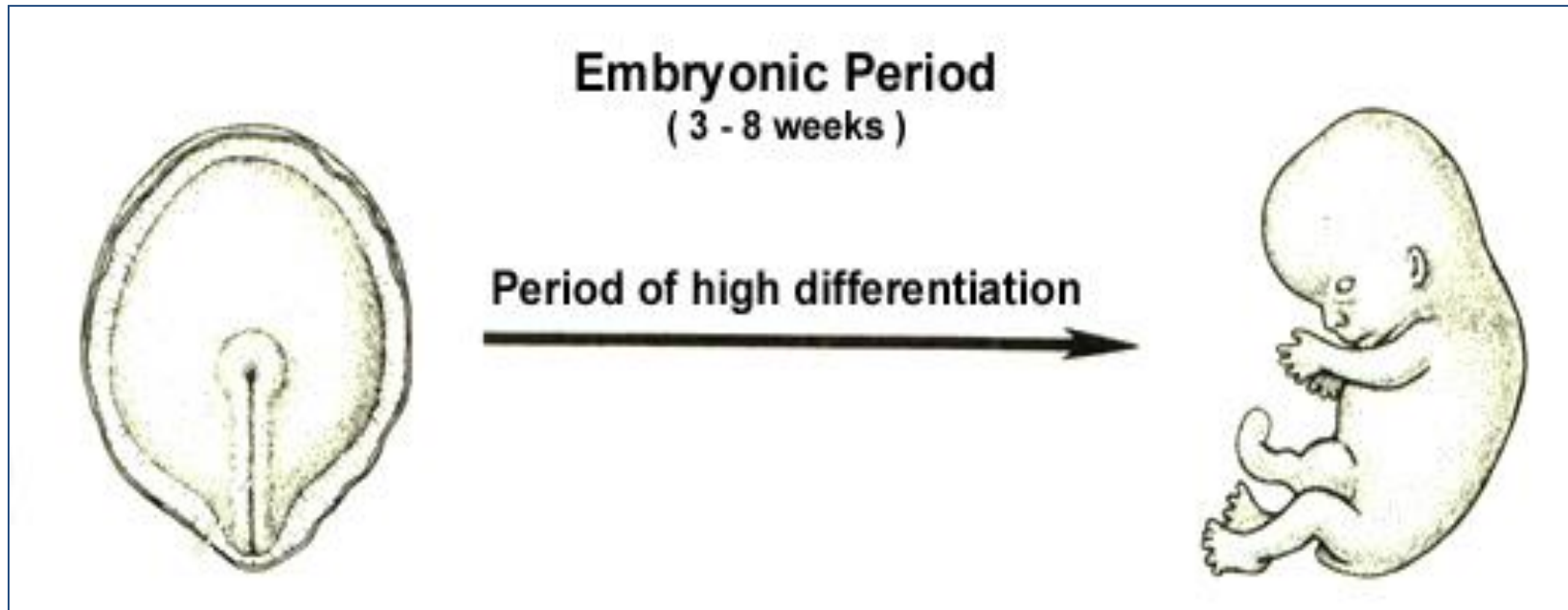
1) Preembryonic period:

- First 2 weeks
- Fertilization to formation of the bilaminar germ disc



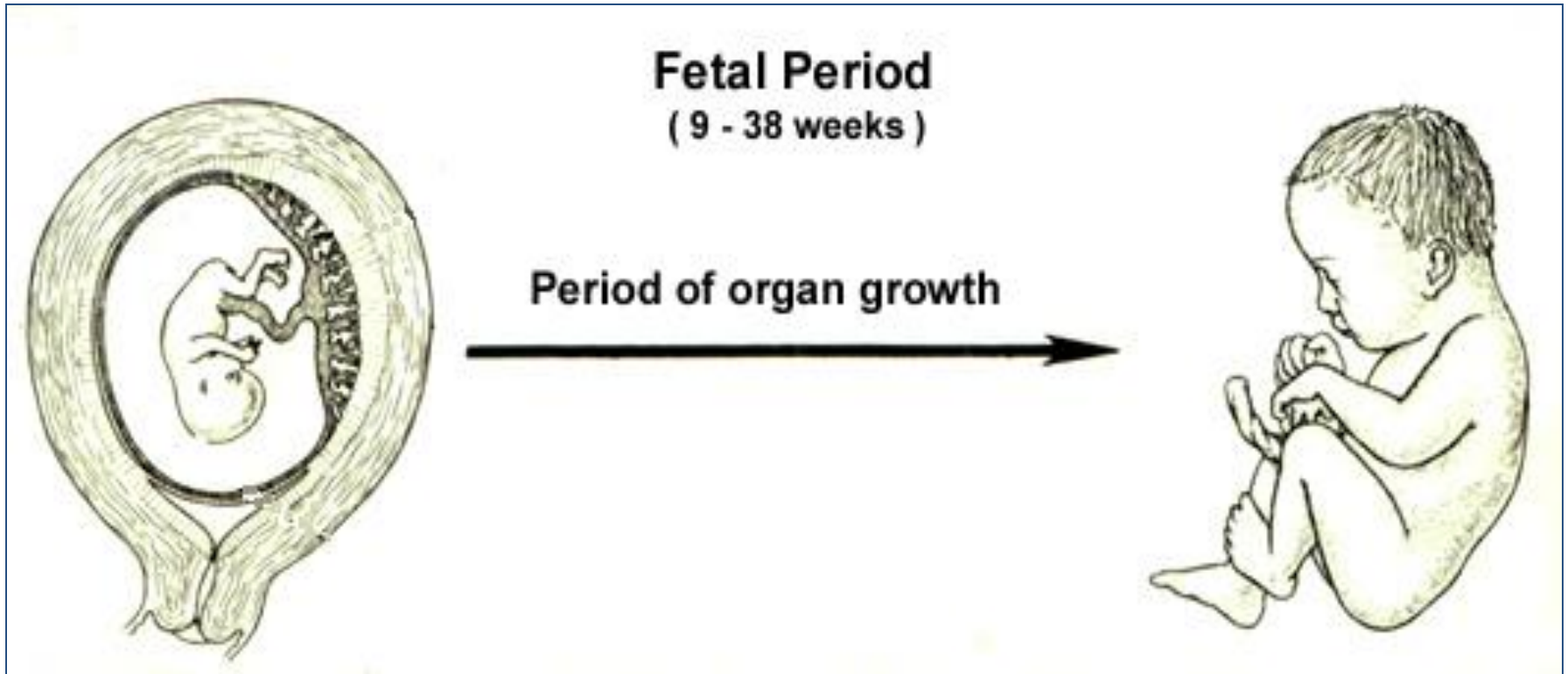
2. Embryonic period:

- Weeks 3 – 8
- Primordia of all major organs develop from the 3 germ layers



3. Fetal period:

- Week 9 – 38/40 or birth
- Growth of the organ systems



Early development of human embryo

Gametogenesis

- a process of formation and maturation of the gametes (sperm and ovum)

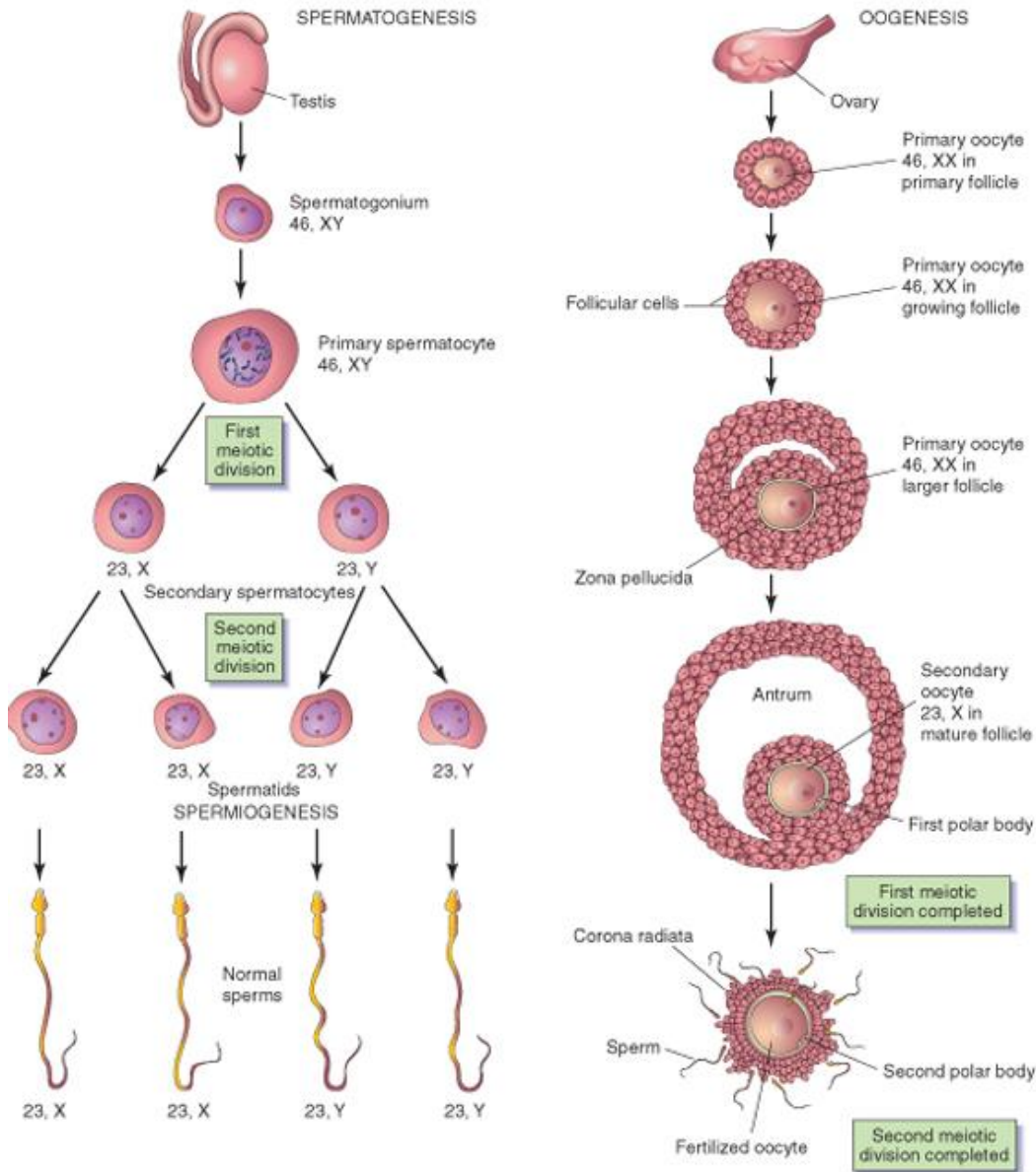
The purpose of gametogenesis

- Reduction of the number of chromosomes.
- Alteration of the shape of germ cells for fertilization

Two types of gametogenesis:

- **Spermatogenesis**
- **Oogenesis**

NORMAL GAMETOGENESIS



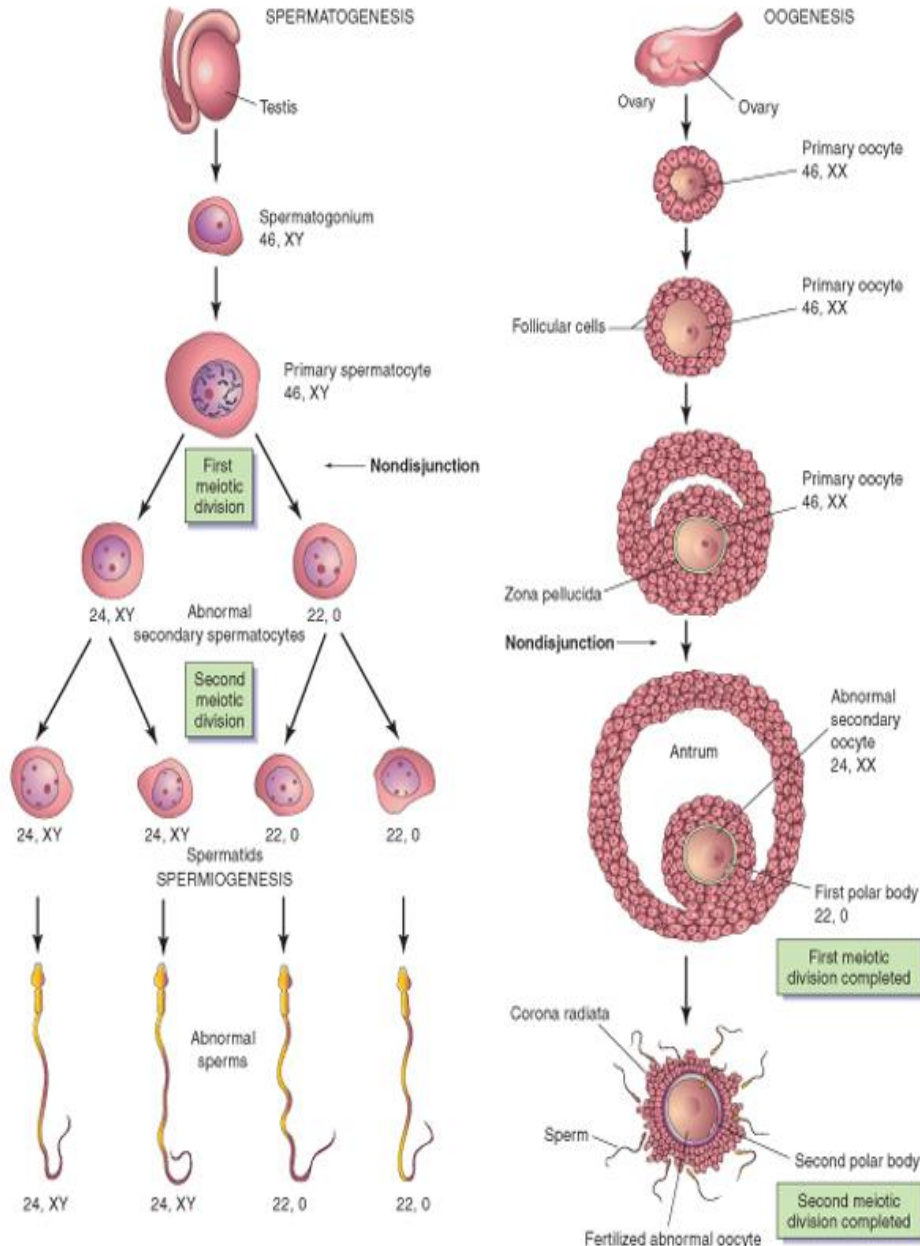
Normal gametogenesis:

- conversion of germ cells into gametes.
- The drawings compare spermatogenesis and oogenesis. Oogonia are not shown in this figure because they differentiate into primary oocytes before birth.

Note that

- (1) following the two meiotic divisions, the diploid number of chromosomes, 46, is reduced to the haploid number, 23;
- (2) four sperms form from one primary spermatocyte, whereas only one mature oocyte results from maturation of a primary oocyte;
- (3) the cytoplasm is conserved during oogenesis to form one large cell, the mature oocyte.
- The polar bodies are small nonfunctional cells that eventually degenerate

ABNORMAL GAMETOGENESIS



Abnormal gametogenesis.

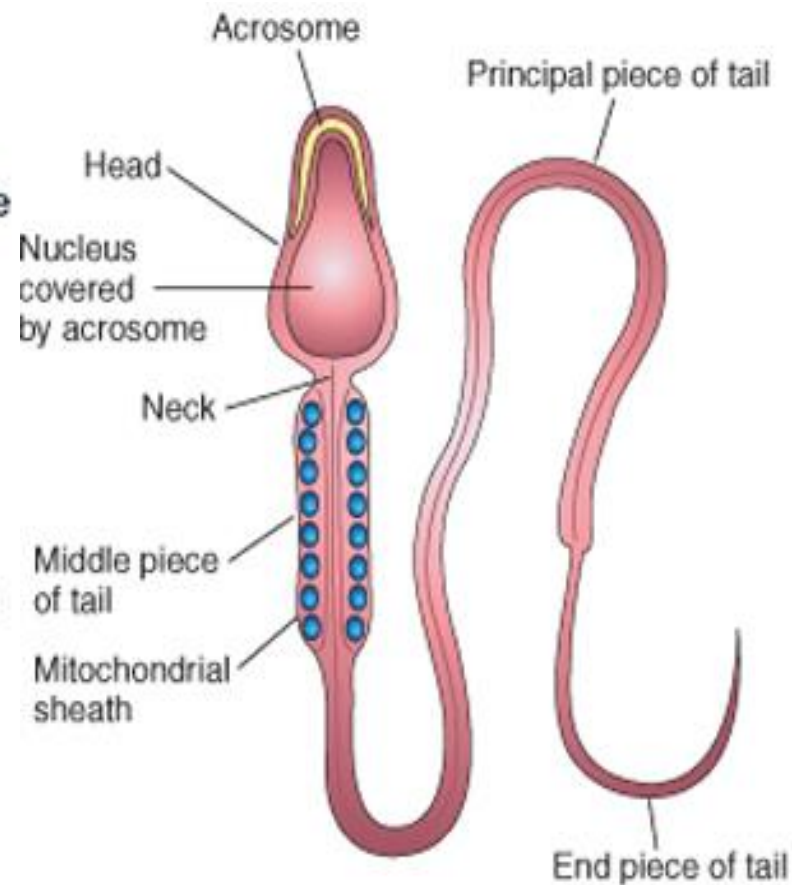
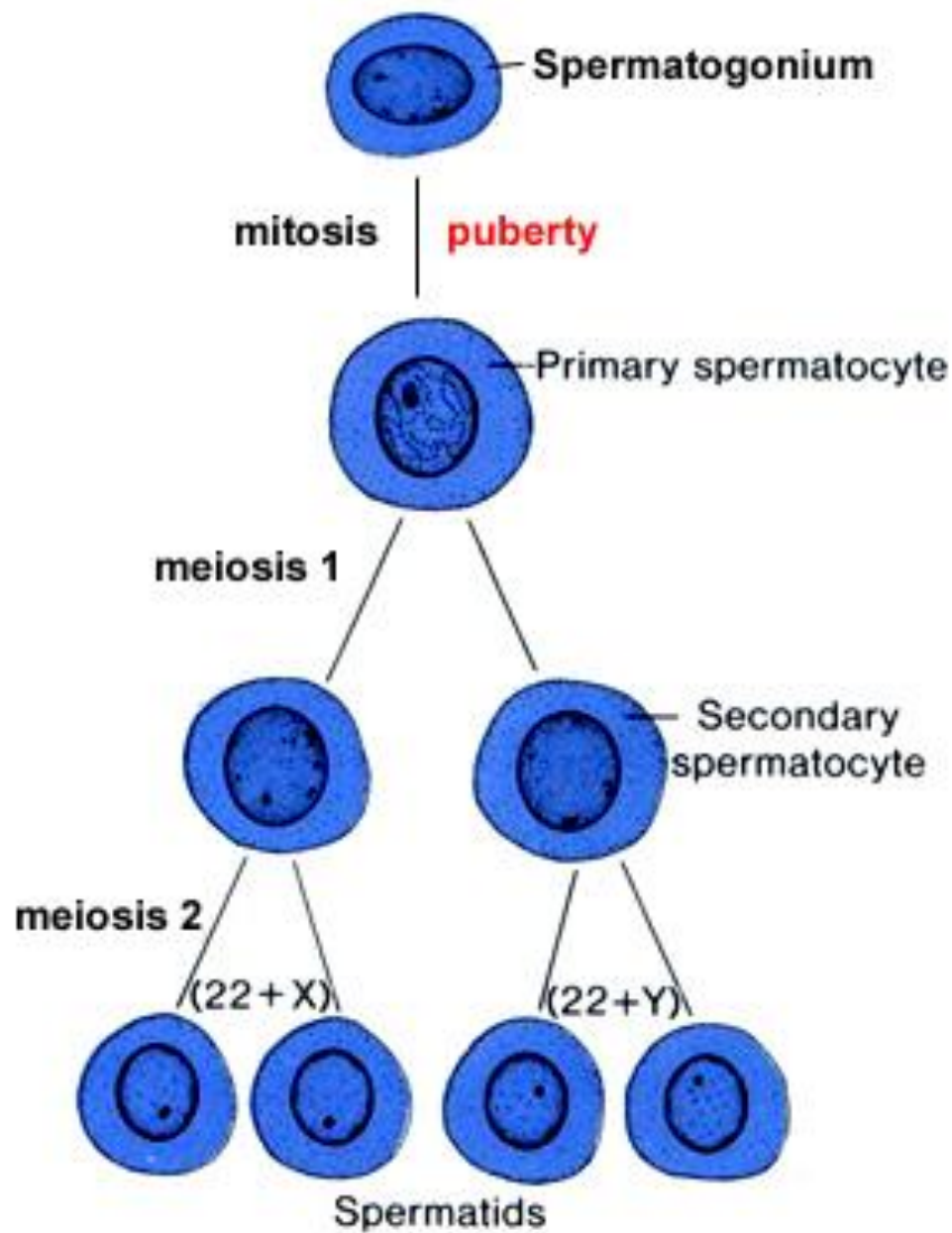
- The drawings show how nondisjunction results in an abnormal chromosome distribution in gametes.*
- When nondisjunction occurs during the first meiotic division of spermatogenesis, one secondary spermatocyte contains 22 autosomes plus an X and a Y chromosome, and the other one contains 22 autosomes and no sex chromosome.*
- Similarly, nondisjunction during oogenesis may give rise to an oocyte with 22 autosomes and two X chromosomes (as shown) or may result in one with 22 autosomes and no sex chromosome.*

Spermatogenesis:

- is the sequence of events by which *spermatogonia* are transformed into **mature sperms**
- maturation process begins at puberty
- **Spermatogonia**, which have been dormant in the seminiferous tubules of the testes since the fetal period, begin to increase in number at puberty
- After several mitotic divisions, the spermatogonia grow and undergo changes

Spermatogonia are:

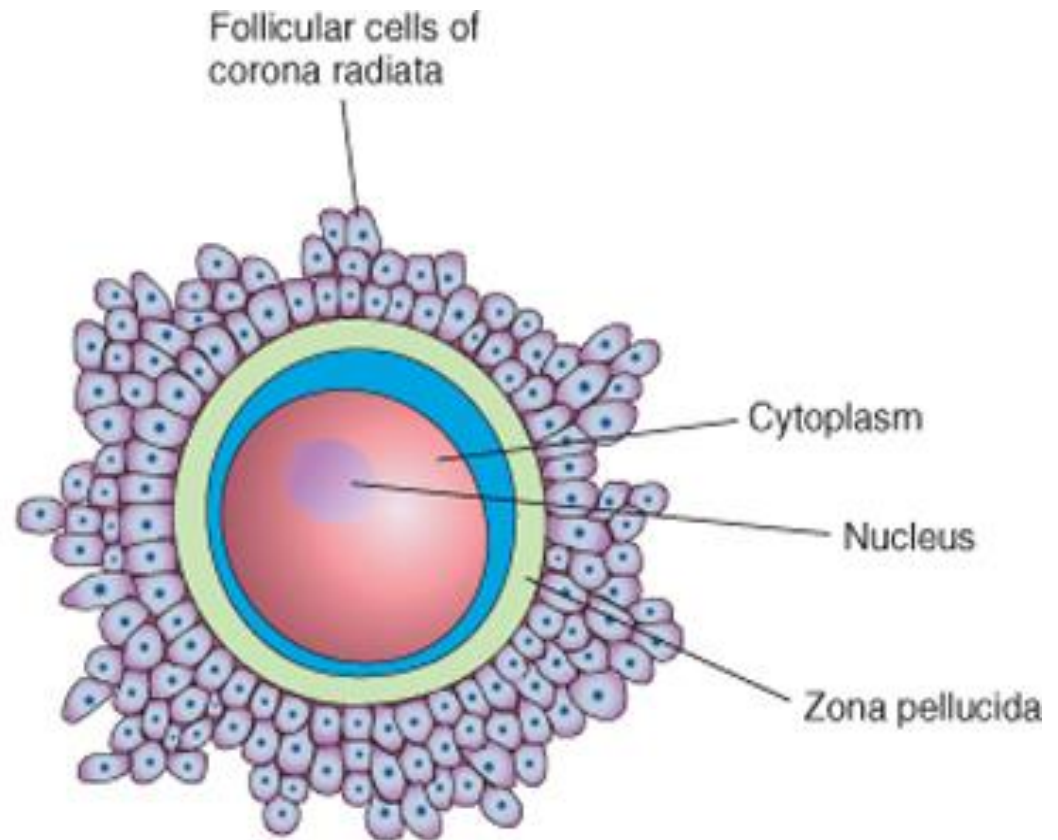
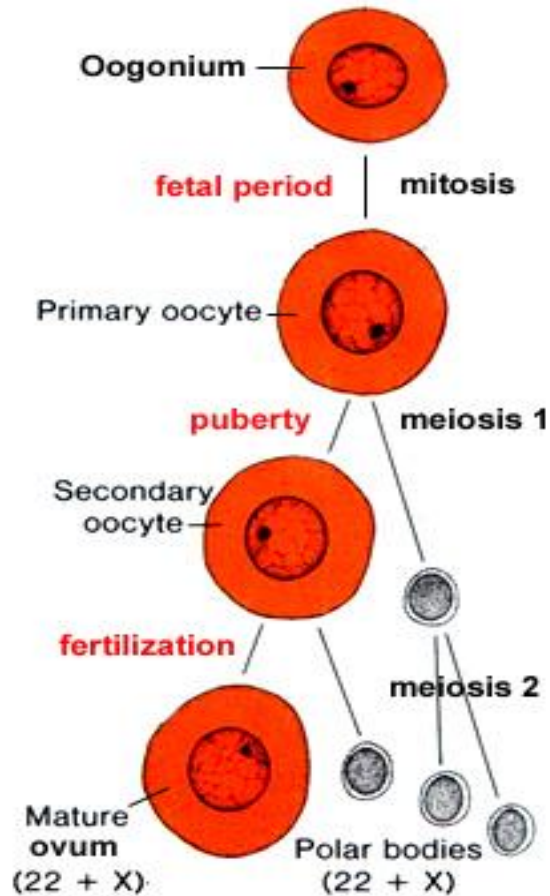
- transformed into **primary spermatocytes**-the largest germ cells in the seminiferous tubules
- Each primary spermatocyte subsequently undergoes a reduction division-the first meiotic division-to form two haploid **secondary spermatocytes**
- **secondary spermatocytes** are approximately half the size of primary spermatocytes
- the secondary spermatocytes undergo a second meiotic division to form four haploid spermatids
- The spermatids are gradually transformed into four mature sperm by a process known as **spermiogenesis**
- The entire process of spermatogenesis, which includes spermiogenesis, takes approximately 2 months
- When spermiogenesis is complete, the sperms enter the lumina of the seminiferous tubules



OÖGENESIS :

- is the sequence of events by which **oögonia** are transformed into **mature oöcytes**
- This maturation process begins before birth and is completed after puberty
- Oögenesis continues to *menopause*, which is permanent cessation of the *menses* (bleeding associated with the menstrual cycles)
- *There are $\approx$ two million primary oöcytes in the ovaries of a newborn female*
- by adolescence no more than 40,000 remain
- Of these, only $\approx$ 400 become secondary oöcytes and are expelled at ovulation during the reproductive period.

- Few of these oocytes, if any, are fertilized and become mature
- The number of oocytes that ovulate is greatly reduced in women who take oral contraceptives because the hormones in them prevent ovulation from occurring



Differences between male and female

Male

- ✓ Has stem cells
- ✓ Highly motile
- ✓ Continuous after puberty
- ✓ 4 sperm from a spermatocyte
- ✓ Equal division
- ✓ Functional mature and
Capacitation

Female

- No stem cells
- Immotile
- Pause twice
- Only 1 ovum from an oocyte
- Unequal division
- No capacitation

Functional mature and capacitation of sperm

Functional mature: gaining ability of **forward movement** and **potential of fertilizing** eggs in the **epididymis**

Capacitation: obtaining **capacity of fertilizing** an egg in the **female reproductive tract** through removal of a glycoprotein coat from the plasma membrane of the acrosomal region by enzyme

FEMALE REPRODUCTIVE CYCLES

- Commencing at puberty,
- involving activities of the **hypothalamus** of the brain, **pituitary gland**, **ovaries**, **uterus**, **uterine tubes**, **vagina**, and **mammary glands**
- These monthly cycles prepare the reproductive system for pregnancy

- OVARIAN CYCLE

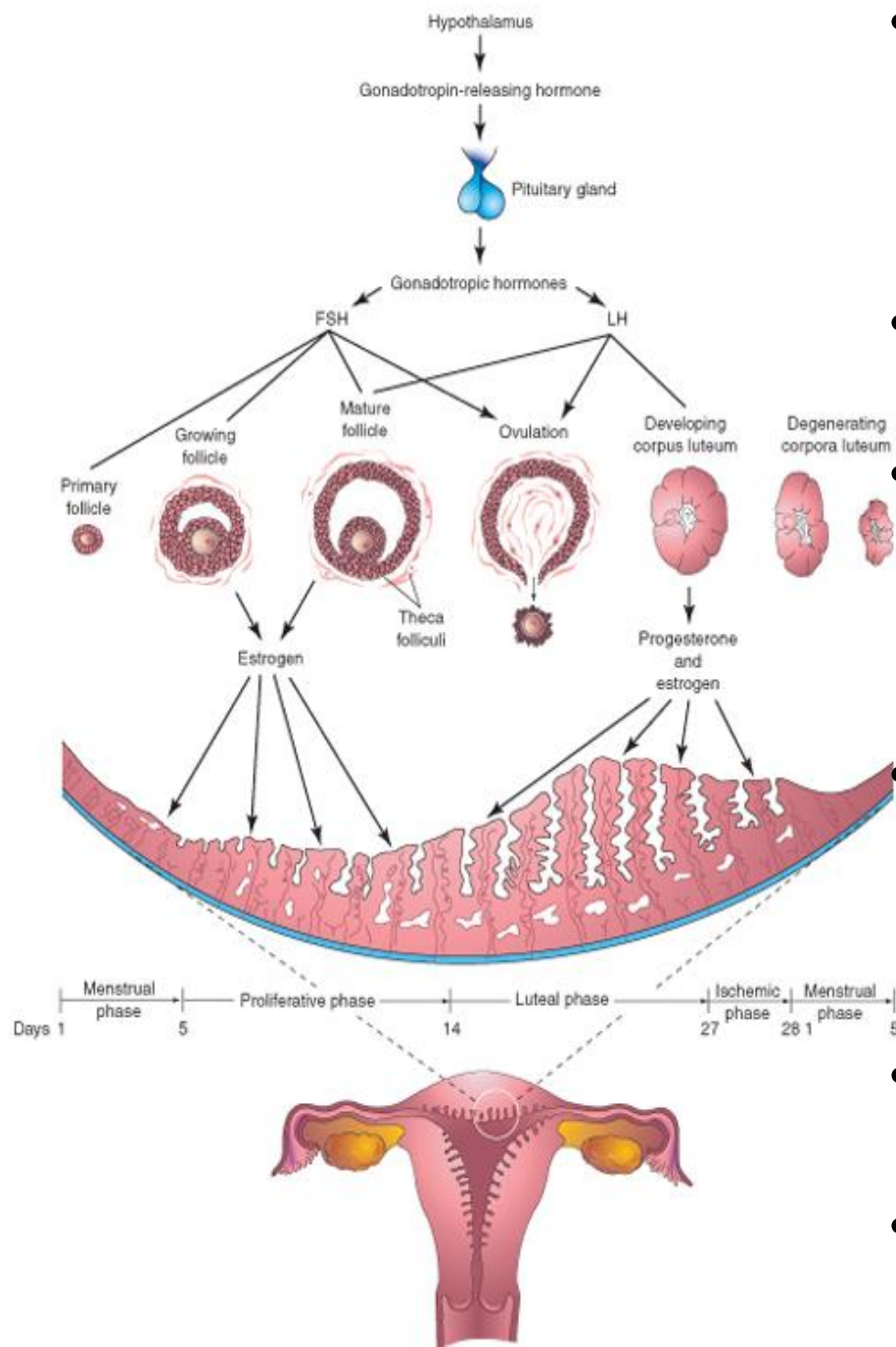
- *FSH and LH produce cyclic changes in the **ovaries-**
the ovarian cycle*

1. Development of follicles

2. Ovulation

3. Corpus luteum formation

- During each cycle, FSH promotes growth of several *primordial follicles* into 5 to 12 primary follicles
 - But, only one primary follicle usually develops into a mature follicle and ruptures through the surface of the ovary, expelling its oocyte



- *Schematic drawings illustrating the interrelations of the **hypothalamus of the brain, pituitary gland, ovaries, and endometrium***
- *One complete menstrual cycle and the beginning of another are shown.*
- *Changes in the ovaries, **the ovarian cycle**, are induced by the gonadotropic hormones (**follicle-stimulating hormone and luteinizing hormone**)*
- *Hormones from the ovaries (estrogens and progesterone) then promote cyclic changes in the structure and function of the **endometrium, the menstrual cycle***
- *Thus, the cyclical activity of the ovary is intimately linked with changes in the uterus.*
- *The ovarian cycles are under the rhythmic endocrine control of the pituitary gland,*

1. Follicular Development

- Development of an ovarian follicle is characterized by:
 - Growth and differentiation of primary oocyte
 - Proliferation of follicular cells
 - Formation of zona pellucida
 - Development of the theca folliculi

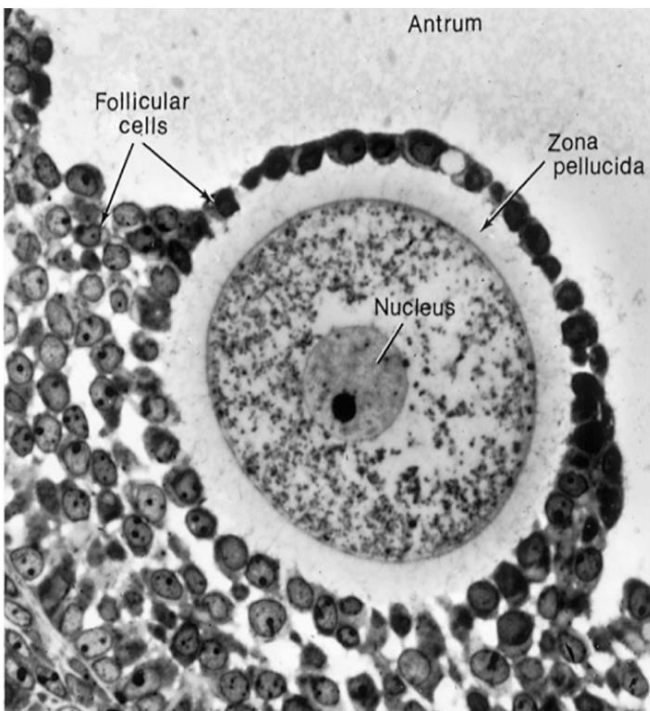


Figure 2-8 Photomicrograph of a human primary oocyte in a secondary follicle, surrounded by the zona pellucida and follicular cells.

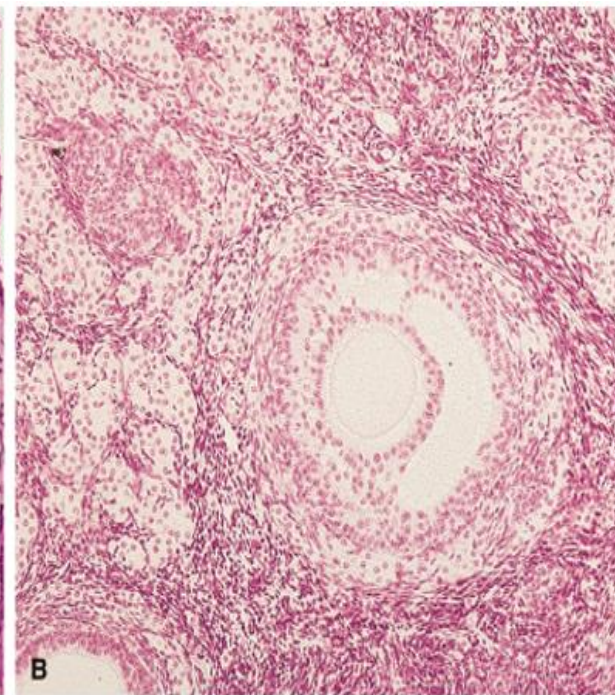
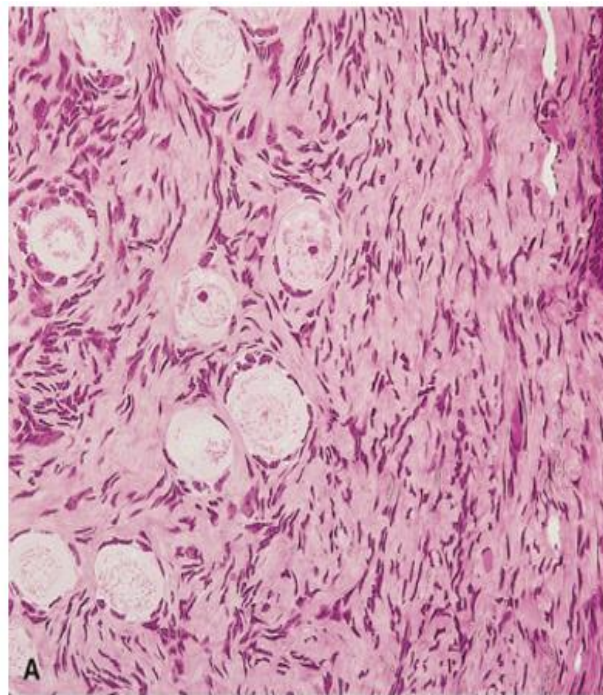


Figure 2-9 Micrographs of ovarian cortex. **A**, Several primordial follicles are visible ($\times 270$). Observe that the primary oocytes are surrounded by follicular cells. **B**, Secondary ovarian follicle. The oocyte is surrounded by granulosa cells of the cumulus oophorus

- **As the primary follicle increases in size**
 - The adjacent connective tissue organizes into a capsule, **the theca folliculi**
 - The theca soon differentiates into two layers
 - an internal vascular and glandular layer, the **theca interna**,
 - a capsule-like layer, the **theca externa**.
 - Thecal cells are thought to produce **an angiogenesis factor** that promotes growth of blood vessels in the theca interna, which provide nutritive support for follicular development.

- The follicular cells divide actively,
 - producing a stratified layer around the oocyte
 - The ovarian follicle soon becomes oval and the oocyte eccentric in position.
 - Subsequently, fluid-filled spaces appear around the follicular cells, which coalesce to form a single large cavity, **the antrum**, which contains **follicular fluid**
 - After the antrum forms, the ovarian follicle is called **a vesicular or secondary follicle**.

- The primary oocyte:
 - is pushed to one side of the follicle
 - is surrounded by a mound of follicular cells, the **cumulus oophorus**, that projects into the antrum
 - The follicle continues to enlarge until it reaches maturity and produces a swelling on the surface of the ovary

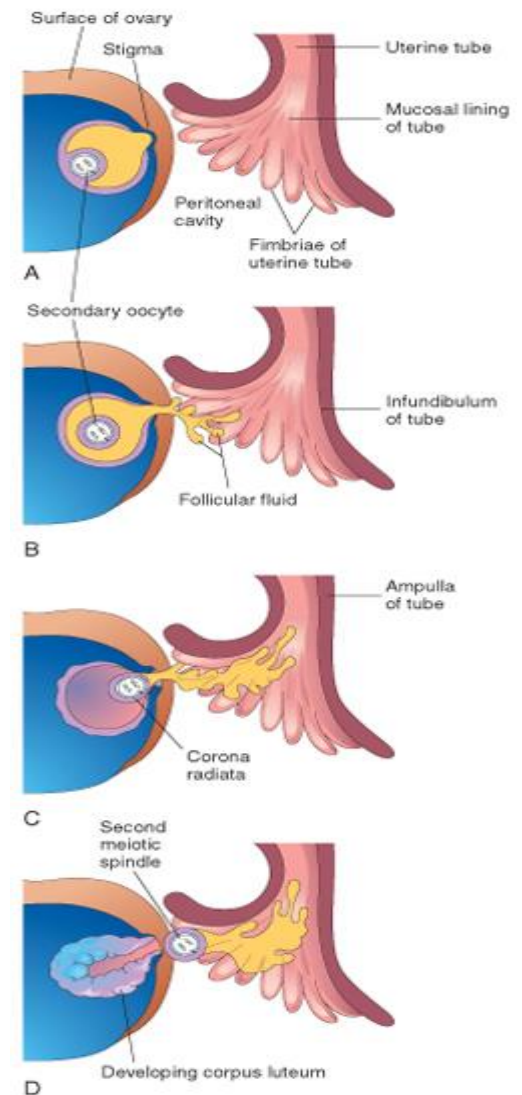


Figure 2-10 Illustrations of ovulation: When the stigma ruptures, the secondary oocyte is expelled from the ovarian follicle with the follicular fluid. After ovulation, the wall of the follicle collapses and is thrown into folds. The follicle is transformed into a glandular structure, the corpus luteum

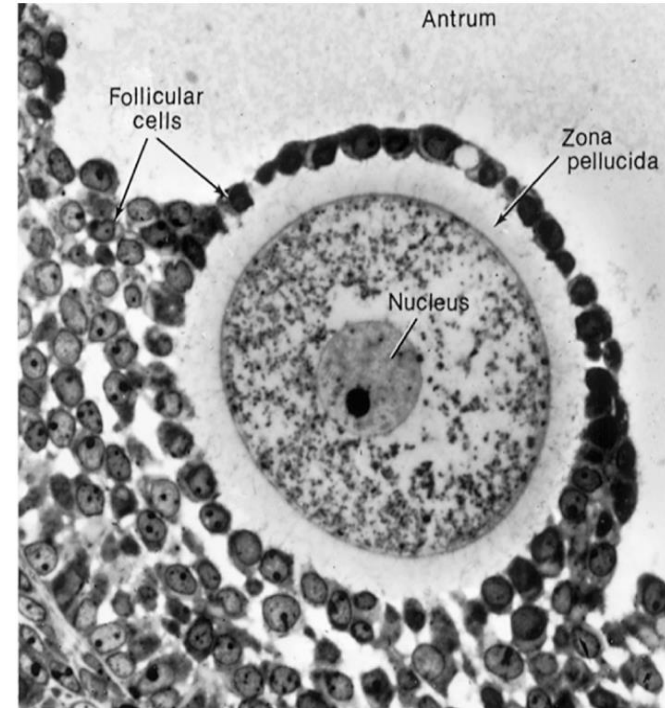
Ovulation

- is triggered by a surge of LH production
- usually follows the LH peak by 12 to 24 hours
- The **LH surge**, elicited by the high estrogen level in the blood, appears to cause the stigma to balloon out, forming a vesicle
- The stigma soon ruptures, expelling the secondary oocyte with the follicular fluid
- Expulsion of the oocyte is the result of intrafollicular pressure and possibly contraction of smooth muscle in the theca externa owing to stimulation by prostaglandins.
- *Enzymatic digestion of the follicular wall* seems to be one of the principal mechanisms leading to ovulation

- The expelled secondary oocyte
- is surrounded by the zona pellucida and one or more layers of follicular cells, which are radially arranged as the **corona radiata**, forming the **oocyte-cumulus complex**
- The LH surge also seems to induce resumption of the first meiotic division of the primary oocyte
- Hence, mature ovarian follicles contain secondary oocytes

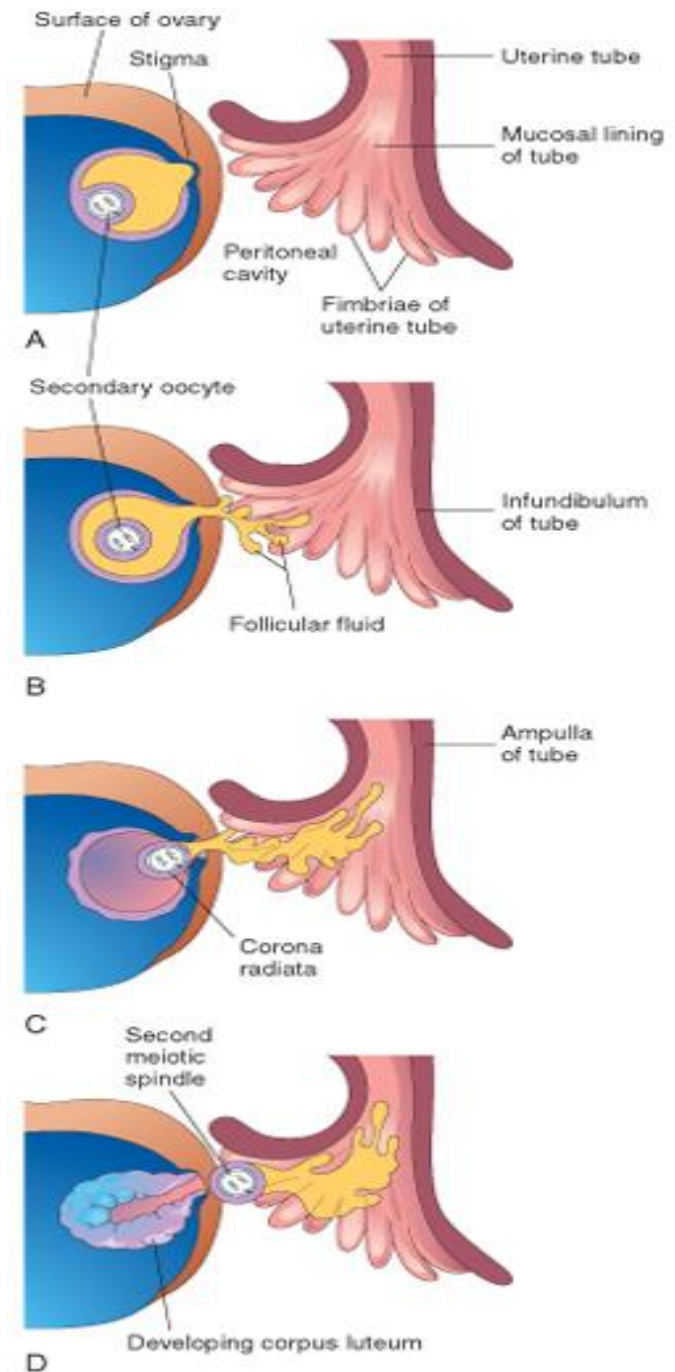
- **The zona pellucida**

- *is composed of three glycoproteins (ZPA, ZPB, ZPC), which usually form a network of filaments with multiple pores*
- Binding of the sperm to the zona pellucida (*sperm-oocyte interactions*) is a complex and critical event during fertilization.



Corpus Luteum

- Shortly after ovulation, the walls of the ovarian follicle and theca folliculi collapse and are thrown into folds (see Fig. 2-10D)
- Under LH influence, they develop into a glandular structure, the **corpus luteum**, which secretes *progesterone* and some *estrogen*, causing the endometrial glands to secrete and prepare the endometrium for implantation of the blastocyst.

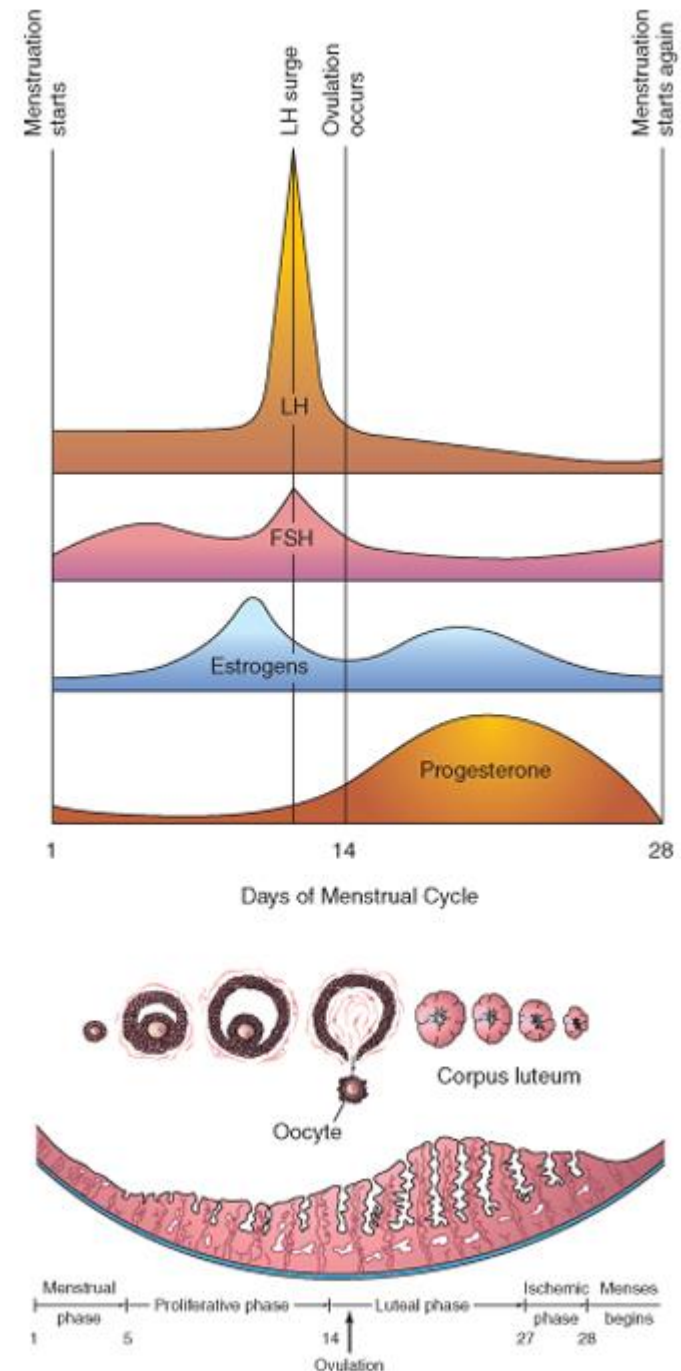


- *If the oocyte is fertilized, the corpus luteum*
 - *enlarges to form a corpus luteum of pregnancy*
 - and increases its hormone production
 - Degeneration of the corpus luteum is prevented by *human chorionic gonadotropin*, a hormone secreted by the syncytiotrophoblast of the blastocyst
 - The corpus luteum of pregnancy remains functionally active throughout the **first 20 weeks of pregnancy**
 - By this time, the placenta has assumed the production of the estrogen and progesterone that is necessary for the maintenance of pregnancy

- *If the oocyte is not fertilized, the corpus luteum*
 - *involutates and degenerates 10 to 12 days after ovulation.*
 - *It is then called a corpus luteum of menstruation.*
- **The corpus luteum:**
 - is subsequently transformed into white scar tissue in the ovary, a ***corpus albicans***
 - Except during pregnancy, ovarian cycles normally persist throughout the reproductive life of women and terminate at **menopause**
 - the permanent cessation of menstruation, usually between the ages of 48 and 55 years
 - The endocrine, somatic (body), and psychological changes occurring at the termination of the reproductive period are called the **climacteric**.

• MENSTRUAL CYCLE

- is the time during which the oocyte matures, is ovulated, and enters the uterine tube
- The hormones produced by the ovarian follicles and corpus luteum (estrogen and progesterone) produce cyclic changes in the endometrium
- These monthly changes in the internal layer of the uterus constitute the **endometrial cycle**, commonly referred to as the **menstrual cycle** or period because menstruation



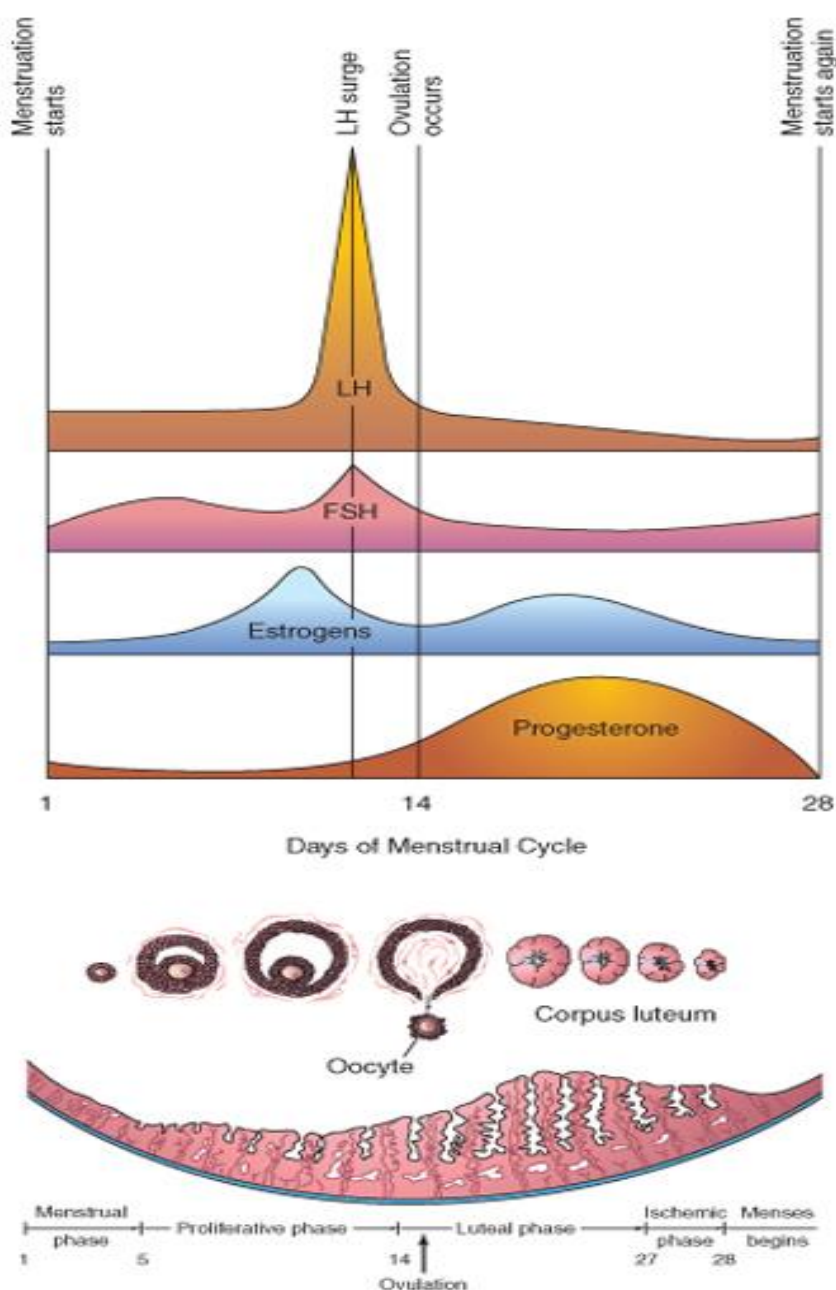


Figure 2-11 Illustration of the blood levels of various hormones during the menstrual cycle.

- Follicle-stimulating hormone (FSH) stimulates the ovarian follicles to develop and produce estrogens.
- The level of estrogens rises to a peak just before the luteinizing hormone (LH) surge.
- Ovulation normally occurs 24 to 36 hours after the LH surge.
- If fertilization does not occur, the blood levels of circulating estrogens and progesterone fall.
- This hormone withdrawal causes the endometrium to regress and menstruation to start again.

Phases of the Menstrual Cycle

- Changes in the estrogen and progesterone levels cause cyclic changes in the structure of the female reproductive tract, notably the endometrium
- the menstrual cycle is divided into three main phases for descriptive purposes
- *the menstrual cycle is a continuous process*
- *each phase gradually passes into the next one*

- **Menstrual Phase**

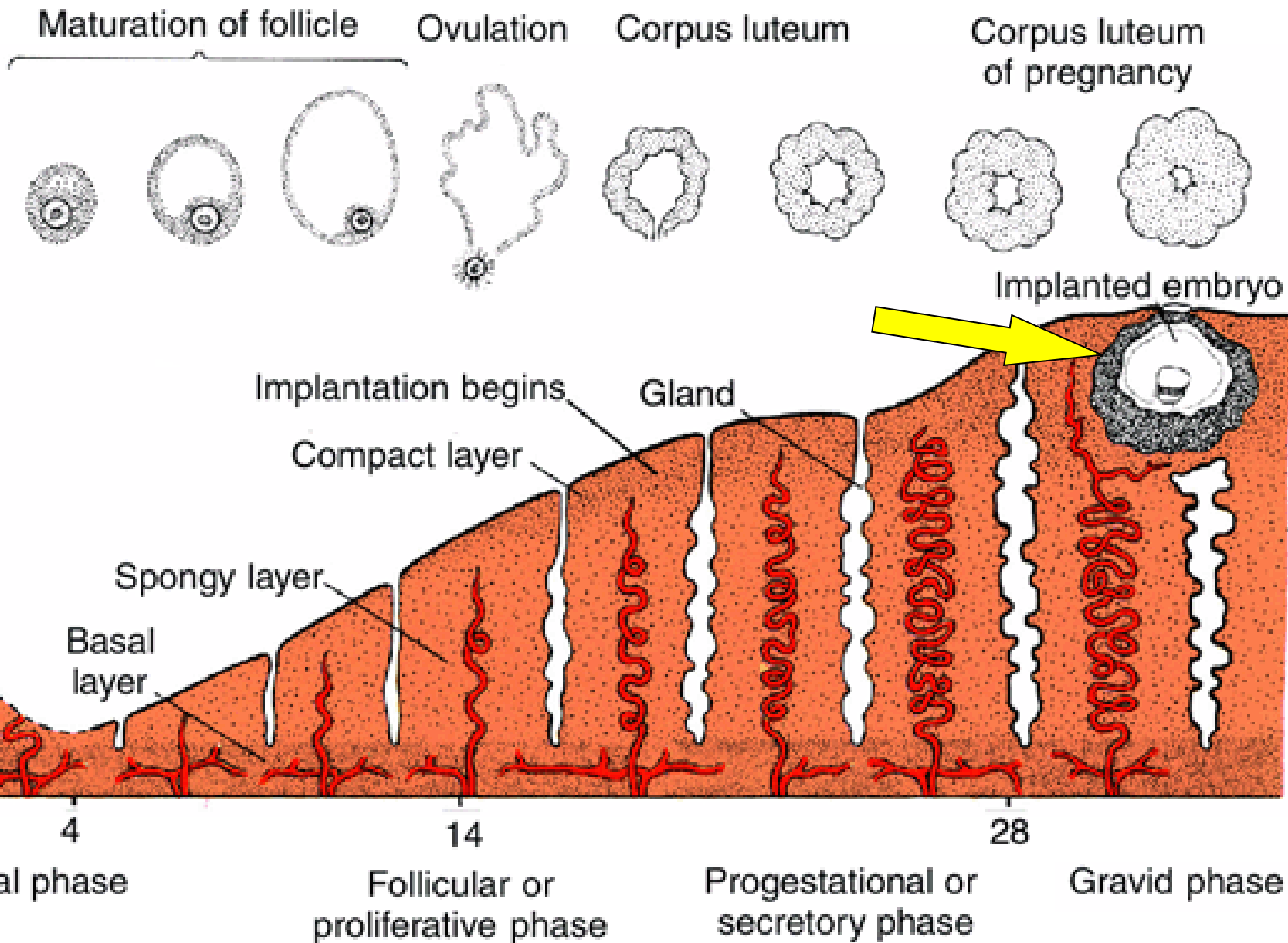
- The functional layer of the uterine wall is sloughed off
- and discarded with the menstrual flow-menses (monthly bleeding),
- usually lasts 4 to 5 days.
- The blood discharged through the vagina is combined with small pieces of endometrial tissue.
- After menstruation, the eroded endometrium is thin.

- **Proliferative Phase**

- the proliferative (follicular, estrogenic) phase
- **lasting approximately 9 days**, coincides with growth of ovarian follicles
- is controlled by estrogen secreted by these follicles
- a two- to threefold increase in the thickness of the endometrium and in its water content during this phase of repair and proliferation
- early during this phase, the surface epithelium reforms and covers the endometrium
- the glands increase in number and length, and the spiral arteries elongate

- **Luteal Phase**

- The luteal (secretory, progesterone) phase
- **lasting approximately 13 days**, coincides with the formation, functioning, and growth of the corpus luteum
- The progesterone produced by the corpus luteum stimulates the glandular epithelium to secrete a glycogen-rich material
- The glands become wide, tortuous, and saccular, and the endometrium thickens because of the influence of progesterone and estrogen from the corpus luteum and because of increased fluid in the connective tissue
- As the **spiral arteries** grow into the superficial compact layer, they become increasingly coiled
- The venous network becomes complex and large *lacunae* (venous spaces) develop.
- *Direct arteriovenous anastomoses* are prominent features of this stage.



Changes in the uterine mucosa correlated with those in the ovary. **Implantation of the blastocyst** has caused development of a large corpus luteum of pregnancy. Secretory activity of the endometrium increases gradually as a result of large amounts of progesterone produced by the

If fertilization does not occur:

- The corpus luteum degenerates.
- Estrogen and progesterone levels fall
- The secretory endometrium enters an ischemic phase.
- Menstruation occurs.

- **Ischemic Phase**

- The ischemic phase occurs when the oocyte is not fertilized.
- *Ischemia* (reduced blood supply) occurs as the spiral arteries constrict, giving the endometrium a pale appearance.

- **If fertilization occurs:**

- Cleavage of the zygote and blastogenesis (formation of blastocyst) occur.
- The blastocyst begins to implant in the endometrium on approximately the **sixth day** of the luteal phase (day 20 of a 28-day cycle).
- Human chorionic gonadotropin, a hormone produced by the syncytiotrophoblast, keeps the corpus luteum secreting estrogens and progesterone.
- The luteal phase continues and menstruation does not occur.

FERTILIZATION

- ***What is Fertilization?***
- *The usual site of fertilization is the ampulla of the uterine tube, its longest and widest part.*
- If the oocyte is not fertilized here, it slowly passes along the tube to the uterus, where it degenerates and is resorbed.
- ***Although fertilization may occur in other parts of the tube, it does not occur in the uterus.***
- Chemical signals (***attractants***), secreted by the oocyte and surrounding follicular cells, guide the capacitated sperms (***sperm chemotaxis***) to the oocyte.

- Fertilization:
 - is a complex sequence of coordinated molecular events that begins with contact between a sperm and an oocyte
 - ends with the intermingling of maternal and paternal chromosomes at ***metaphase of the first mitotic division*** of the zygote, a unicellular embryo.
 - Defects at any stage in the sequence of these events might cause the zygote to die
 - The fertilization process takes approximately **24 hours**.
 - ***Transgenic and gene knockout studies*** in animals have shown that carbohydrate binding molecules and ***gamete-specific proteins*** on the surface of the sperms are involved in ***sperm-egg recognition*** and their union.

Phases of Fertilization

Fertilization is a sequence of coordinated events:

1. Passage of a sperm through the corona radiata.

- Dispersal of the follicular cells of the corona radiata surrounding the oocyte and zona pellucida appears to result mainly from the action of the enzyme *hyaluronidase* released from the acrosome of the sperm
- *Tubal mucosal enzymes* also appear to assist the dispersal.
- Movements of the tail of the sperm are also important in its penetration of the corona radiata.

2. Penetration of the zona pellucida.

- is the important phase in the initiation of fertilization
- Formation of a pathway also results from the action of enzymes released from the acrosome.
- The enzymes ***esterases***, ***acrosin***, and ***neuraminidase*** appear to cause lysis of the zona pellucida, thereby forming a path for the sperm to follow to the oocyte.
- The most important of these enzymes is **acrosin**, a proteolytic enzyme.
- Once the sperm penetrates the zona pellucida, a **zona reaction**-a change in the properties of the zona pellucida-occurs that makes it impermeable to other sperms.
- The composition of this extracellular glycoprotein coat changes after fertilization.

The zona reaction:

- is believed to result from the action of lysosomal enzymes released by cortical granules near the plasma membrane of the oocyte
- The contents of these granules, which are released into the perivitelline space, also cause changes in the plasma membrane that make it impermeable to other sperms.

3. Fusion of plasma membranes of the oocyte and sperm.

- The plasma or cell membranes of the oocyte and sperm fuse break down at the area of fusion.
- The head and tail of the sperm enter the cytoplasm of the oocyte, but the sperm's plasma membrane remains behind.

4. Completion of the second meiotic division of oocyte and formation of female pronucleus.

- Penetration of the oocyte by a sperm activates the oocyte into completing the second meiotic division and forming a mature oocyte **and a second polar body**.
- Following decondensation of the maternal chromosomes, the nucleus of the mature oocyte becomes **the female pronucleus**.

5. Formation of the male pronucleus

- Within the cytoplasm of the oocyte, the nucleus of the sperm enlarges to form the **male pronucleus** and the tail of the sperm **degenerates**.
- *Morphologically, the male and female pronuclei are indistinguishable.*
- During growth of the pronuclei, they replicate their DNA-1 n (haploid), 2 c (two chromatids).
- The oocyte containing two haploid pronuclei is called an **ootid**
- **As the pronuclei fuse into a single diploid aggregation of chromosomes, the ootid becomes a zygote**
- The chromosomes in the zygote become arranged on a **cleavage spindle** in preparation for cleavage of the zygote.

Fertilization, the process by which male and female gametes fuse,

- occurs in the [ampullary region of the uterine tube](#).
- Only 1% of sperm deposited in the vagina enter the cervix, where they may survive for many hours
- Movement of sperm from the cervix to the uterine tube occurs primarily by their own propulsion, although they may be assisted by movements of fluids created by uterine cilia
- The trip from cervix to oviduct requires a minimum of 2 to 7 hours, and after reaching the isthmus, sperm become less motile and cease their migration.

- At ovulation, sperm again become motile, perhaps because of **chemoattractants** produced by cumulus cells surrounding the egg, and swim to the ampulla, where fertilization usually occurs.
- Spermatozoa are not able to fertilize the oocyte immediately upon arrival in the female genital tract but must undergo
 1. Capacitation and
 2. Acrosome reaction to acquire this capability.

- **Capacitation:**

- is a period of conditioning in the female reproductive tract that in the human lasts **approximately 7 hours**
- occurs in the uterine tube, entails epithelial interactions between the sperm and mucosal surface of the tube
- During this time, a ***glycoprotein coat and seminal plasma proteins are removed from the plasma membrane that overlies the acrosomal region of the spermatozoa.***
- Only capacitated sperm can pass through the corona cells and undergo the acrosome reaction.

- The **acrosome reaction**

- occurs after binding to the zona pellucida,
- is induced by zona proteins.
- This reaction culminates in the ***release of enzymes*** needed to penetrate the zona pellucida, including acrosin- and trypsin-like substances.

Phase 1:

Penetration of the Corona Radiata

- Of the **200 to 300** million spermatozoa deposited in the female genital tract, only **300 to 500** reach the site of fertilization.
- Only **one** of these fertilizes the egg.
- It is thought that the others aid the fertilizing sperm in penetrating the barriers protecting the female gamete.
- Capacitated sperm pass freely through corona cells.

Phase 2:

- **Penetration of the Zona Pellucida**

- The zona is a glycoprotein shell surrounding the egg that facilitates and maintains sperm binding and induces the *acrosome reaction*.
- Both binding and the acrosome reaction are mediated by the ligand ZP3, a zona protein.
- Release of acrosomal enzymes (acrosin) allows sperm to penetrate the zona, thereby coming in contact with the plasma membrane of the oocyte.

Phase 3:

Fusion of the Oocyte and Sperm Cell Membranes

- The initial adhesion of sperm to the oocyte is mediated in part by the interaction of integrins on the oocyte and their ligands, disintegrins, on sperm.
- After adhesion, the plasma membranes of the sperm and egg fuse.
- Because the plasma membrane covering the acrosomal head cap disappears during the acrosome reaction, actual fusion is accomplished between the oocyte membrane and the membrane that covers the posterior region of the sperm head.
- In the human, both the head and tail of the spermatozoon enter the cytoplasm of the oocyte, but the plasma membrane is left behind on the oocyte surface.

- As soon as the spermatozoon has entered the oocyte, the egg responds in three ways:

a. Cortical and zona reactions.

As a result of the release of cortical oocyte granules, which contain lysosomal enzymes,

- the oocyte membrane becomes impenetrable to other spermatozoa
- the zona pellucida alters its structure and composition to prevent sperm binding and penetration.
- These reactions prevent polyspermy (penetration of more than one spermatozoon into the oocyte).

b. Resumption of the second meiotic division.

- The oocyte finishes its second meiotic division immediately after entry of the spermatozoon.
- One of the daughter cells, which receives hardly any cytoplasm, is known as the second polar body; the other daughter cell is the definitive oocyte.
- Its chromosomes (22 plus X) arrange themselves in a vesicular nucleus known as the female pronucleus.

b. Metabolic activation of the egg.

- The activating factor is probably carried by the spermatozoon.
- Postfusion activation may be considered to encompass the initial cellular and molecular events associated with early embryogenesis.

Illustrations of fertilization, the procession of events beginning when the sperm contacts the secondary oocyte's plasma membrane and ending with the intermingling of maternal and paternal chromosomes at **metaphase of the first mitotic division** of the zygote.

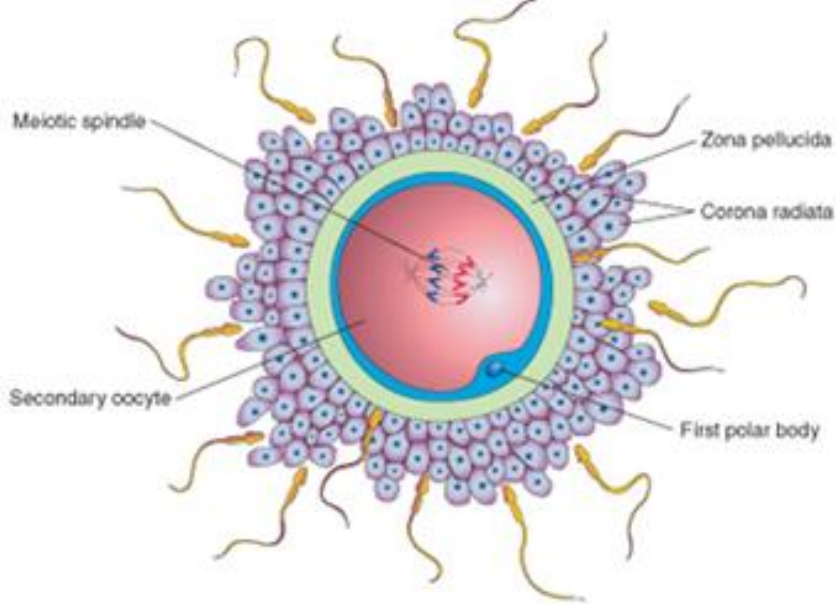
A, Secondary oocyte surrounded by several sperms, two of which have penetrated the corona radiata. (Only four of the 23 chromosome pairs are shown.)

B, The corona radiata has disappeared, a sperm has entered the oocyte, and the second meiotic division has occurred, forming a mature oocyte. The nucleus of the oocyte is now the **female pronucleus**.

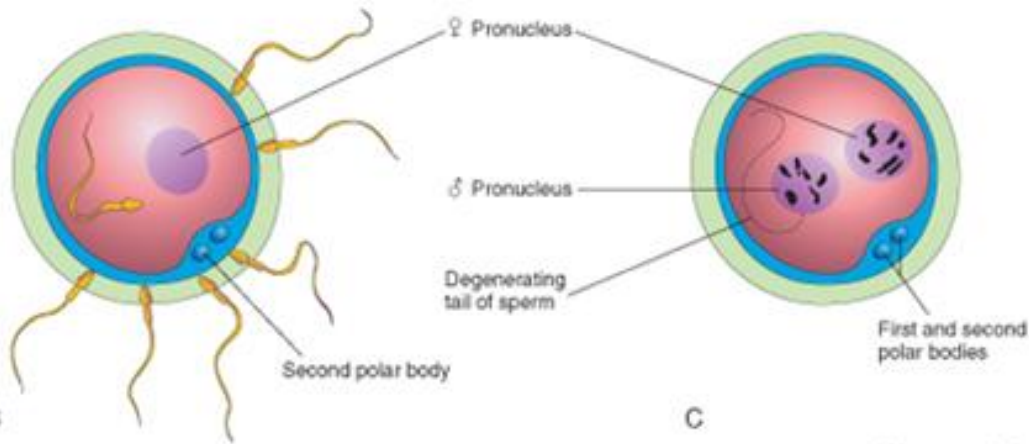
C, The sperm head has enlarged to form the **male pronucleus**.

This cell, an ootid, contains the male and female pronuclei.

D, The pronuclei are fusing.



A



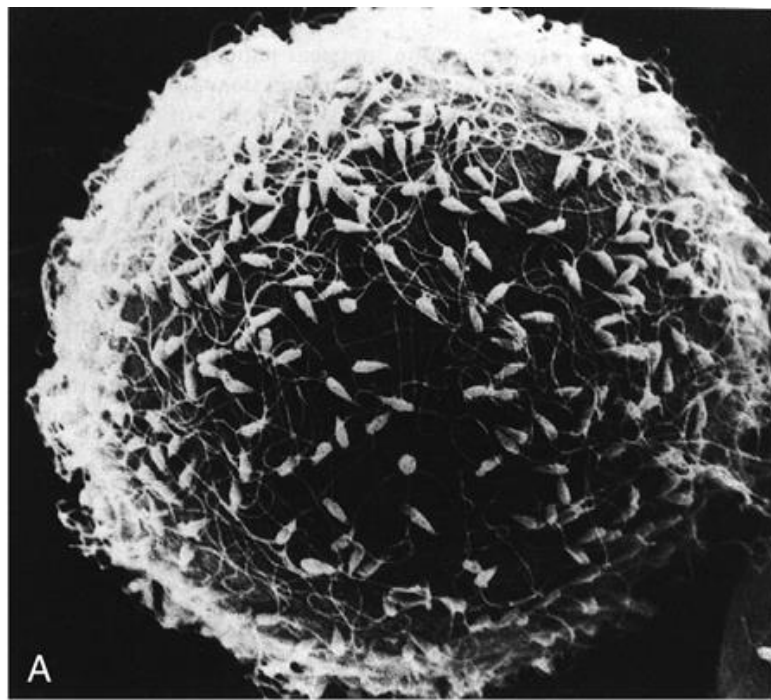
B



C



D



A. Scanning electron micrograph of sperm binding to the zona pellucida.

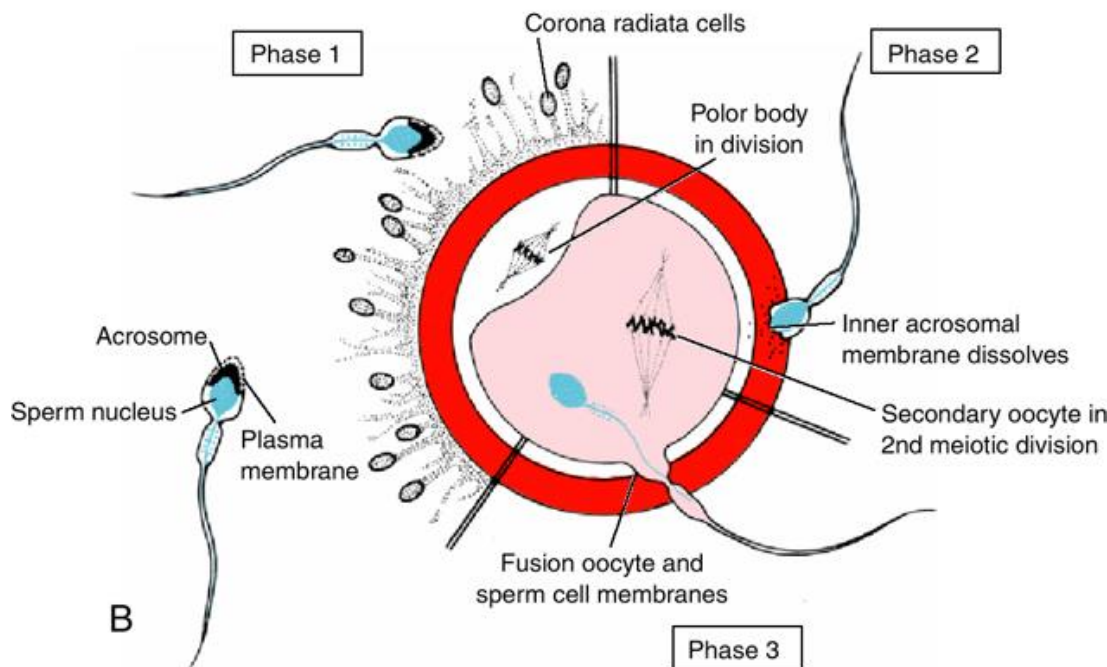
B. The three phases of oocyte penetration.

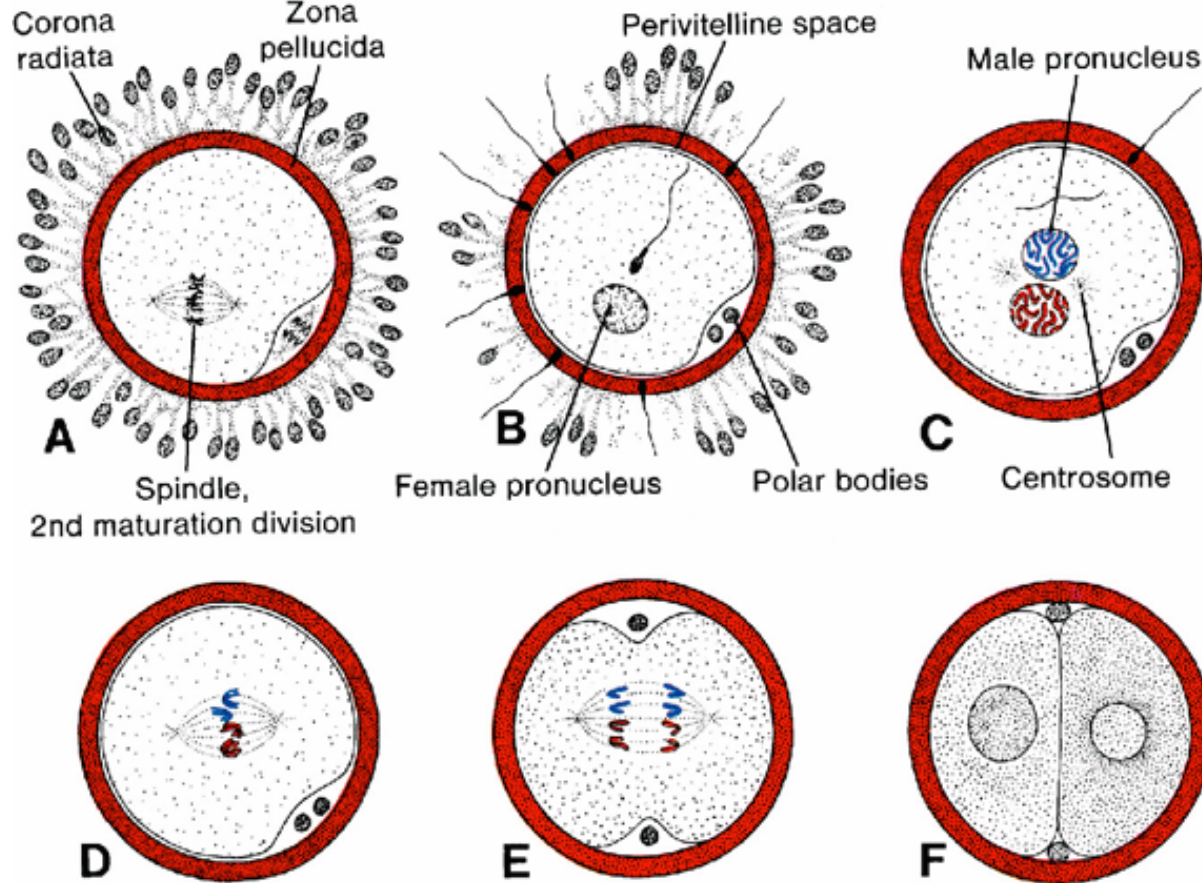
► **In phase 1,**
spermatozoa pass through the corona radiata barrier

► **In phase 2,**
one or more
spermatozoa
penetrate the zona
pellucida

► **In phase 3,**
one spermatozoon
penetrates the oocyte
membrane while losing its
own plasma membrane.

**Inset shows normal
spermatocyte with acrosomal
head cap.**





- A.** Oocyte immediately after ovulation, showing the spindle of the second meiotic division.
- B.** A spermatozoon has penetrated the oocyte, which has finished its second meiotic division. Chromosomes of the oocyte are arranged in a vesicular nucleus, the female pronucleus. Heads of several sperm are stuck in the zona pellucida.
- C.** Male and female pronuclei.
- D,E.** Chromosomes become arranged on the spindle, split longitudinally, and move to opposite poles.
- F.** Two-cell stage.

The zygote

- is genetically unique because half of its chromosomes came from the mother and half from the father.
- contains a new combination of chromosomes that is different from that in the cells of either of the parents.
- This mechanism forms the basis of biparental inheritance and variation of the human species.

Meiosis

- Allows ***independent assortment*** of maternal and paternal chromosomes among the germ cells.
- ***Crossing over*** of chromosomes, by relocating segments of the maternal and paternal chromosomes, "shuffles" the genes, thereby producing a recombination of genetic material.
- The ***embryo's chromosomal sex*** is determined at fertilization by the kind of sperm (X or Y) that fertilizes the oocyte.
- Fertilization by an X-bearing sperm produces a 46, XX zygote, which normally develops into a female, whereas
- fertilization by a Y-bearing sperm produces a 46, XY zygote, which normally develops into a male.

Significance of Fertilization

1. Stimulates the penetrated oocyte to complete the second meiotic division
2. Restores the normal diploid number of chromosomes (46) in the zygote
3. Results in variation of the human species through mingling of maternal and paternal chromosomes
4. Determines chromosomal sex of the embryo
5. Causes metabolic activation of the ootid and initiates cleavage (cell division) of the zygote.

- Determination of sex

$X+X \rightarrow$



$Y+X \rightarrow$



Important events in fertilization

Before:

- **Capacitation**
- **Acrosome reaction:** release of enzymes from acrosome induced by, and needed for sperm **penetration** of, the zona pellucida

During:

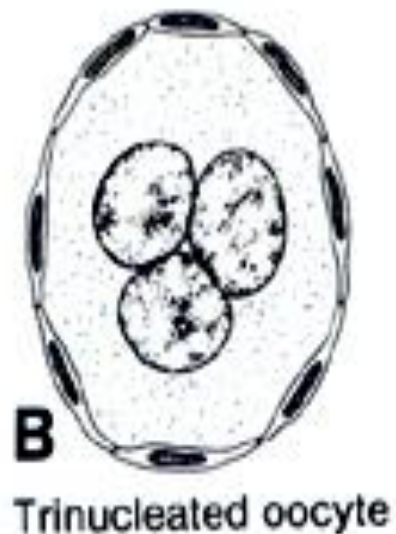
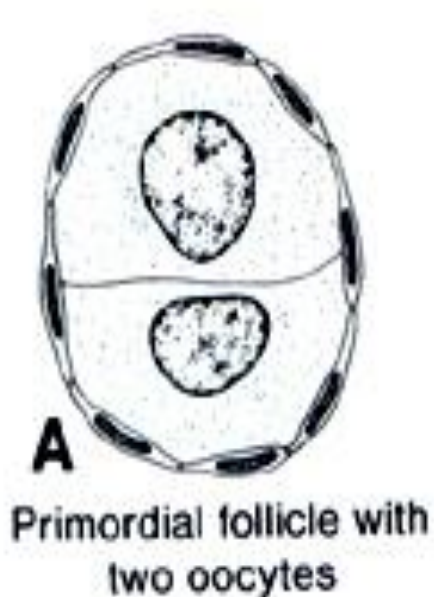
- **Penetration** of corona radiata, zona pellucida, oocyte cell membrane
- **Recognition:** a zona protein **ZP3** is responsible for **species-specific** fertilization

After entry:

- **Zona reaction:** zona becomes impenetrable to other sperm (**monospermy**), through enzyme release from cortical granules of oocyte to change structure and composition of the zona.

Requirements

- Normal **number** and **viability** of male and female gametes (sperm > 20 million/ml, abnormal $< 20\%$, viability $> 70\%$)
- Normal **transport** of gametes (occluded oviduct)
- Two gametes **meet in due time**



Preselection of the Embryo's Sex

Because X and Y sperms are formed in equal numbers, the expectation is that the sex ratio at fertilization (primary sex ratio) would be 1.00 (100 boys per 100 girls).

It is well known, however, that ***there are more male babies than female babies born in all countries.***

- In North America, for example, the sex ratio at birth (secondary sex ratio) is approximately 1.05 (105 boys per 100 girls).

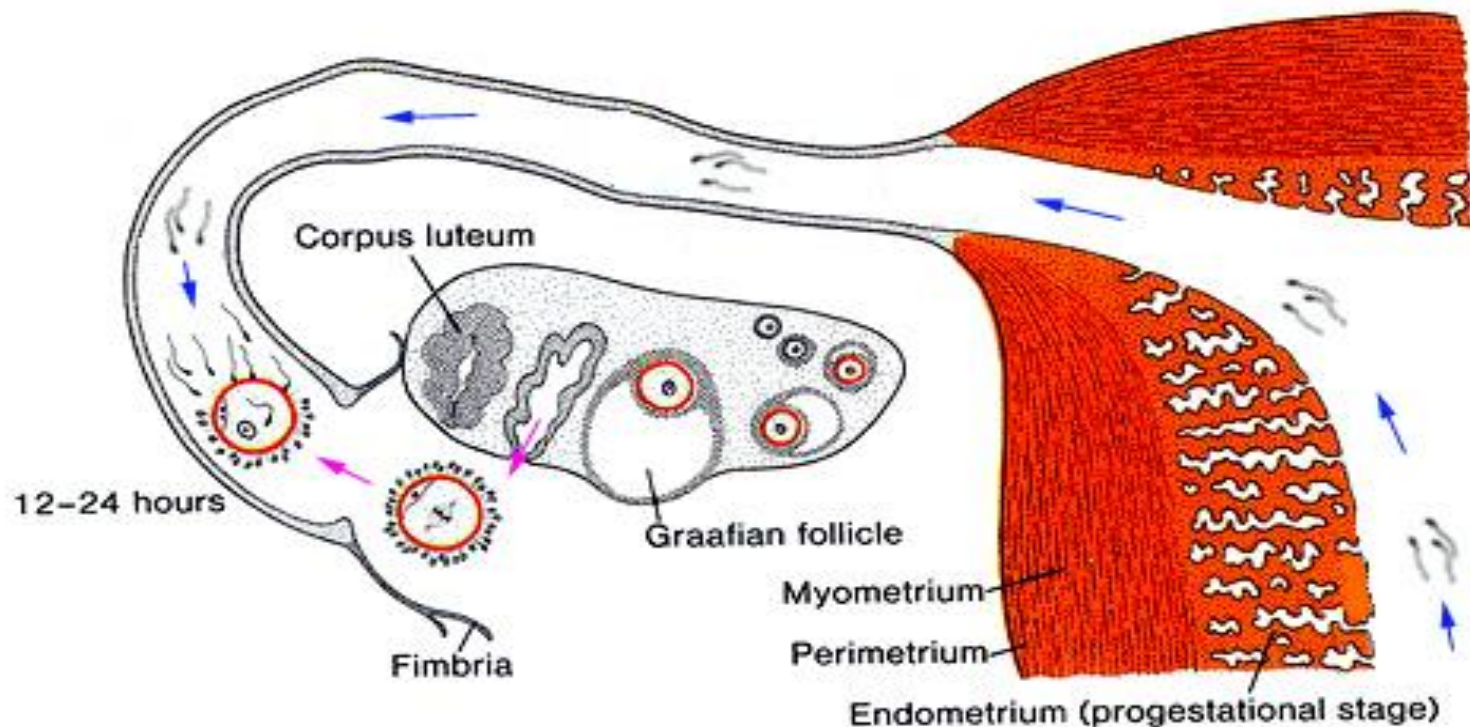
Various microscopic techniques have been developed in an attempt to separate X and Y sperms (gender selection) using:

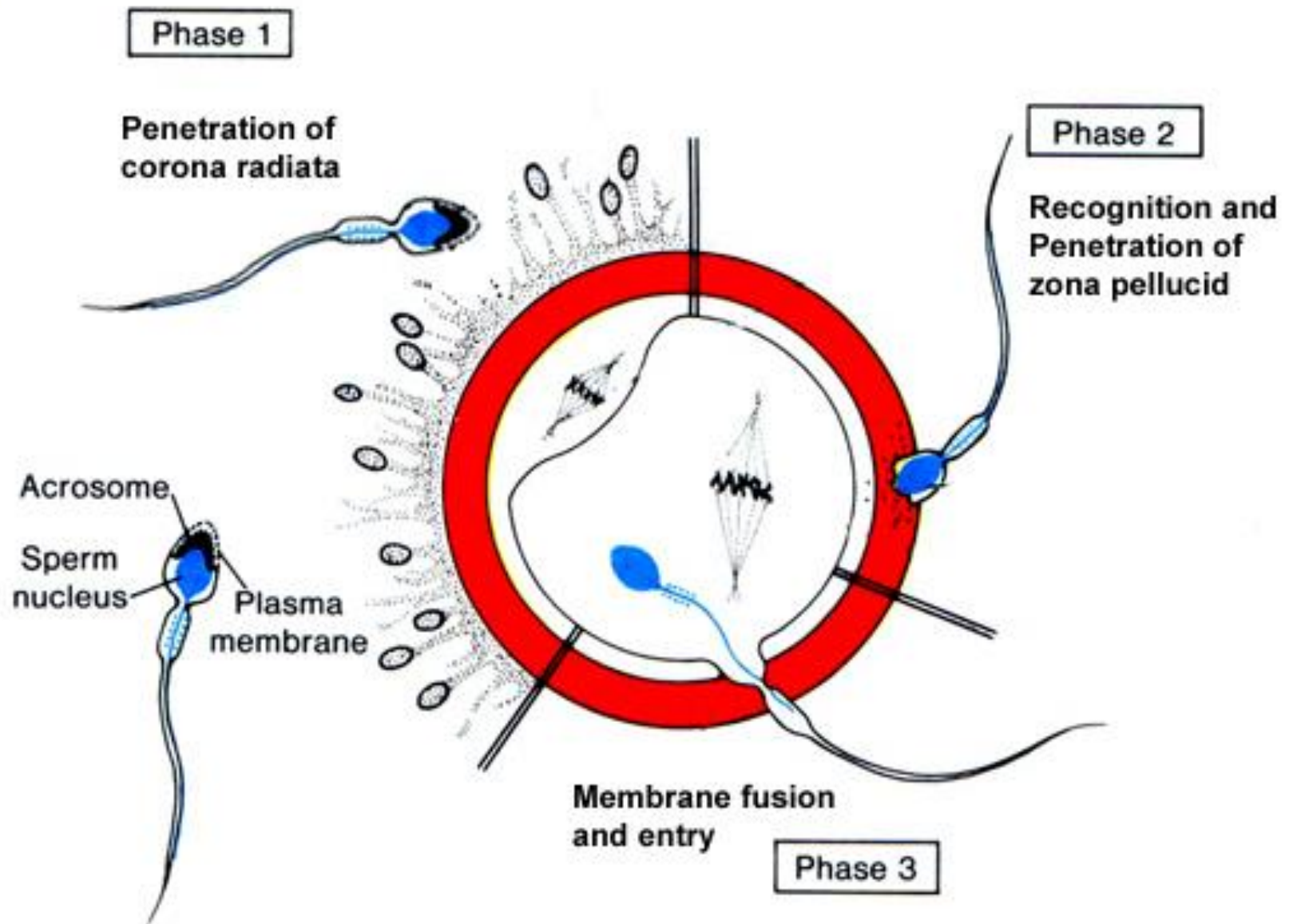
- The differential swimming abilities of the X and Y sperms.
- Different speeds of migration of sperms in an electric field.
- Differences in the appearance of X and Y sperms.
- DNA difference between X (2.8% more DNA) and Y sperms.

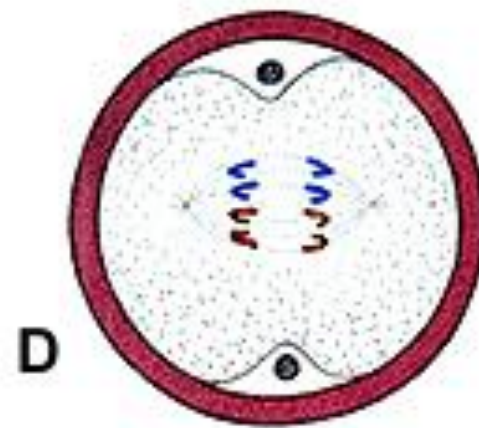
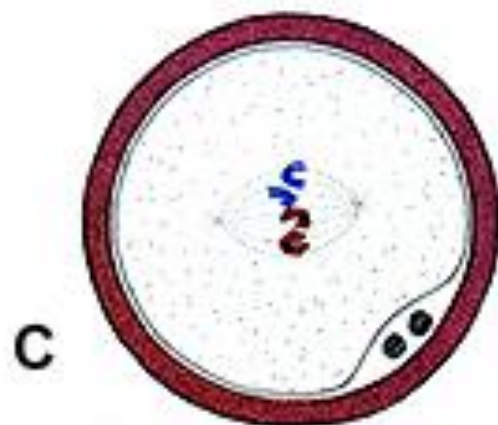
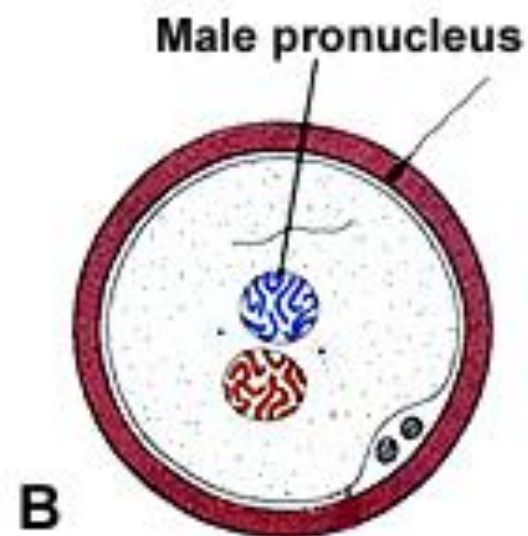
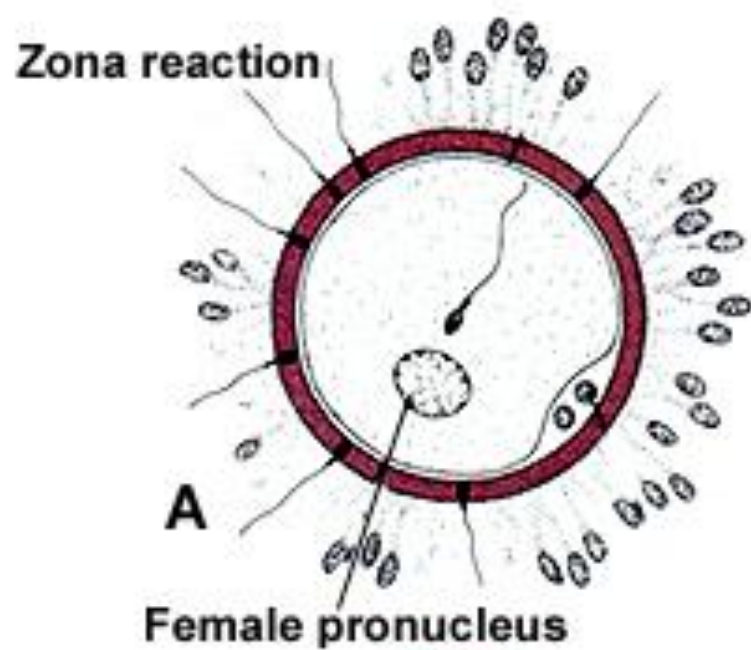
The use of a selected sperm sample in artificial insemination may produce the desired sex.

Fertilization

- union of a **sperm** and an **oocyte** to form a fertilized egg (**zygote**)
- **Time: 12 - 24 hours** after ovulation
- **Site: ampulla** of uterine tube



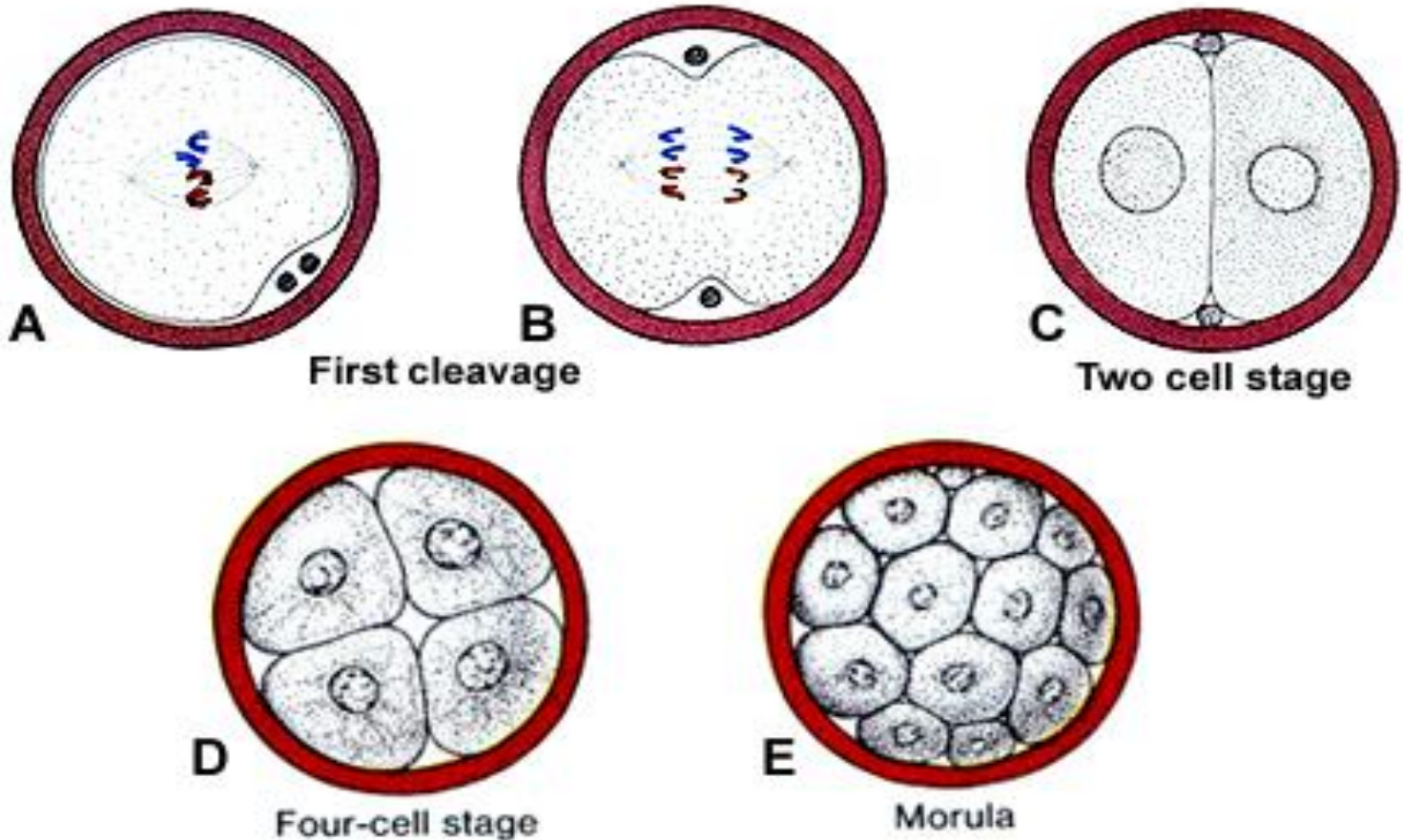




Phase 4 **Formation and fusion of pronuclei**

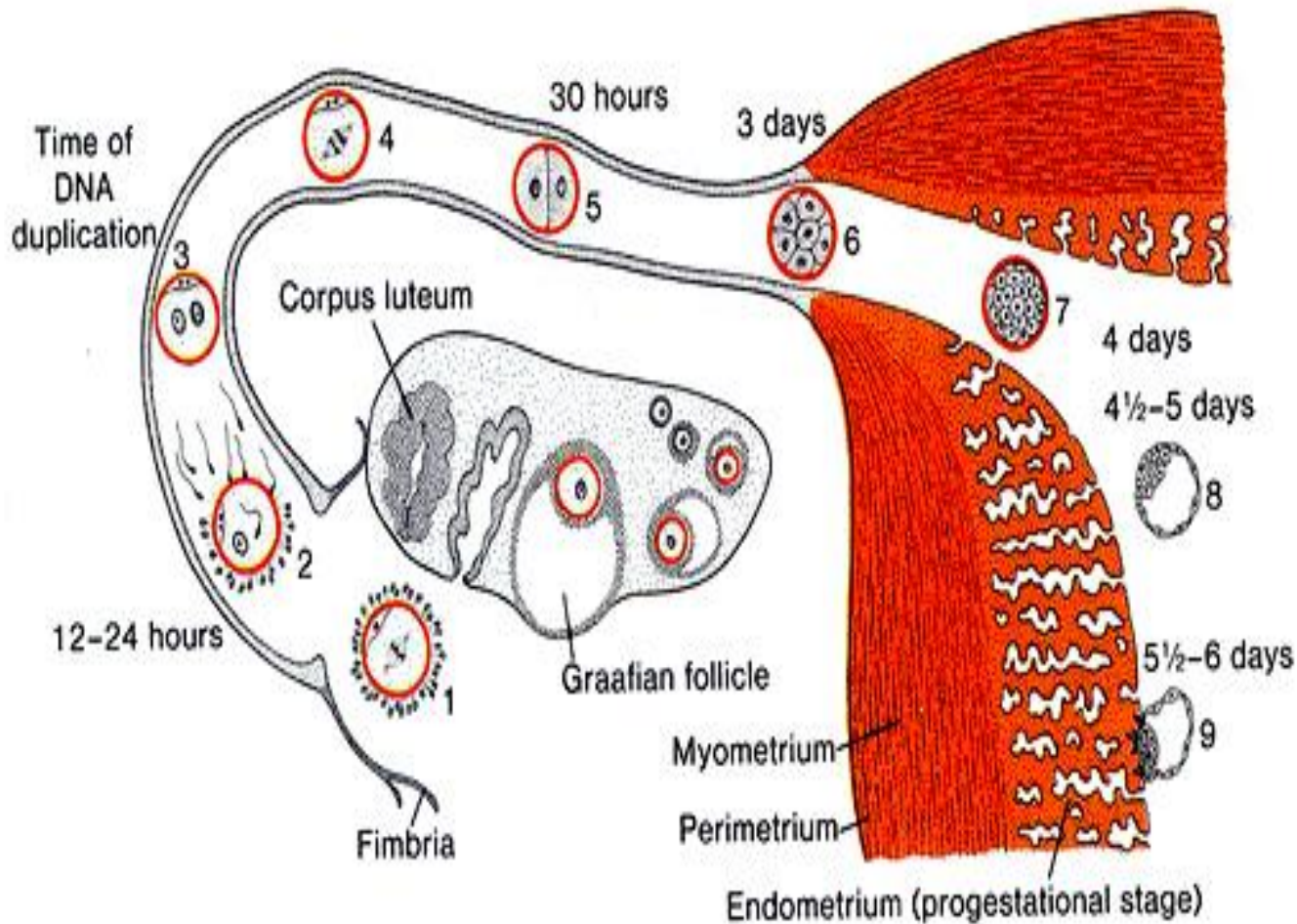
Cleavage

- a series of mitotic divisions of fertilized egg →
 - a number of smaller daughter cells, **blastomeres**
- **morula**



- **CLEAVAGE OF THE ZYGOTE**

- *cleavage consists of repeated mitotic divisions of the zygote*
- resulted in a rapid increase in the number of embryonic cells.
- these embryonic cells-**blastomeres**-become smaller with each successive cleavage division
- cleavage normally occurs as the zygote passes along the uterine tube toward the uterus



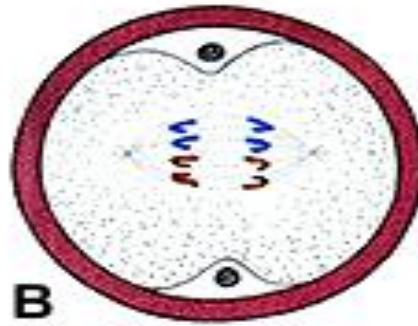
During cleavage, the zygote is within the rather thick zona pellucida. Division of the zygote into blastomeres begins approximately 30 hours after fertilization

- Subsequent cleavage divisions follow one another, forming progressively smaller blastomeres
- **After the 8-cell stage,**
 - the blastomeres change their shape
 - tightly align themselves against each other to form a compact ball of cells called **compaction**- is probably mediated by cell surface adhesion glycoproteins.
 - **Compaction permits:**
 - greater cell-to-cell interaction
 - is a prerequisite for segregation of the internal cells that form the inner cell mass or *embryoblast* of the blastocyst

- When there are 12 to 32 blastomeres, the developing human is called a **morula** (L., *morus*, mulberry)
- Internal cells of the morula (*inner cell mass*) are surrounded by a layer of cells that form the outer cell layer.
- The spherical morula forms approximately 3 days after fertilization and enters the uterus.

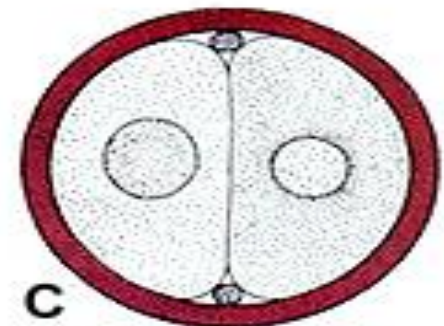


A



B

First cleavage



C

Two cell stage



D

Four-cell stage



E

Morula

Cleavage

The zygote now begins to cleave, with each division occurring into two cells called blastomeres.

The zygote's first cell division begins a series of divisions, with each division occurring approximately every twenty hours.

When cell division ungenerated about sixteen cells, the zygote becomes a **morula** (mulberry shaped).

It leaves the fallopian tube and enters the uterine cavity three to four days after fertilization.

0.1 - 0.2 mm

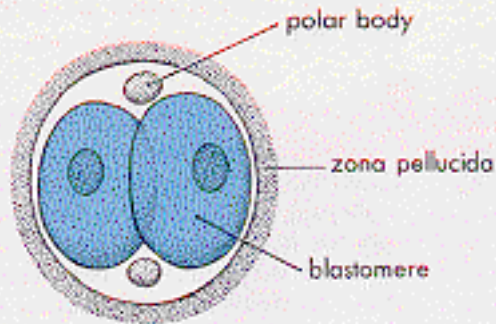
4 days post-ovulation

Blastocyst formation

Two cell types are forming:

embryoblast (inner cell mass on the inside of the blastocoele)

trophoblast (the cells on the outside of the blastocoele).



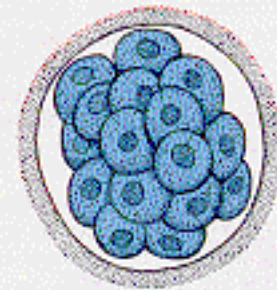
A 2-cell stage



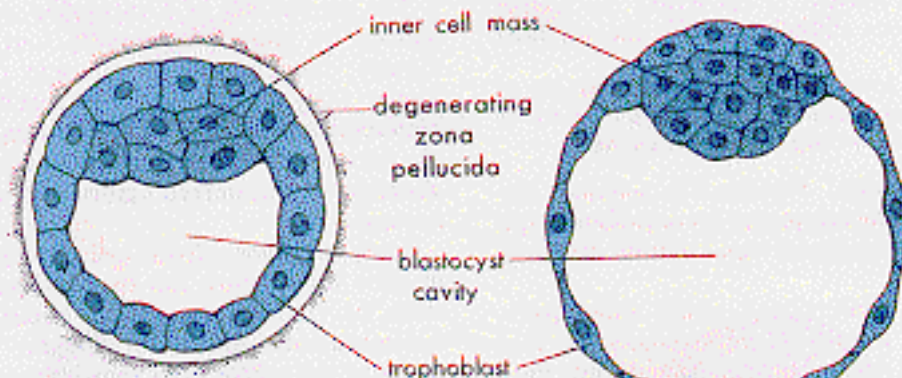
B 4-cell stage



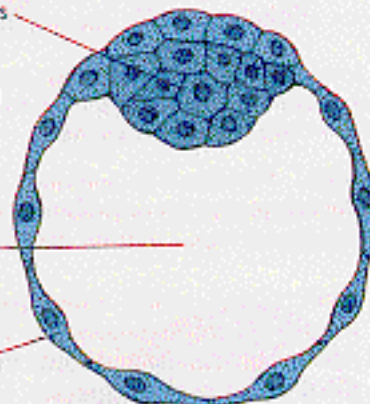
C 8-cell stage



D morula



E early blastocyst



F late blastocyst

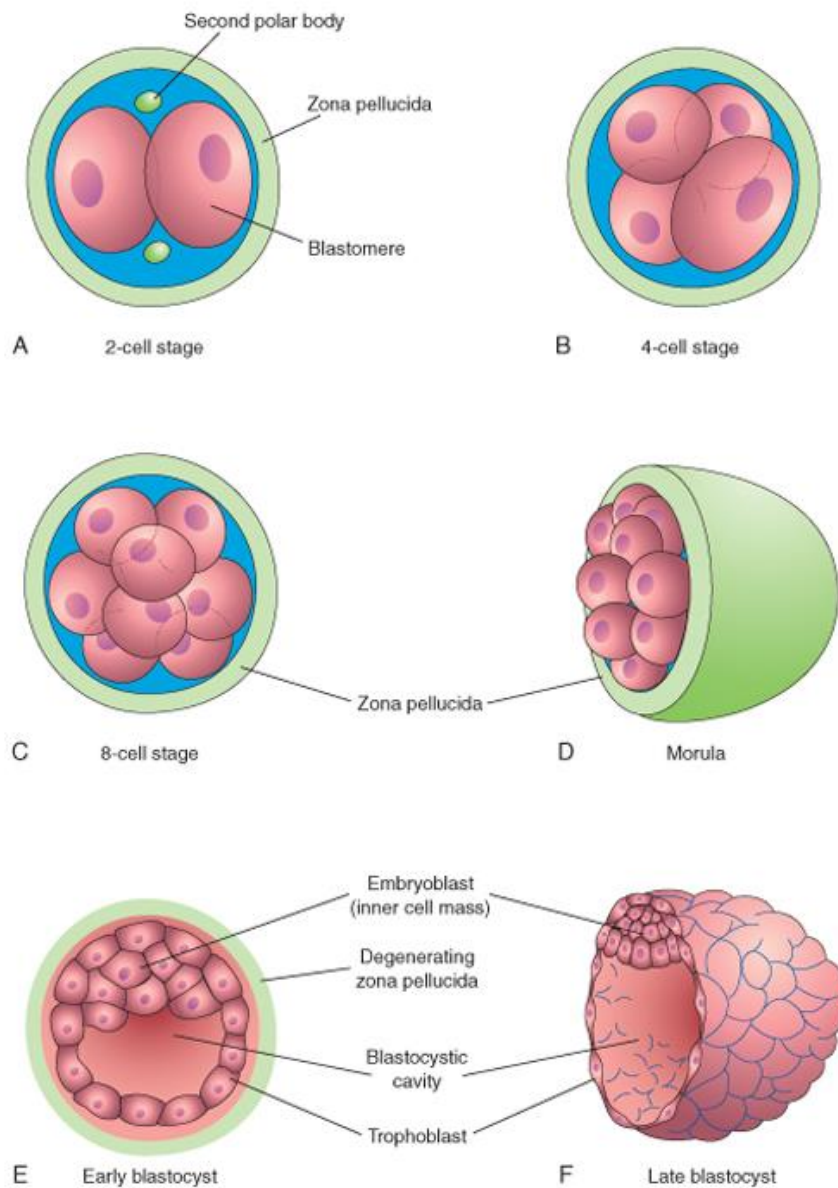


Figure 2-18 Illustrations of cleavage of the zygote and formation of the blastocyst. **A** to **D**, Various stages of cleavage. The period of the morula begins at the 12- to 16-cell stage and ends when the blastocyst forms. **E** and **F**, Sections of blastocysts. The zona pellucida has disappeared by the late blastocyst stage (5 days). The second polar bodies shown in **A** are small, nonfunctional cells that soon degenerate. Cleavage of the zygote and formation of the morula occur as the dividing zygote passes along the uterine tube. Blastocyst formation normally occurs in the uterus. Although cleavage increases the number of blastomeres, note that each of the daughter cells is smaller than the parent cells. As a result, there is no increase in the size of the developing embryo until the zona pellucida degenerates. The blastocyst then enlarges considerably

Blastocyst formation

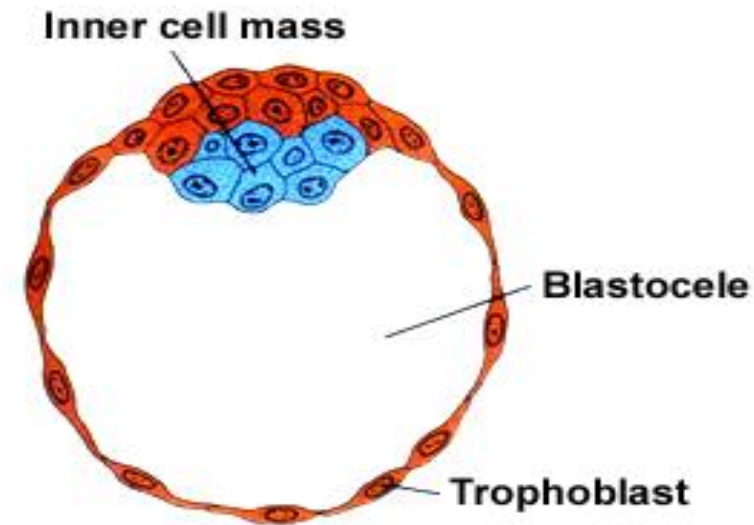
- **Formation:** **morula** → cell number ↑ → spaces appearing → **blastocyst**
- Shortly after the morula enters the uterus (approximately 4 days after fertilization), a fluid-filled space called the **blastocystic cavity** appears **inside the morula**
- fluid passes from the uterine cavity through the zona pellucida to form this space:

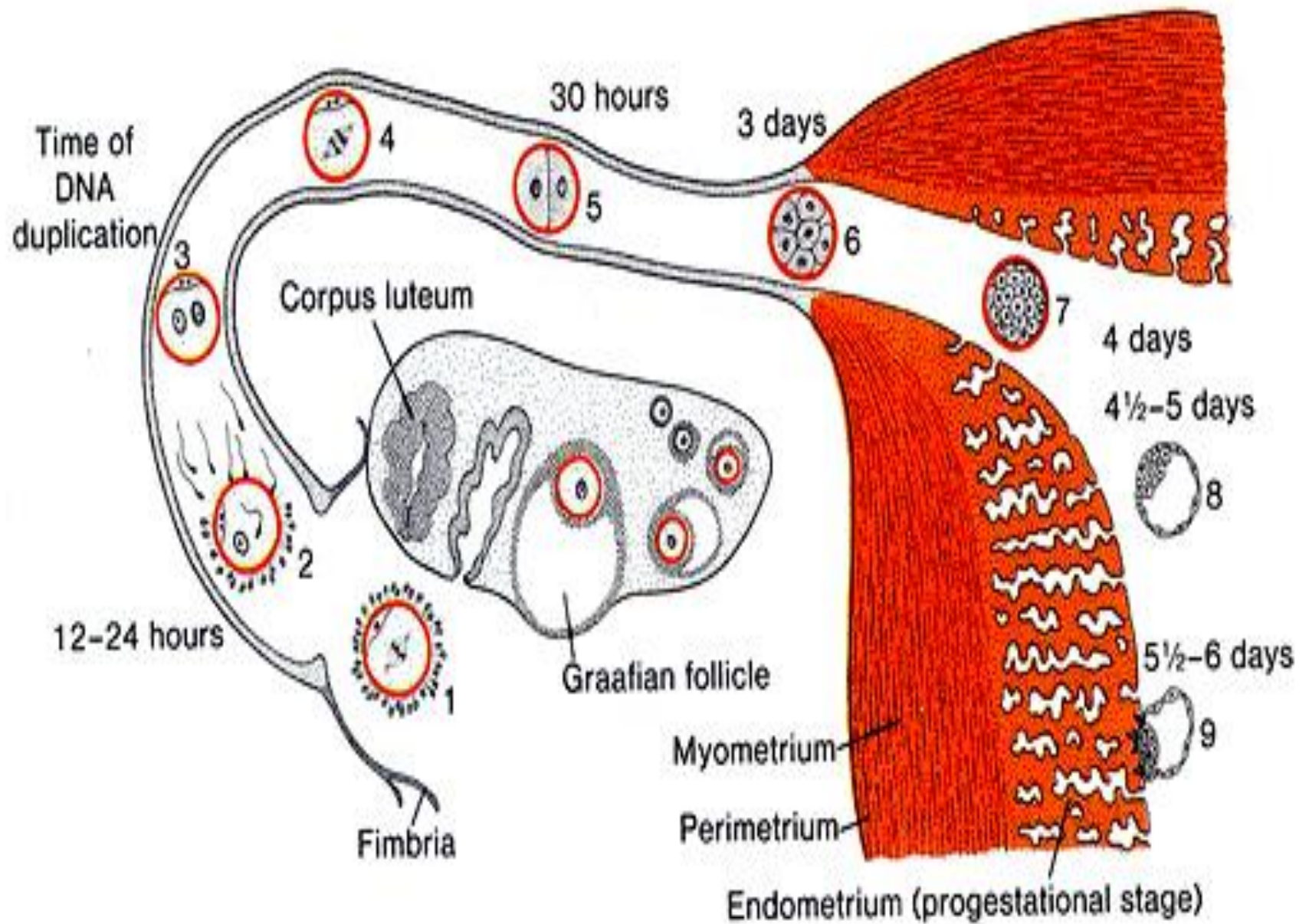
it separates the blastomeres into two parts

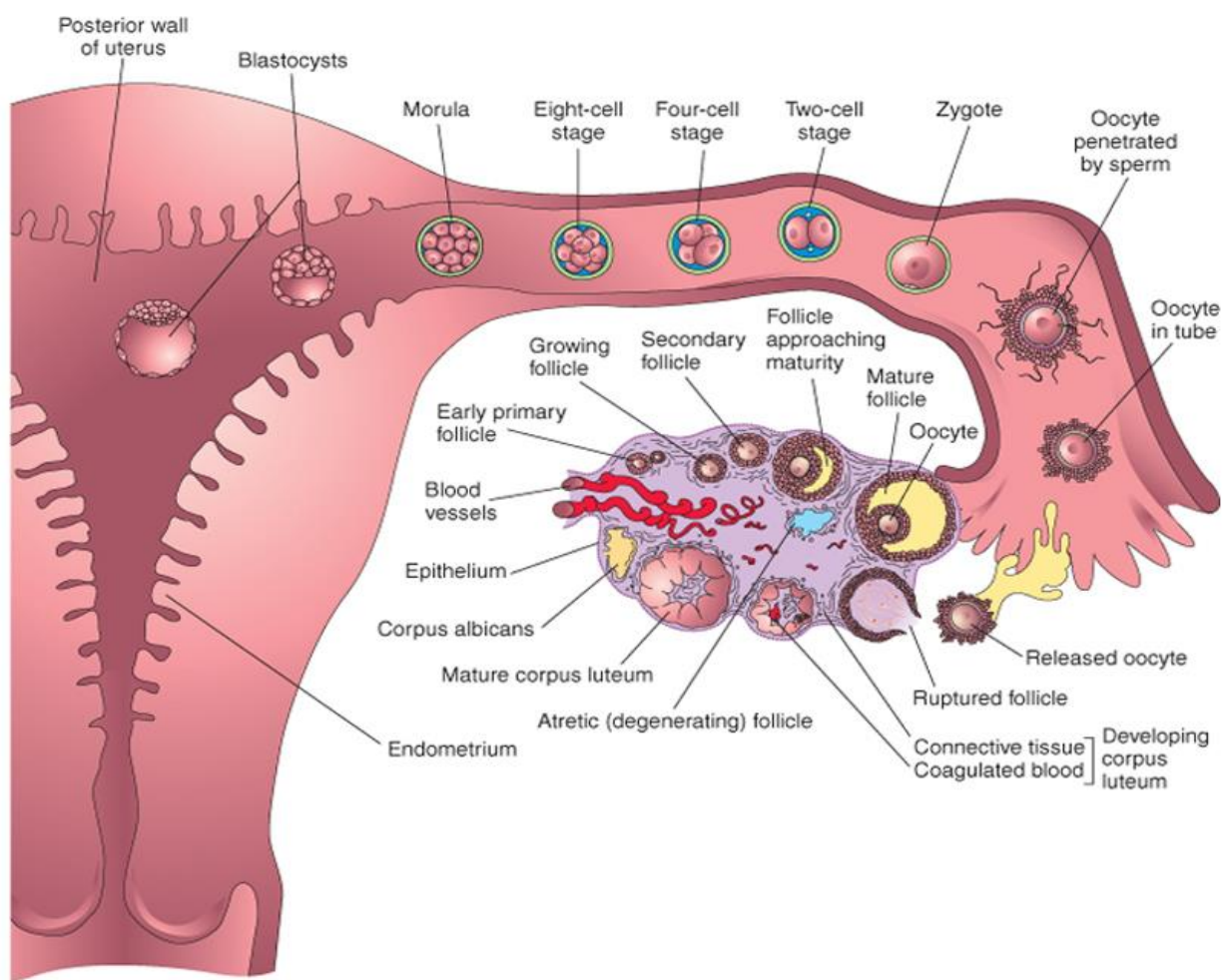
Inner cell mass (or **embryoblast**):
develops into **embryo proper**.

Outer cell mass (trophoblast): a
surrounding layer of flattened cells,
absorbs nutrients and fetal part of
the placenta

Blastocele: a single cavity







Summary of the ovarian cycle, fertilization, and human development during the first week. **Stage 1** of development begins with fertilization in the uterine tube and ends when the **zygote forms**. **Stage 2 (days 2 to 3)** comprises the early stages of cleavage (from 2 to approximately 32 cells, the morula). **Stage 3 (days 4 to 5)** consists of the free (unattached) blastocyst. **Stage 4 (days 5 to 6)** is represented by the blastocyst attaching to the posterior wall of the uterus, the usual site of implantation

- During this stage of development-**blastogenesis**-the conceptus is called a **blastocyst**
- The embryoblast now projects into the blastocystic cavity and the trophoblast forms the wall of the blastocyst.

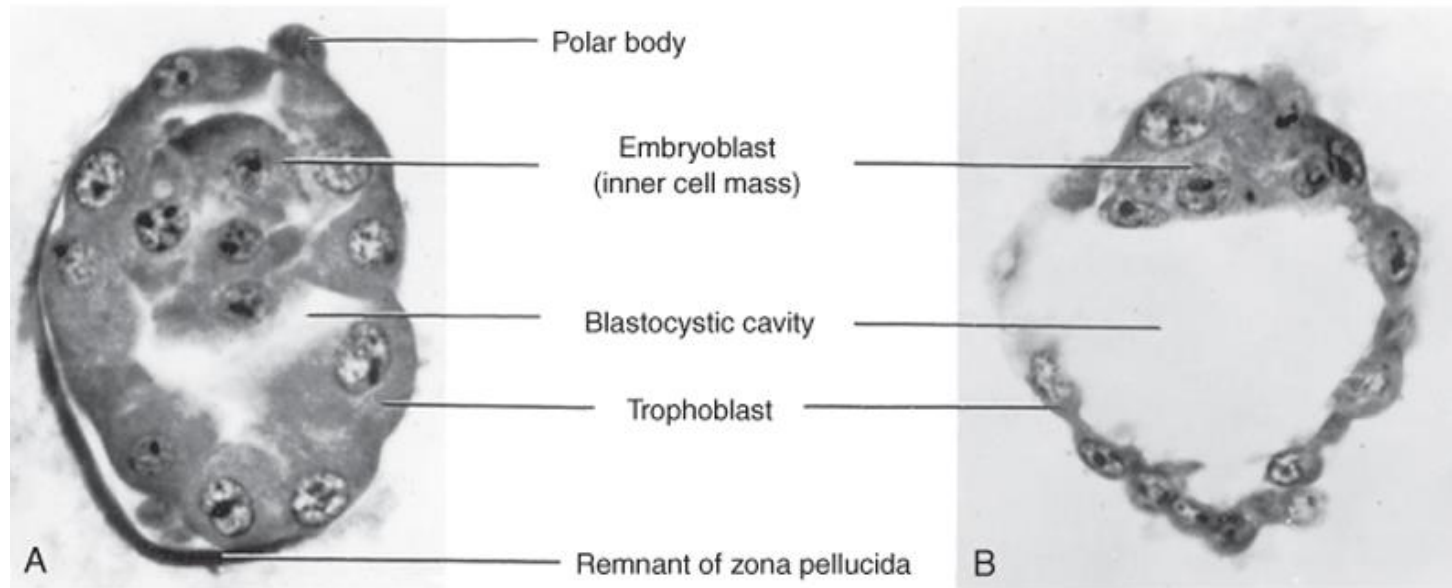
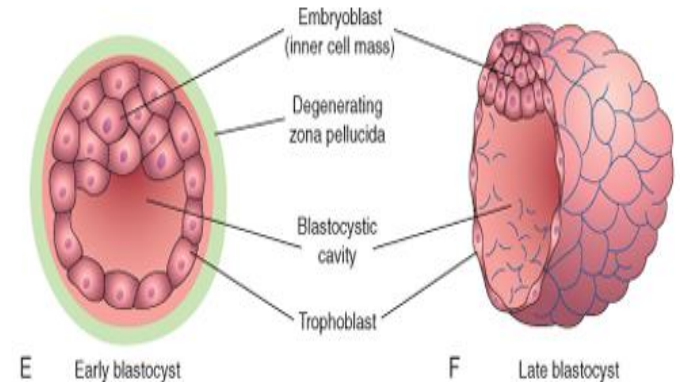
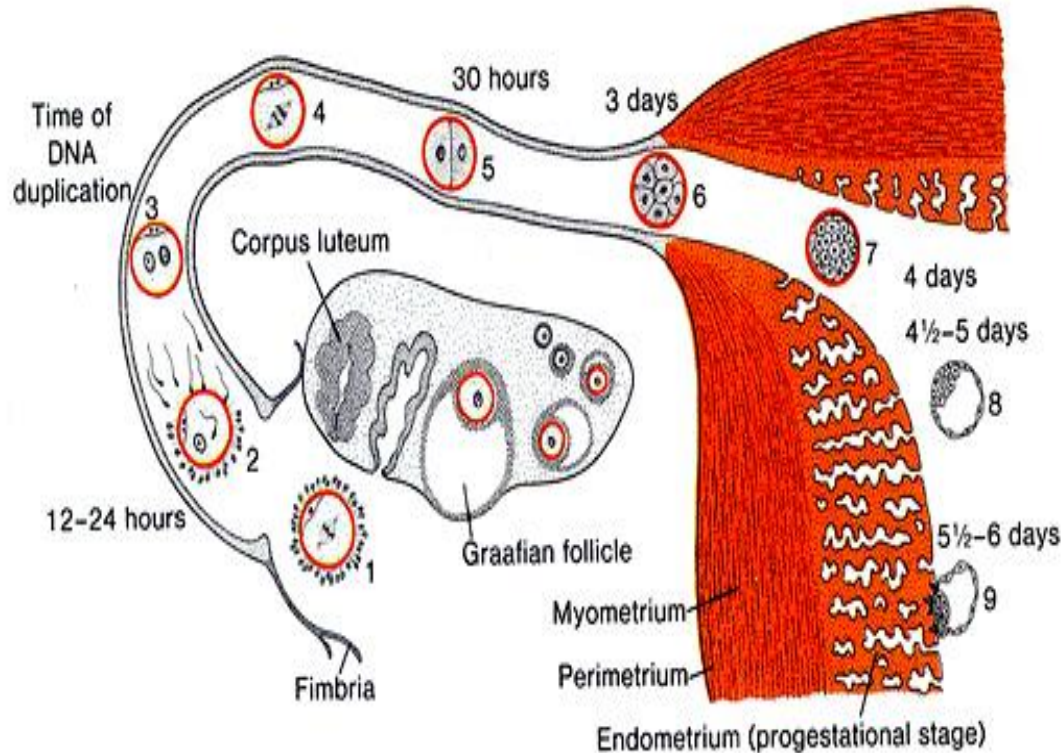


Figure 2-20 Photomicrographs of sections of human blastocysts recovered from the uterine cavity. **A**, At 4 days: the blastocystic cavity is just beginning to form and the zona pellucida is deficient over part of the blastocyst. **B**, At 4½ days; the blastocystic cavity has enlarged and the embryoblast and trophoblast are clearly defined. **The zona pellucida has disappeared.**

- After the free blastocyst has floated in the uterine secretions for approximately 2 days, the zona pellucida gradually degenerates and disappears



Shedding of the zona pellucida permits

- the hatched blastocyst to increase rapidly in size.
- While floating in the uterus, this early embryo derives nourishment from secretions of the uterine glands.

- *Approximately 6 days after fertilization* (day 20 of a 28-day menstrual cycle),
 - the blastocyst attaches to the endometrial epithelium, usually adjacent to the **embryonic pole**
 - As soon as it attaches to the endometrial epithelium,
 - the trophoblast starts to proliferate rapidly and gradually differentiates into two layers
 1. An inner layer of **cytotrophoblast**
 2. An outer layer of **syncytiotrophoblast** consisting of a multinucleated protoplasmic mass in which no cell boundaries can be observed

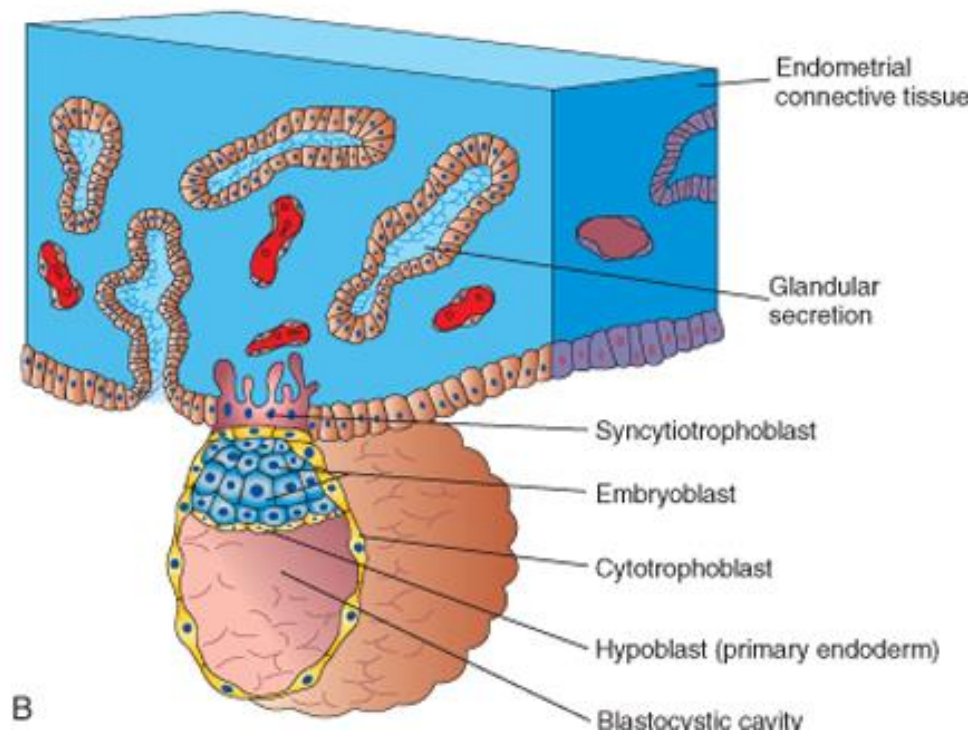
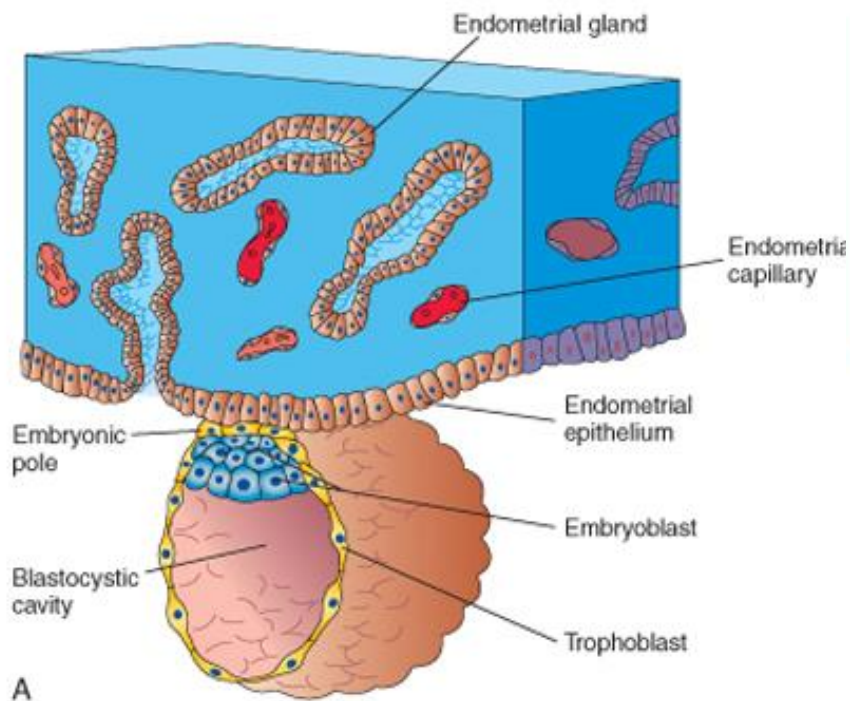
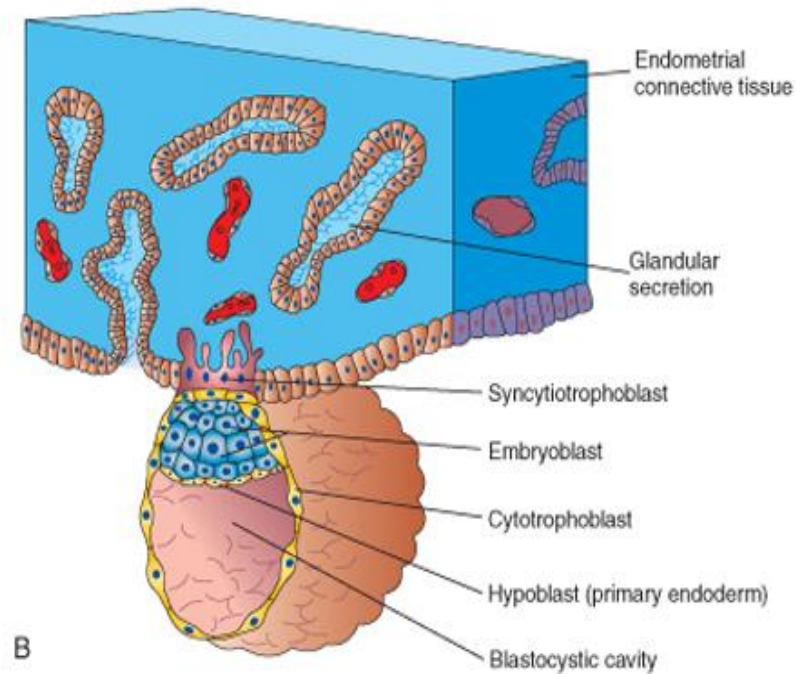


Figure 2-21 Attachment of the blastocyst to the endometrial epithelium during the early stages of its implantation. **A**, At 6 days: the trophoblast is attached to the endometrial epithelium at the embryonic pole of the blastocyst. **B**, At 7 days: the syncytiotrophoblast has penetrated the epithelium and has started to invade the endometrial connective tissue.

- **By the end of the first week:**
 - the blastocyst is superficially implanted in the compact layer of the endometrium
 - is deriving its nourishment from the eroded maternal tissues
 - The highly invasive syncytiotrophoblast expands quickly adjacent to the embryoblast, the area known as the **embryonic pole**.
 - The syncytiotrophoblast produces enzymes that erode the maternal tissues, enabling the blastocyst to burrow into the endometrium.

At approximately 7 days,

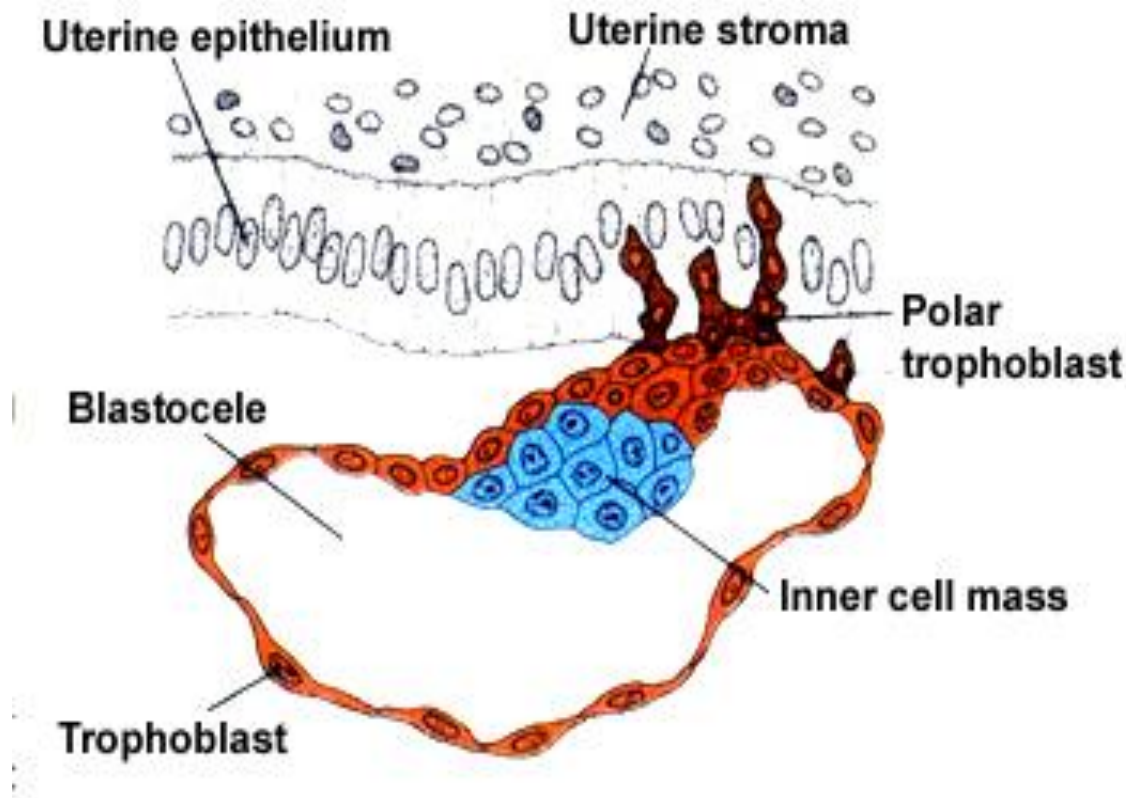
- a layer of cells, the **hypoblast** (primary endoderm) appears on the surface of the embryoblast facing the blastocystic cavity
- Comparative embryologic data suggest that the hypoblast arises by delamination of **blastomeres** from the **embryoblast**.



B, At 7 days: the syncytiotrophoblast has penetrated the epithelium and has started to invade the endometrial connective tissue.

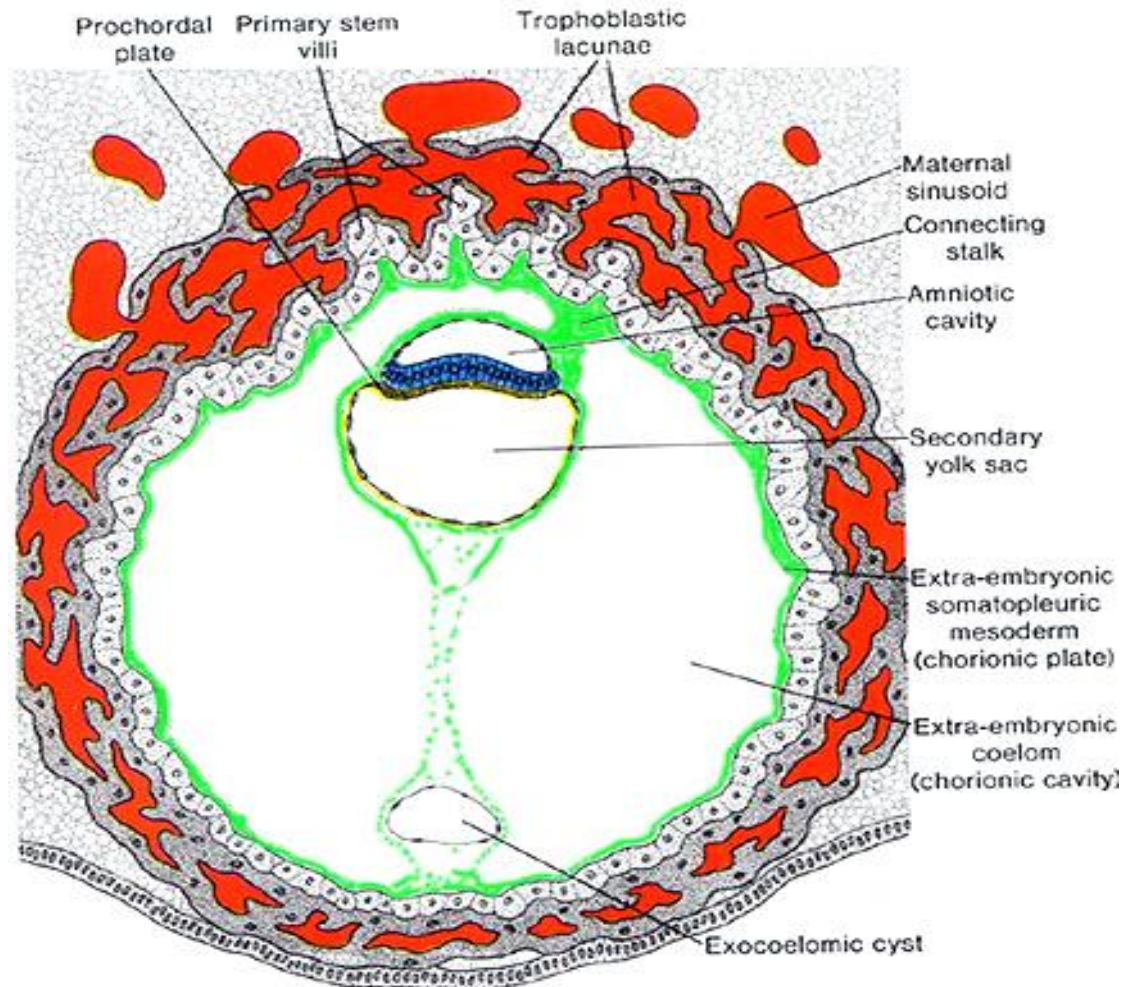
Implantation

- process of **blastocyst** embedding into **endometrium**
 - **Time**: Day 5 - 6 to Day 11 - 12.
 - **Site**: along posterior or anterior wall of uterine body.
- **Process**



Requirements

- 1) **Zona pellucida disappears** in time.
- 2) **Normal development** and **transport** of young embryo
- 3) Endometrium in **secretory phase**.
- 4) Normal **endocrine regulation**



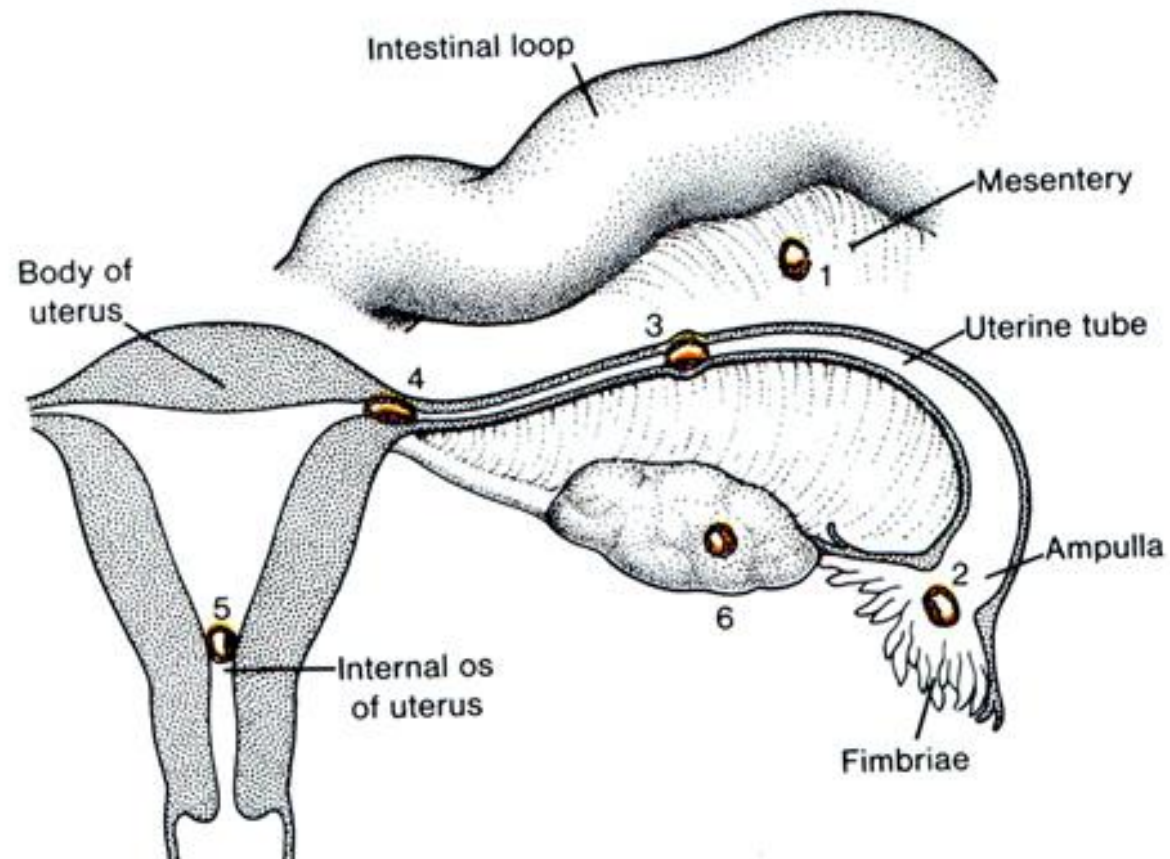
Abnormal implantation

1) Placenta previa:

Implantation near the internal os.

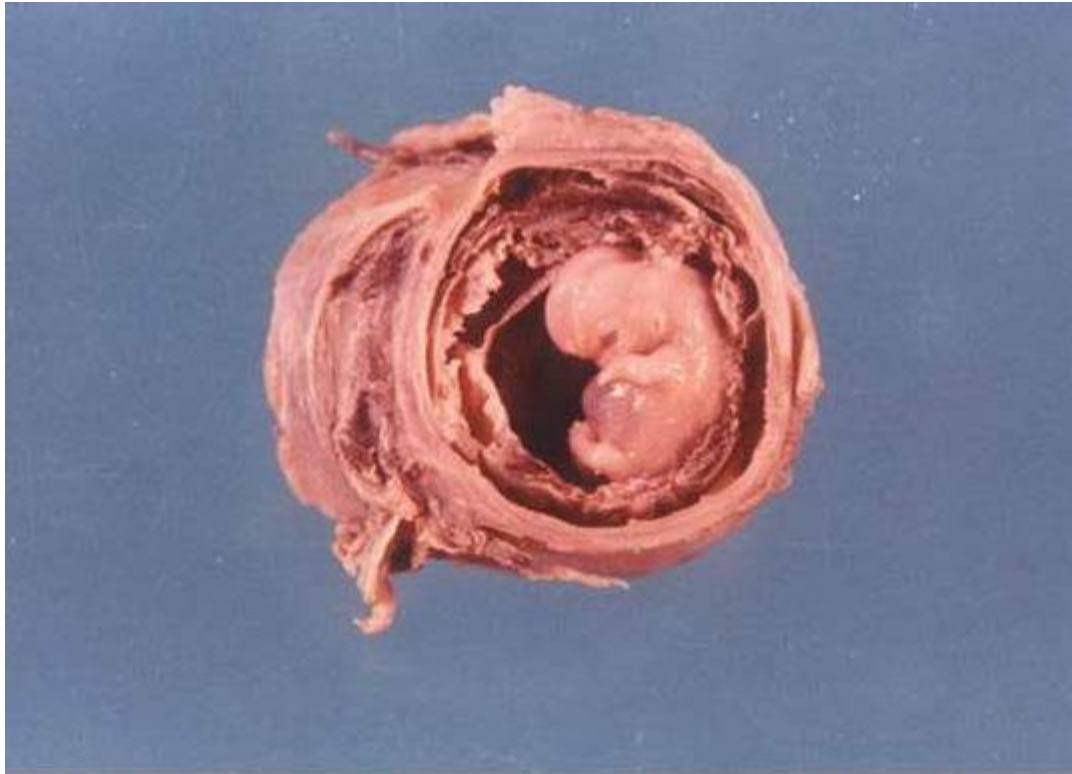
The placenta overbridges the os and causes

- severe bleeding during later pregnancy and
- delivery.



Ectopic pregnancy:

- Implantation outside the uterus: the uterine tube, abdominal cavity, or ovary.
- Abortion and severe hemorrhage during the 2nd month of gestation



SUMMARY OF THE FIRST WEEK

- Oocytes are produced by the ovaries (oogenesis) and expelled from them during ovulation
- The fimbriae of the uterine tube sweep the oocyte into the ampulla where it may be fertilized
- Sperms are produced in the testes (spermatogenesis) and are stored in the epididymis
- Ejaculation of semen during sexual intercourse results in the deposit of millions of sperms in the vagina
- Several hundred sperms pass through the uterus and enter the uterine tubes.
- When an oocyte is contacted by a sperm, it completes the second meiotic division
- As a result, a mature oocyte and a second polar body are formed.
- The nucleus of the mature oocyte constitutes the female pronucleus.

- After the sperm enters the oocyte, the head of the sperm separates from the tail and enlarges to become the male pronucleus
- Fertilization is complete when the male and female pronuclei unite and the maternal and paternal chromosomes intermingle during metaphase of the first mitotic division of the zygote.
- As it passes along the uterine tube toward the uterus, the zygote undergoes cleavage (a series of mitotic cell divisions) into a number of smaller cells-blastomeres
- Approximately 3 days after fertilization, a ball of 12 or more blastomeres-a morula-enters the uterus.

- A cavity forms in the morula, converting it into a blastocyst consisting of the embryoblast, a blastocystic cavity, and the trophoblast
- The trophoblast encloses the embryoblast and blastocystic cavity and later forms extraembryonic structures and the embryonic part of the placenta.
- Four to 5 days after fertilization, the zona pellucida is shed and the trophoblast adjacent to the embryoblast attaches to the endometrial epithelium.
- The trophoblast at the embryonic pole differentiates into two layers, an outer syncytiotrophoblast and an inner cytotrophoblast
- The syncytiotrophoblast invades the endometrial epithelium and underlying connective tissue
- Concurrently, a cuboidal layer of hypoblast forms on the deep surface of the embryoblast
- By the end of the first week, the blastocyst is superficially implanted in the endometrium.

NO THANKS!

Second Week of Development

Bilaminar Germ Disc

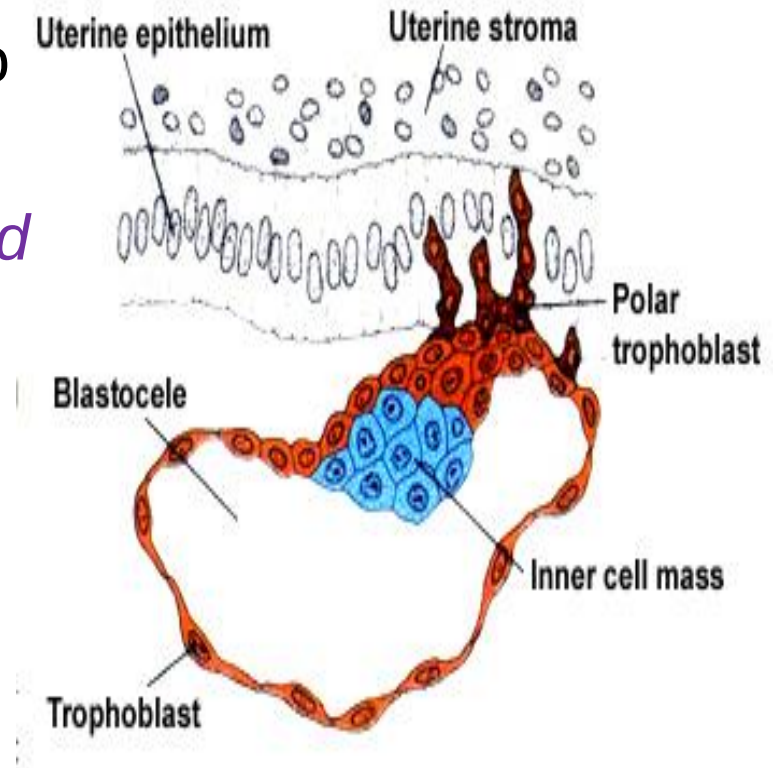
Day 8

- At the eighth day of development, the blastocyst is partially:

- embedded in the endometrial stroma

In the area over the embryoblast the trophoblast has differentiated into two layers:

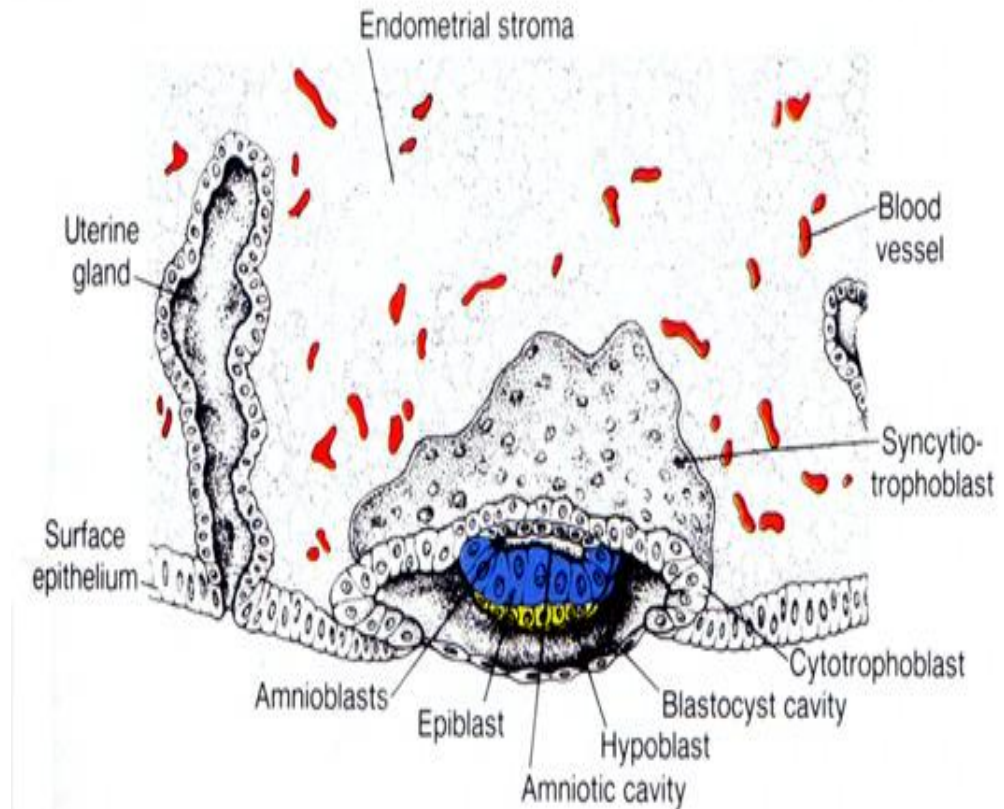
- 1. an inner layer of mononucleated cells, **the cytotrophoblast**,*
- 2. an outer multinucleated zone without distinct cell boundaries, **the syncytiotrophoblast***



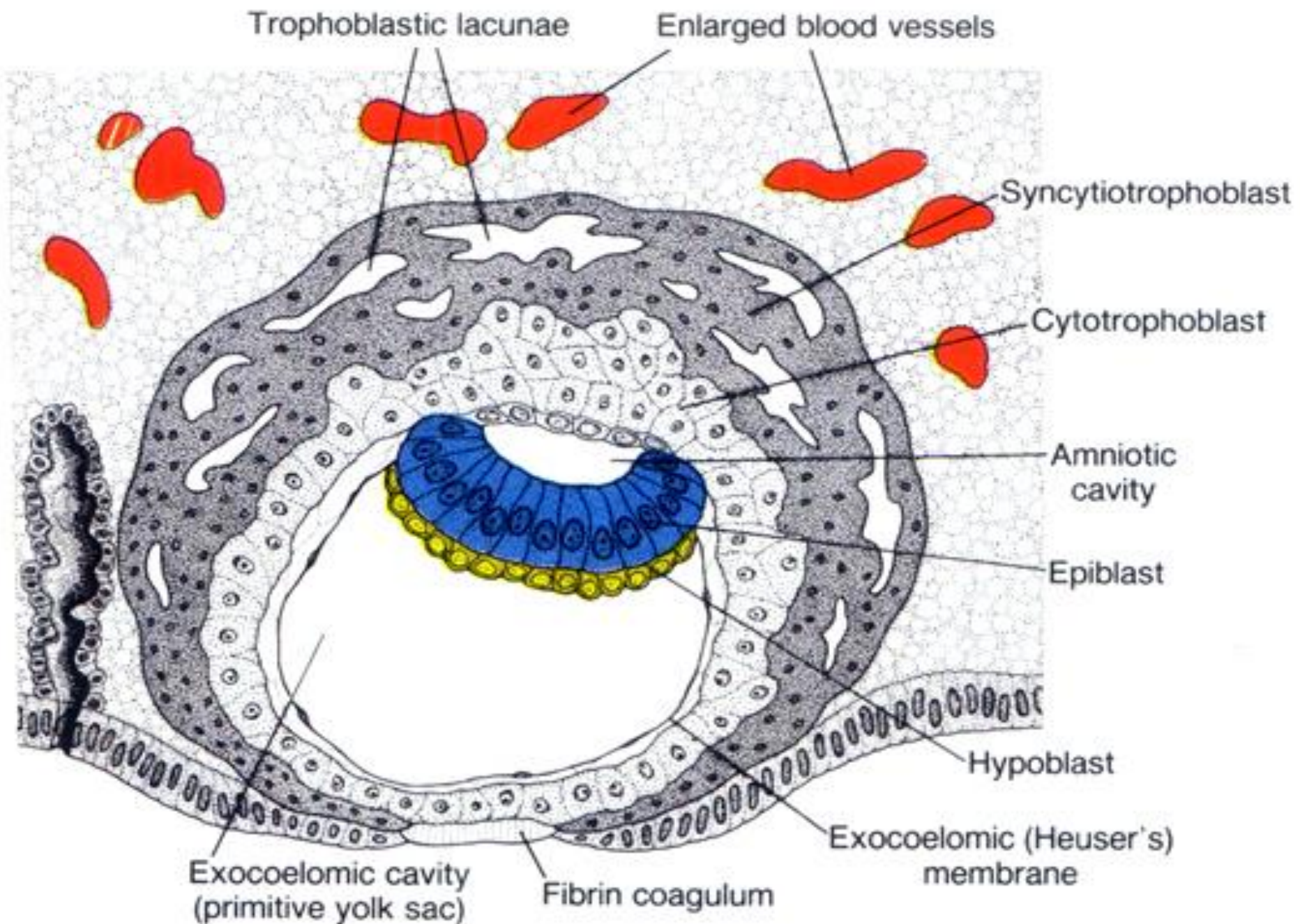
- The cytotrophoblast:
 - Inner layer
 - Mononucleated
 - Is mitotically active and forms new cells that migrate into the increasing mass of syncytiotrophoblast,
 - where they fuse and lose their cell membranes
- The syncytiotrophoblast:
 - a rapidly expanding to invade maternal endometrium
 - Is multinucleated mass without cell boundaries are discernible
 - Considered to be the most aggressive cell

- The erosive **syncytiotrophoblast**:
 - ✓ invades the endometrial connective tissue
 - ✓ the blastocyst slowly embeds itself in the endometrium.
 - ✓ Syncytiotrophoblastic cells displace endometrial cells at the implantation site.
 - ✓ The endometrial cells undergo **apoptosis** (programmed cell death), which facilitates the invasion
 - ✓ The molecular mechanisms of implantation involve synchronization between the invading blastocyst and a receptive endometrium.
 - ✓ The connective tissue cells around the implantation site accumulate glycogen and lipids and assume a polyhedral appearance.
 - ✓ Some of these cells-**decidual cells**-degenerate adjacent to the penetrating syncytiotrophoblast.
 - ✓ The syncytiotrophoblast engulfs these degenerating cells, providing a rich source of embryonic nutrition.

- Mitotic figures are found in the cytotrophoblast but not in the syncytiotrophoblast.
- Thus, cells in the **cytotrophoblast** divide and migrate into the syncytiotrophoblast, where they fuse and lose their individual cell membranes.



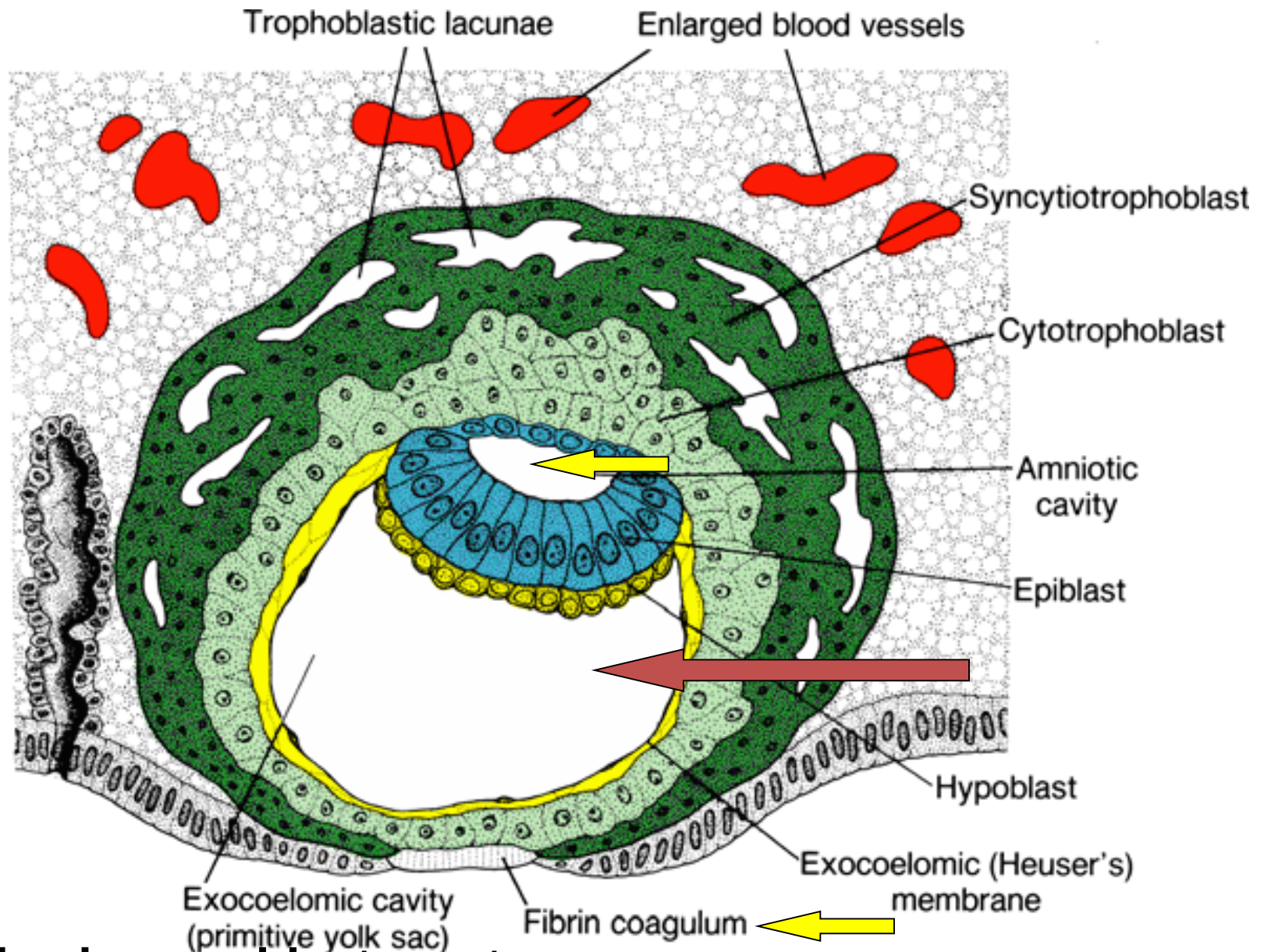
- **Embryoblast also differentiate into two layers:**
 - ***layer of small cuboidal cells*** adjacent to the blastocyst cavity (*Making the roof of it*), known as the **hypoblast**
 - ***a layer of high columnar cells*** adjacent to the amniotic cavity (making the floor of it), the **epiblast layer**
 - Together, the layers form a flat disc.
 - At the same time, a small cavity appears within the epiblast and enlarges to become the **amniotic cavity**
 - Epiblast cells adjacent to the cytotrophoblast are called **amnioblasts**
 - together with the rest of the epiblast, they line the **amniotic cavity**



- The endometrial stroma adjacent to the implantation site is edematous and highly vascular.
- The large, tortuous glands secrete abundant glycogen and mucus.
 - **The syncytiotrophoblast** produces a hormone-*human chorionic gonadotrophin* (hCG), which enters the maternal blood via lacunae in the **syncytiotrophoblasts**
 - **hCG** maintains the hormonal activity of the corpus luteum in the ovary during pregnancy.
 - Highly sensitive radioimmunoassays are available for detecting hCG and pregnancy and forms the basis for ***pregnancy tests***.
 - Enough hCG is produced by the **syncytiotrophoblast** at the end of the second week to give a positive pregnancy test, even though the woman is probably unaware that she is pregnant

Day 9

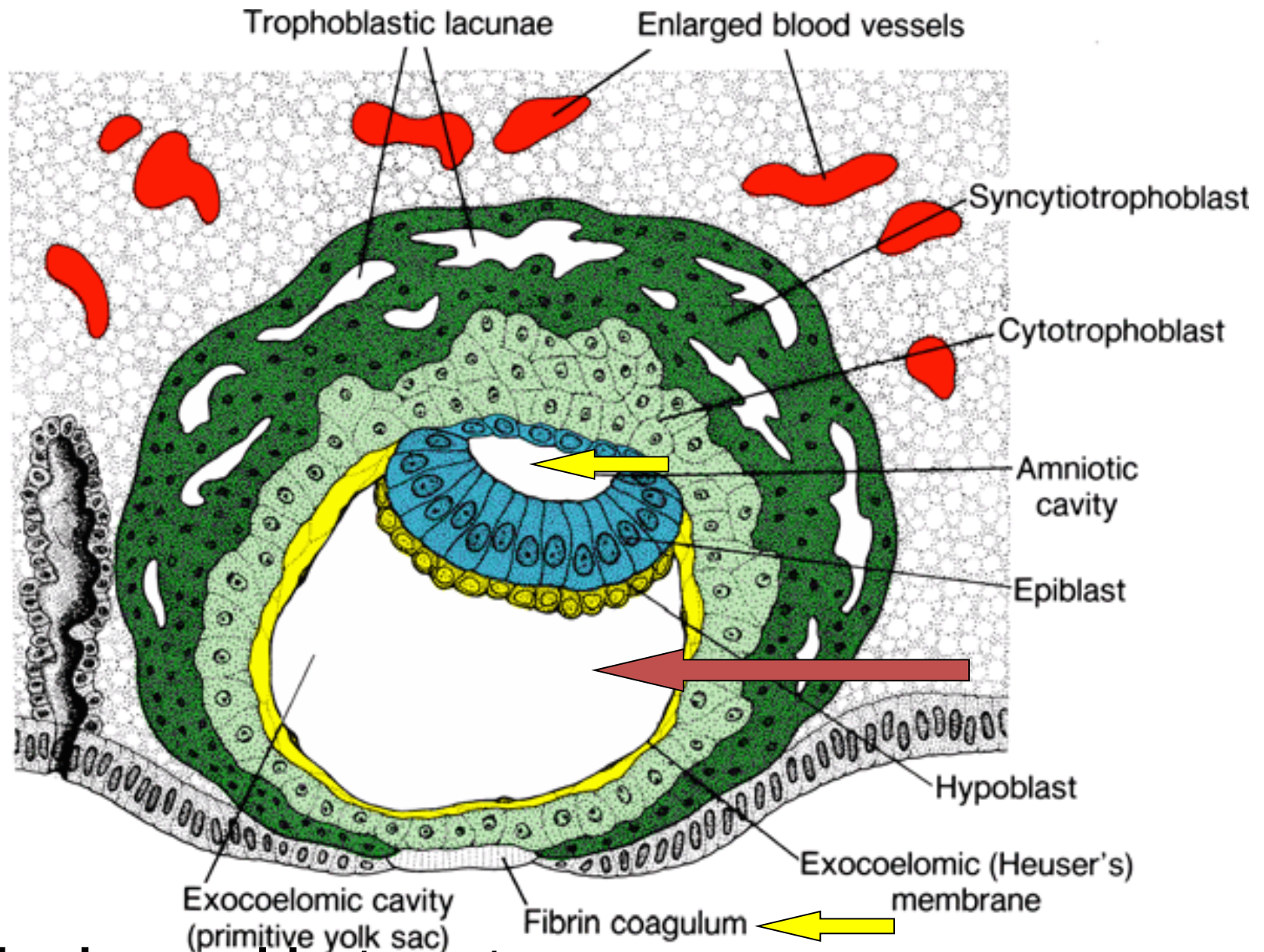
- The blastocyst is more deeply embedded in the endometrium
- The penetration defect in the surface epithelium is closed by a fibrin coagulum (**blood clot and cellular debris**)
- The trophoblast shows considerable progress in development, particularly at the embryonic pole, where vacuoles appear in the syncytium known as lacunae
- When these vacuoles fuse, they form large lacunae, and this phase of trophoblast development is thus known as the **lacunar stage**



A 9-day human blastocyst.

The syncytiotrophoblast shows a large number of lacunae. Flat cells form the exocoelomic membrane. The bilaminar disc consists of a layer of columnar epiblast cells and a layer of cuboidal hypoblast cells. The original surface defect is closed by

- At the **abembryonic pole**, flattened cells probably originating from the hypoblast form a thin membrane, **the exocoelomic (Heuser's) membrane**,
 - lines the inner surface of the cytotrophoblast
 - The membrane, together with the hypoblast, forms the lining of the **exocoelomic cavity**, or **primitive yolk sac**

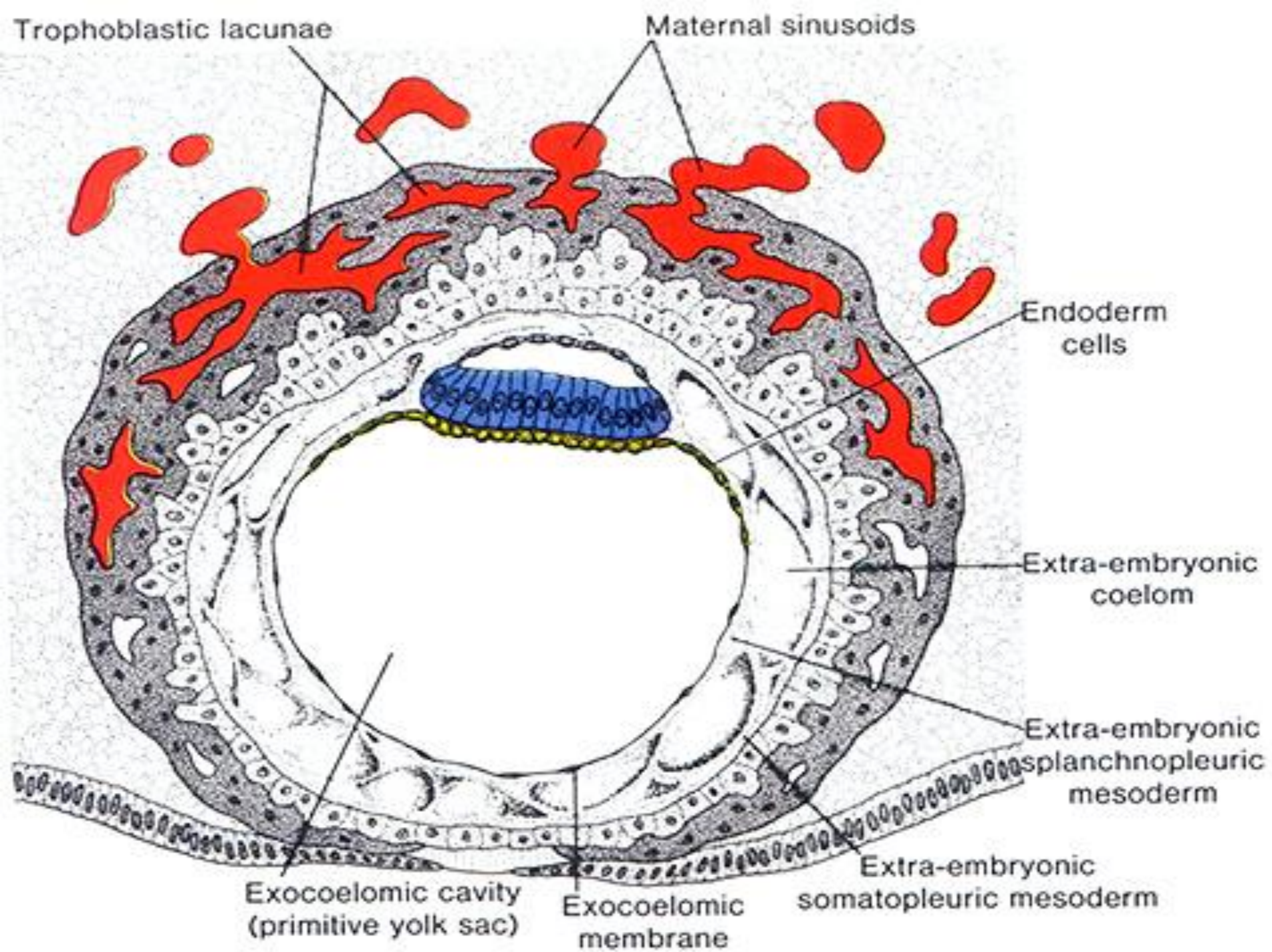


A 9-day human blastocyst.

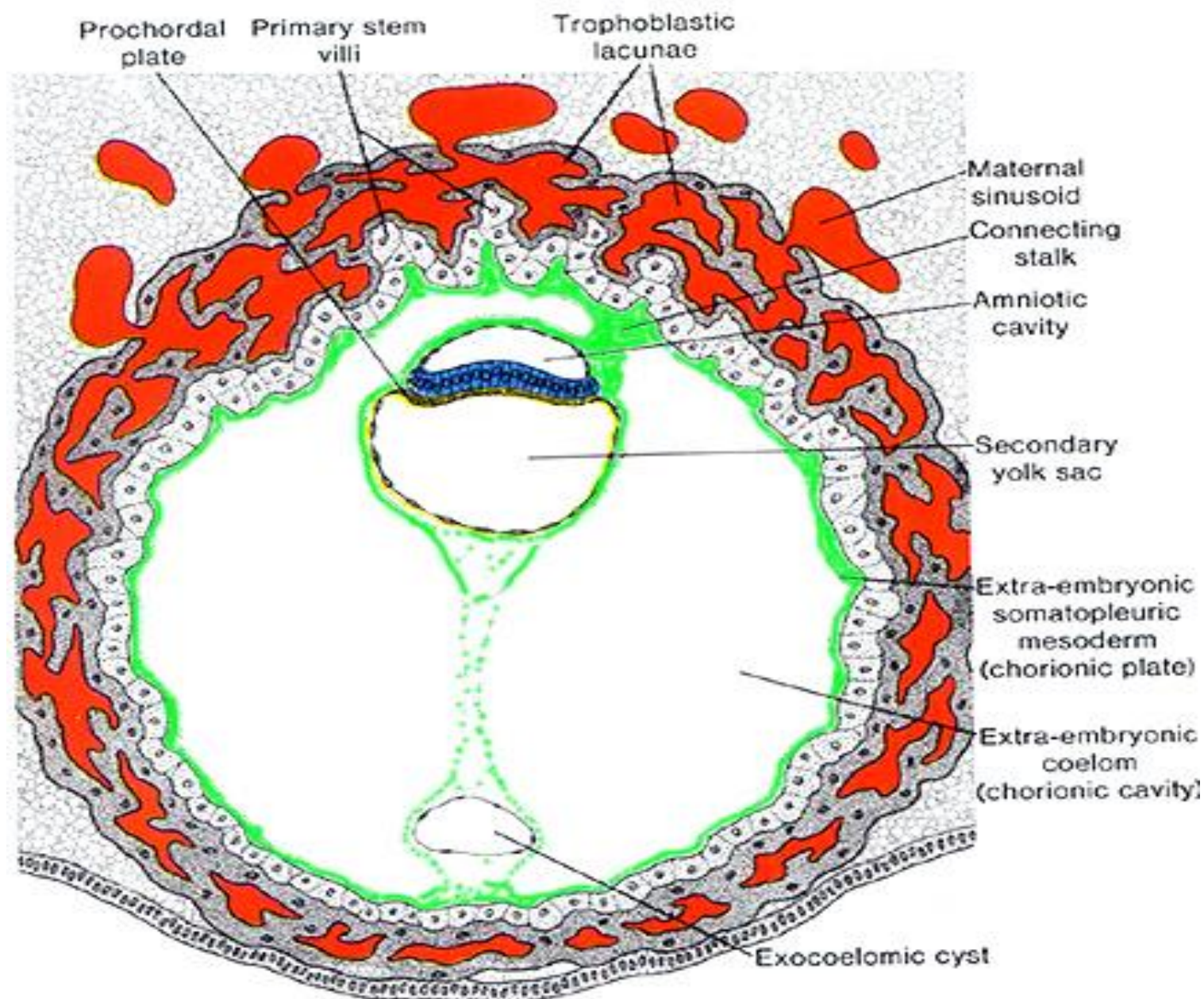
The syncytiotrophoblast shows a large number of lacunae. Flat cells form the exocoelomic membrane. The bilaminar disc consists of a layer of columnar epiblast cells and a layer of cuboidal hypoblast cells. The original surface defect is closed by

Days 11 and 12

- By the 11th to 12th day of development,
 - the blastocyst is completely embedded in the endometrial stroma,
 - and the surface epithelium almost entirely covers blastocyst
- The trophoblast
 - is characterized by lacunar spaces in the syncytium that form an intercommunicating network
- Cells of the syncytiotrophoblast penetrate deeper into the stroma and erode the endothelial lining of the maternal capillaries known as sinusoids



- the syncytial lacunae become continuous with the sinusoids
- maternal blood enters the lacunar system
- as the trophoblast continues to erode more and more sinusoids, maternal blood begins to flow through the trophoblastic system, establishing **the uteroplacental circulation**.
- a new population of cells appears between the inner surface of the cytotrophoblast and the outer surface of the **exocoelomic cavity**.
- these cells, derived from yolk sac cells, form a fine, loose connective tissue, the **extraembryonic mesoderm**

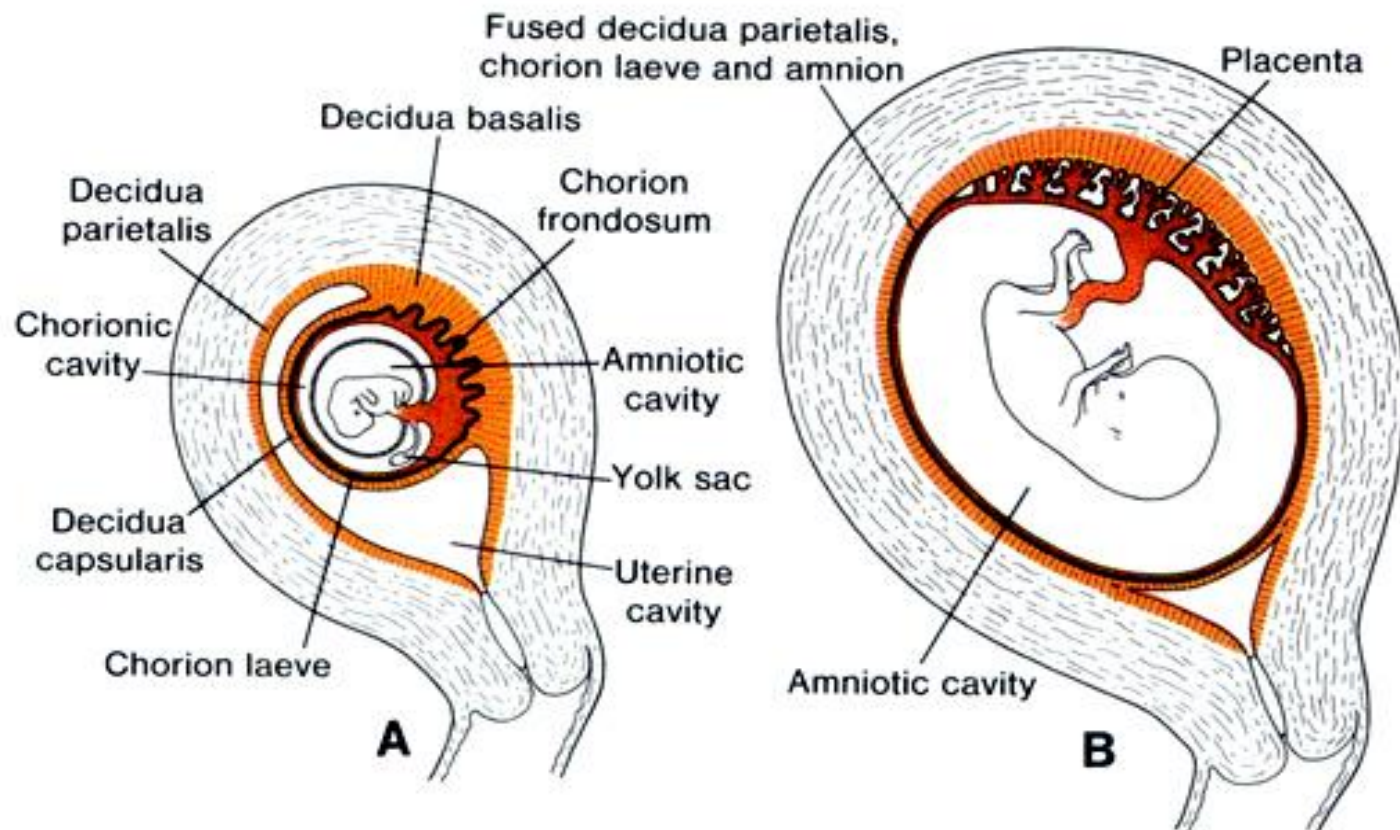


- **extraembryonic mesoderm** fills the space between the trophoblast externally and the amnion and exocoelomic membrane internally.
- Soon large cavities develop in the extraembryonic mesoderm
- when these become confluent, they form a new space known as the **extraembryonic coelom**, or **chorionic cavity**
 - This space surrounds the primitive yolk sac and amniotic cavity except where the germ disc is connected to the trophoblast by the **connecting stalk**
- The extra-embryonic mesoderm lining the cytotrophoblast and amnion is said to be the **extraembryonic somatopleuric mesoderm**
- the lining covering the yolk sac is known as the **extraembryonic splanchnopleuric mesoderm**

- Growth of the bilaminar disc is relatively slow compared with that of the trophoblast; consequently, the disc remains very small (**0.1 to 0.2 mm**)
- Cells of the endometrium, meanwhile, become polyhedral and loaded with glycogen and lipids
- intercellular spaces are filled with extravasate, and the tissue is edematous.
- These changes, known as the **decidua reaction**
 - at first are confined to the area immediately surrounding the implantation site but soon occur throughout the endometrium.

Decidual reaction: endometrium → decidua

- **Decidua basalis**: beneath implanted blastocyst and forming the maternal component of **placenta**.
- **Decidua capsularis**: over the blastocyst.
- **Decidua parietalis**: the remaining endometrium



FORMATION OF THE AMNIOTIC CAVITY, EMBRYONIC DISC, AND UMBILICAL VESICLE.

- As implantation of the blastocyst progresses, a small space appears in the embryoblast. This space is the primordium of the **amniotic cavity**. Soon amniogenic (amnion-forming) cells-*amnioblasts*-separate from the epiblast and form the **amnion**, which encloses the amniotic cavity. Concurrently, morphologic changes occur in the embryoblast that result in the formation of a flat, almost circular bilaminar plate of cells, the **embryonic disc**, consisting of two layers:

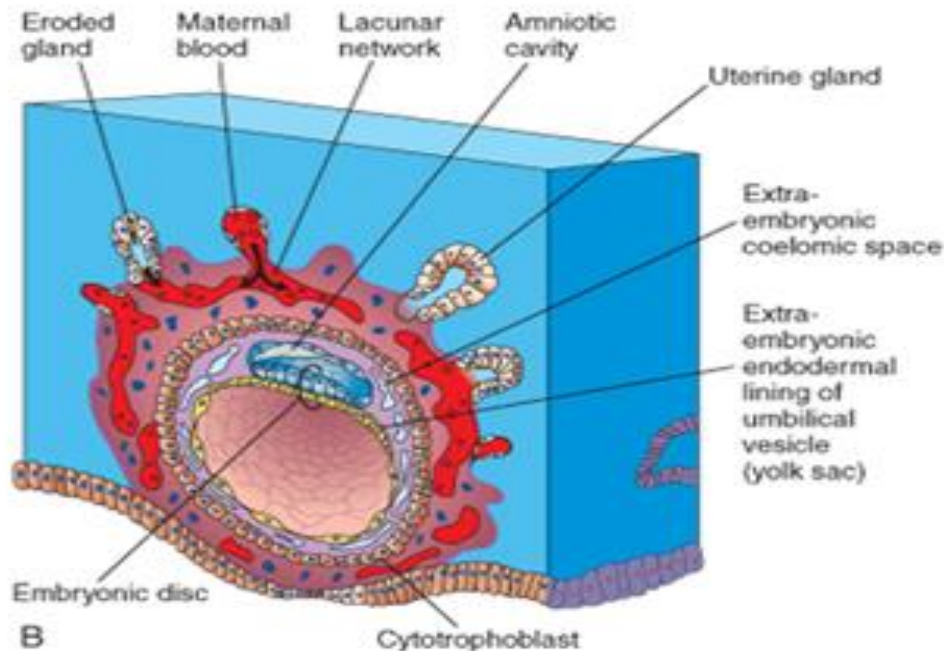
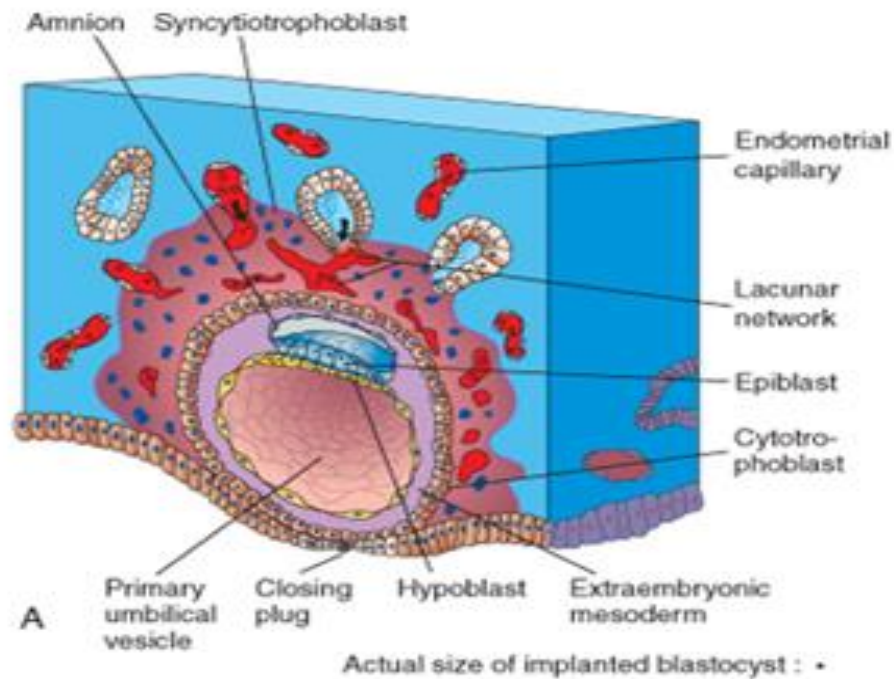
- **Epiblast**, the thicker layer, consisting of high columnar cells related to the amniotic cavity.
- **Hypoblast**, consisting of small cuboidal cells adjacent to the exocoelomic cavity

- The epiblast forms the floor of the amniotic cavity and is continuous peripherally with the amnion.
- The hypoblast forms the roof of the **exocoelomic cavity** and is continuous with the thin exocoelomic **membrane**.

- This membrane, together with the hypoblast, lines the **primary umbilical vesicle**.

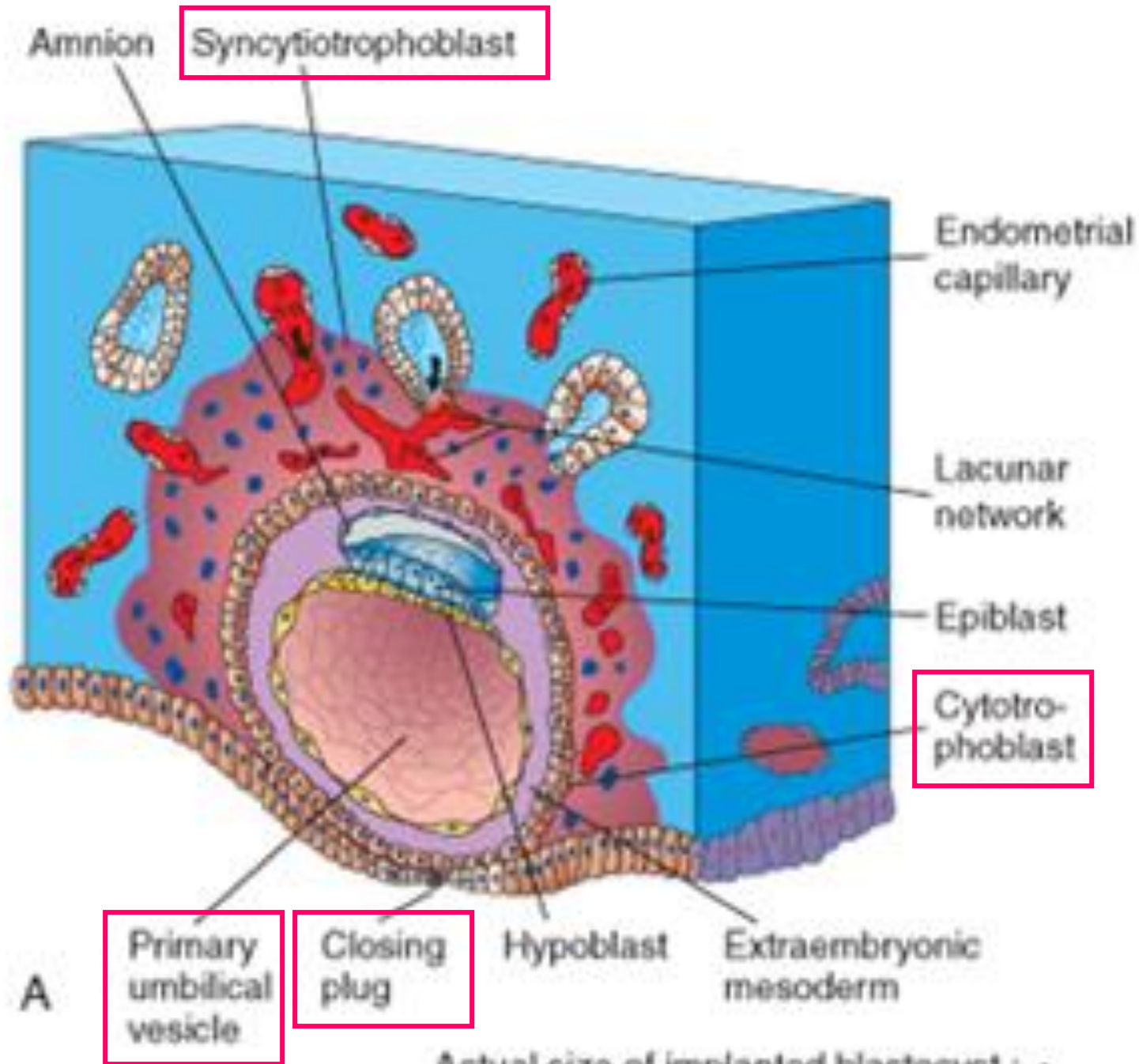
- The embryonic disc now lies between the amniotic cavity and the umbilical vesicle.
 - Cells from the vesicle endoderm form a layer of connective tissue, the **extraembryonic mesoderm**, which surrounds the amnion and umbilical vesicle.
 - This mesoderm continues to form from cells that arise from the primitive

- As the amnion, embryonic disc, and primary umbilical vesicle form, isolated cavities- **lacunae**-appear in the syncytiotrophoblast.
- The lacunae soon become filled with a mixture of maternal blood from ruptured endometrial capillaries and cellular debris from eroded uterine glands.
- The fluid in the lacunar spaces-*embryotroph* (Greek, *trophe*, nourishment)-passes to the embryonic disc by diffusion and provides nutritive material to the embryo.
- The communication of the eroded endometrial capillaries with the lacunae establishes the **primordial uteroplacental circulation**.
- When maternal blood flows into the lacunae, oxygen and nutritive substances are available to the embryo.
 - *Oxygenated blood* passes into the lacunae from the *spiral endometrial arteries*, and *poorly oxygenated blood* is removed from them through the endometrial veins.
- *The 10-day human conceptus (embryo and extraembryonic membranes) is completely embedded in the endometrium.*
 - For approximately 2 days, there is a defect in the endometrial epithelium that is filled by a **closing plug**, a fibrinous coagulum of blood.
 - By day 12, an almost completely regenerated uterine epithelium covers the closing plug.
- As the conceptus implants, the endometrial connective tissue cells undergo a transformation, the **decidual reaction**.
 - After the cells swell because of the accumulation of glycogen and lipid in their cytoplasm, they are known as **decidual cells**.
 - The primary function of the decidual reaction is to provide nutrition for the early embryo and an immunologically privileged site for the conceptus.

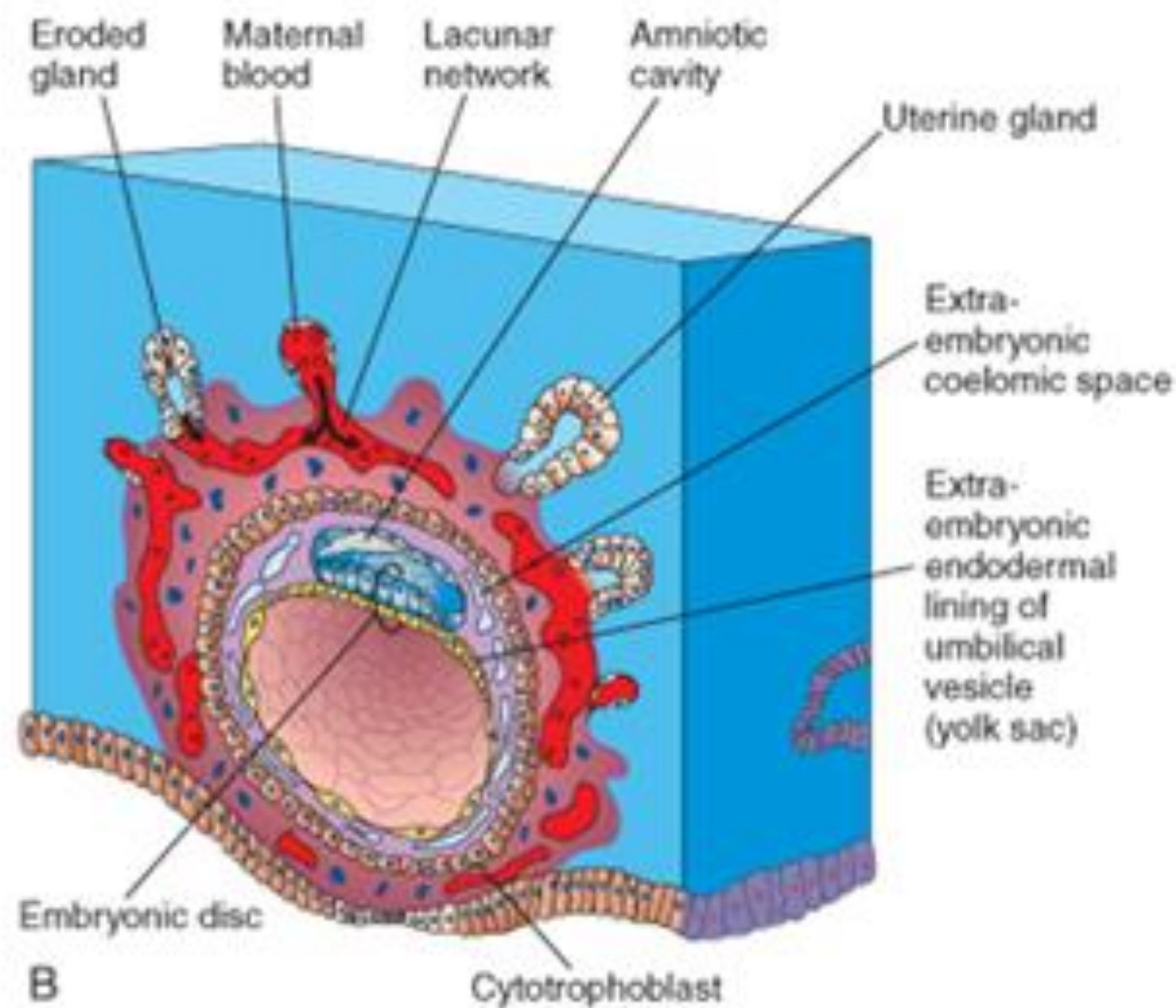


Implanted blastocysts.

A, At 10 days; **B**, at 12 days. This stage of development is characterized by communication of the blood-filled lacunar networks. Note in **B** that coelomic spaces have appeared in the extraembryonic mesoderm, forming the beginning of the extraembryonic coelom.



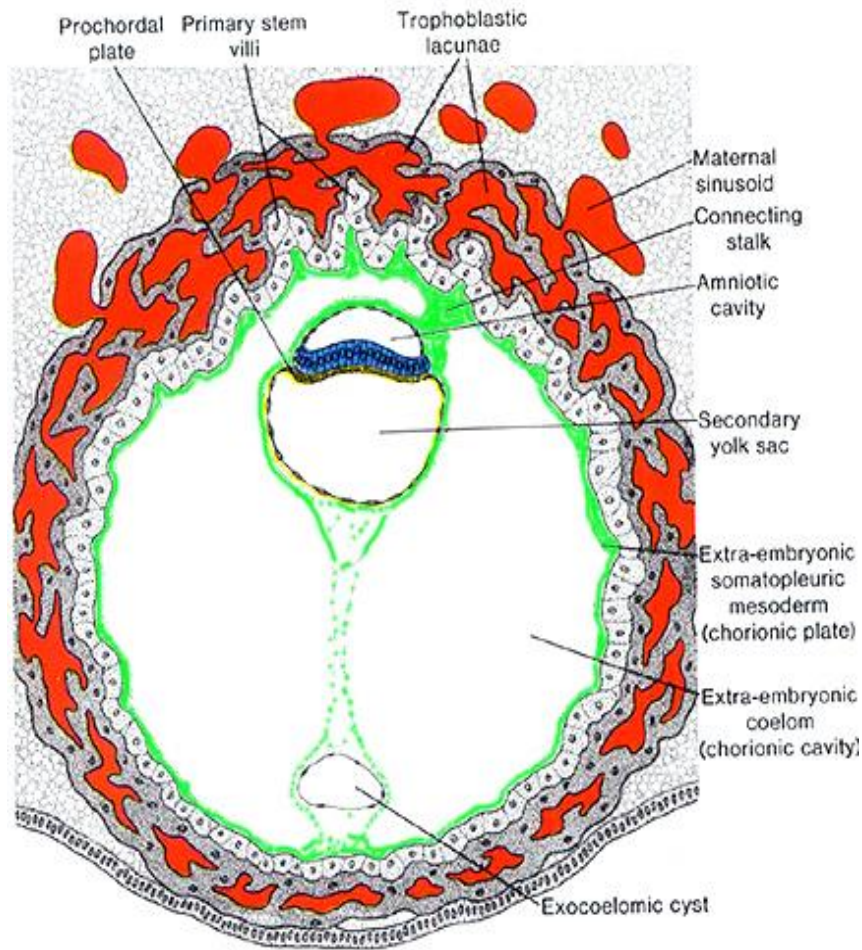
Actual size of implanted blastocyst : •



Day 13

- By the 13th day of development, the surface defect in the endometrium has usually healed.
- Occasionally, however, bleeding occurs at the implantation site as a result of increased blood flow into the lacunar spaces.
- Because this bleeding occurs near the **28th day** of the menstrual cycle, it may be confused with normal menstrual bleeding and, therefore, cause inaccuracy in determining the expected delivery date

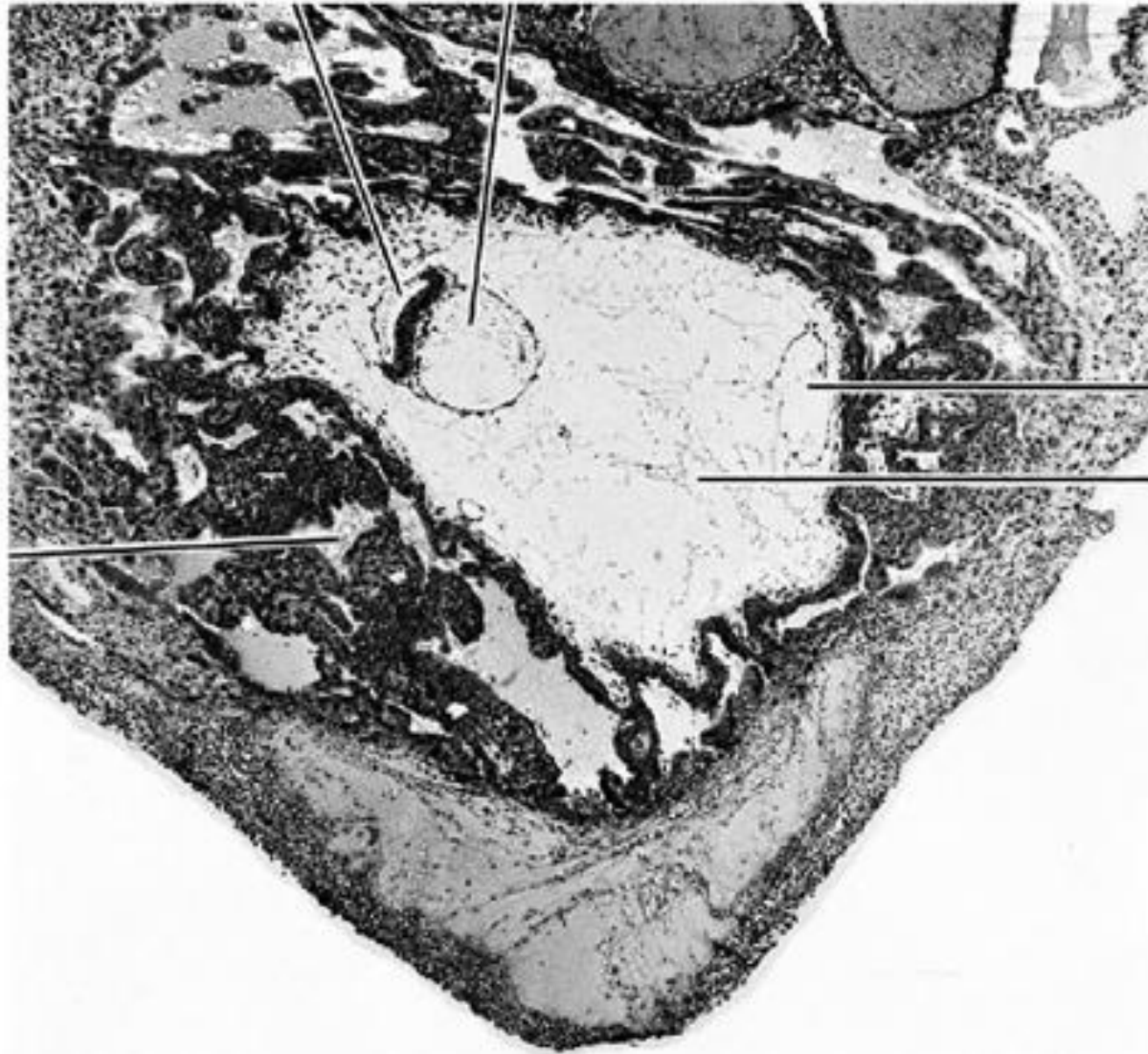
- **The trophoblast:**
 - is characterized by villous structures
 - Cells of the cytotrophoblast proliferate locally and penetrate into the syncytiotrophoblast, forming cellular columns surrounded by **syncytium**.
 - Cellular columns with the syncytial covering are known as **primary villi**
- the hypoblast produces additional cells that migrate along the inside of the exocoelomic membrane.
- These cells proliferate and gradually form a new cavity within the exocoelomic cavity.
- This new cavity is known as the **secondary yolk sac** or **definitive yolk sac**
- the extraembryonic coelom expands and forms a large cavity, the **chorionic cavity**



- The secondary yolk sac is much smaller than the original exocoelomic cavity, or primitive yolk sac.
 - during its formation, large portions of the exocoelomic cavity are pinched off.
 - These portions are represented by exocoelomic cysts, which are often found in the extraembryonic coelom or chorionic cavity.
- Meanwhile, **the extraembryonic coelom** expands and forms a large cavity, **the chorionic cavity**.
- The extraembryonic mesoderm lining the inside of the cytotrophoblast is then known as the **chorionic plate**. The only place where extraembryonic mesoderm traverses the chorionic cavity is in the connecting stalk.
- With development of blood vessels, the stalk becomes the umbilical cord.

Amniotic cavity

Secondary yolk sac



Exocoelomic
cyst
Chorionic
cavity

Trophoblast
with
lacunae

Section through the implantation site of a 13-day embryo. Note the amniotic cavity, yolk sac, and exocoelomic cyst in the chorionic cavity. Most of the lacunae are

DEVELOPMENT OF THE CHORIONIC SAC

- The end of the 2nd week is characterized by the appearance of **primary chorionic villi**.
- Proliferation of cytotrophoblastic cells produces **cellular extensions that grow into the syncytiotrophoblast**.
- The growth of these extensions is thought to be induced by the underlying **extraembryonic somatic mesoderm**
- The cellular projections form primary chorionic villi, the first stage in the development of the chorionic villi of the placenta
- The extraembryonic coelom splits the extraembryonic mesoderm into two layers:
 - ✓ **Extraembryonic somatic mesoderm**, lining the trophoblast and covering the amnion

- The extraembryonic somatic mesoderm and the two layers of trophoblast form the chorion.
- *The chorion forms the wall of the chorionic sac*, within which the embryo and its amniotic sac and umbilical vesicle are suspended by the connecting stalk.
- The extraembryonic coelom is now called the chorionic cavity.
- The amniotic sac and the umbilical vesicle can be thought of as two balloons pressed together (at the site of embryonic disc) and suspended by a cord (connecting stalk) from the inside of a larger balloon (chorionic sac).
- .

****Transvaginal ultrasonography* (endovaginal sonography)

is used for measuring the chorionic (gestational) sac diameter

- This measurement is valuable for evaluating early embryonic development and pregnancy outcome

- The **14-day embryo**:

- still has the form of a flat bilaminar embryonic disc
- the hypoblastic cells in a localized area are now columnar and form a thickened circular area-the **prechordal plate** , which indicates the future site of the **mouth** and an important organizer of the head region.

Site of implantation

- Implantation of blastocysts usually occurs in the endometrium of the uterus, superior in the body of the uterus, slightly more often on the posterior than on the anterior wall.
- Implantation of a blastocyst can be detected by ultrasonography and highly sensitive radioimmunoassays of hCG as early as the end of the second week.

Extrauterine Implantation

Blastocyst may implant outside the uterus (**ectopic pregnancy**)

- 95% - 98% of ectopic implantations occur in the **uterine tubes**, *most often in the ampulla and isthmus*.
- The incidence of ectopic pregnancy has increased in most countries, ranging from **1 in 80** to **1 in 250** pregnancies, depending on the socioeconomic level of the population.
- **For example:** In the **United States**, the frequency of ectopic pregnancy is approximately **2%** of all pregnancies, and it is the main cause of maternal deaths during the first trimester.

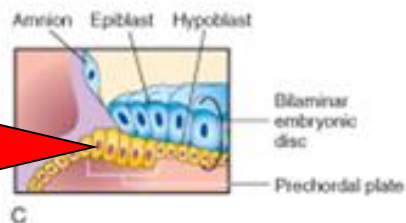
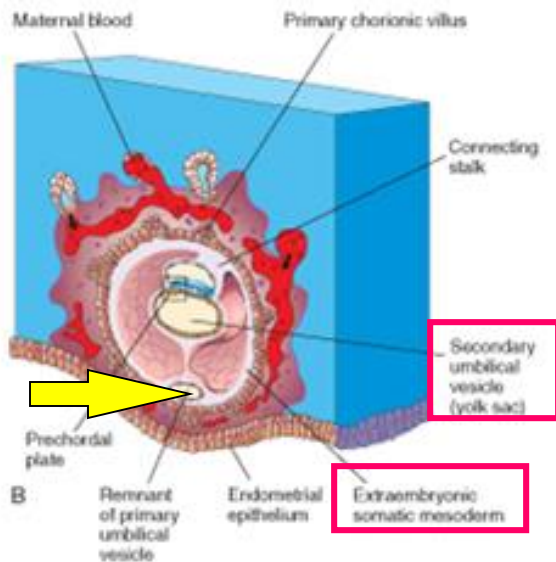
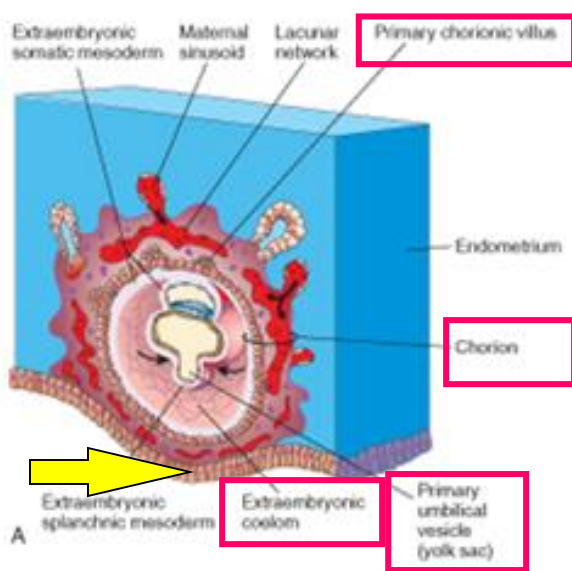
- ✓ A woman with a **tubal pregnancy** has signs and symptoms of pregnancy (e.g., misses her menstrual period).
- ✓ She may also experience **abdominal pain and tenderness** because of distention of the uterine tube, **abnormal bleeding**, and irritation of the pelvic peritoneum (**peritonitis**).
- ✓ The pain may be confused with **appendicitis** if the pregnancy is in the right uterine tube.
- ✓ Ectopic pregnancies produce human chorionic gonadotropin at a slower rate than normal pregnancies
- ✓ (hCG assays may give false-negative results if performed too early!!!)

● **Transvaginal ultrasonography is very helpful in the early detection of ectopic tubal pregnancies.**

- ✓ There are several causes of tubal pregnancy, but they are often related to factors that *delay or prevent transport of the cleaving zygote to the uterus*, for example, by mucosal adhesions in the uterine tube or from blockage of it that is caused by scarring resulting from *pelvic inflammatory disease*.
- ✓ Ectopic tubal pregnancies usually result in *rupture of the uterine tube* and hemorrhage into the peritoneal cavity during the first 8 weeks, followed by death of the embryo.
- ✓ Tubal rupture and hemorrhage constitute a threat to the mother's life
- ✓ The affected tube and conceptus are usually surgically removed.

- When blastocysts implant in the ***isthmus*** of the uterine tube, the tube tends to rupture early because this narrow part of the tube is relatively ***unexpandable***, and often with ***extensive bleeding*** (rich anastomoses between ovarian and uterine vessels).
- When blastocysts implant in the ***uterine (intramural) part of the tube***, they may develop beyond 8 weeks before expulsion occurs. When a uterine tubal pregnancy ruptures, it usually ***bleeds profusely***.
- Blastocysts that implant in the ***ampulla or on fimbriae of the uterine*** tube may be expelled into the peritoneal cavity where they commonly implant in the rectouterine pouch.
- Usually the ectopic pregnancy can be terminated by surgical removal of the involved uterine tube without interfering with the intrauterine pregnancy.
- ***Cervical implantations are unusual***; in some cases, the placenta becomes firmly attached to fibrous and muscular tissues of the cervix, often resulting in bleeding and requiring subsequent surgical intervention, such as ***hysterectomy*** (excision of uterus).

- In exceptional cases, an **abdominal pregnancy** may continue to full term and the fetus may be delivered alive through an abdominal incision.
- Usually, however, the placenta attaches to abdominal organs and causes considerable ***intraperitoneal bleeding***.
- Abdominal pregnancy increases the risk of maternal death from hemorrhage by a **factor of 90** when compared with intrauterine pregnancy, and **seven times** more than that for tubal pregnancy.
- In very unusual cases, an abdominal conceptus dies and is not detected; the fetus becomes calcified, forming a "stone fetus"-***lithopedion*** (Greek, *lithos*, stone, + *paidion*, child).
- ● **Simultaneous intrauterine and extrauterine pregnancies** are unusual, occurring approximately **1 in 7000**.
- ● The ectopic pregnancy is masked initially by the presence of the uterine pregnancy.



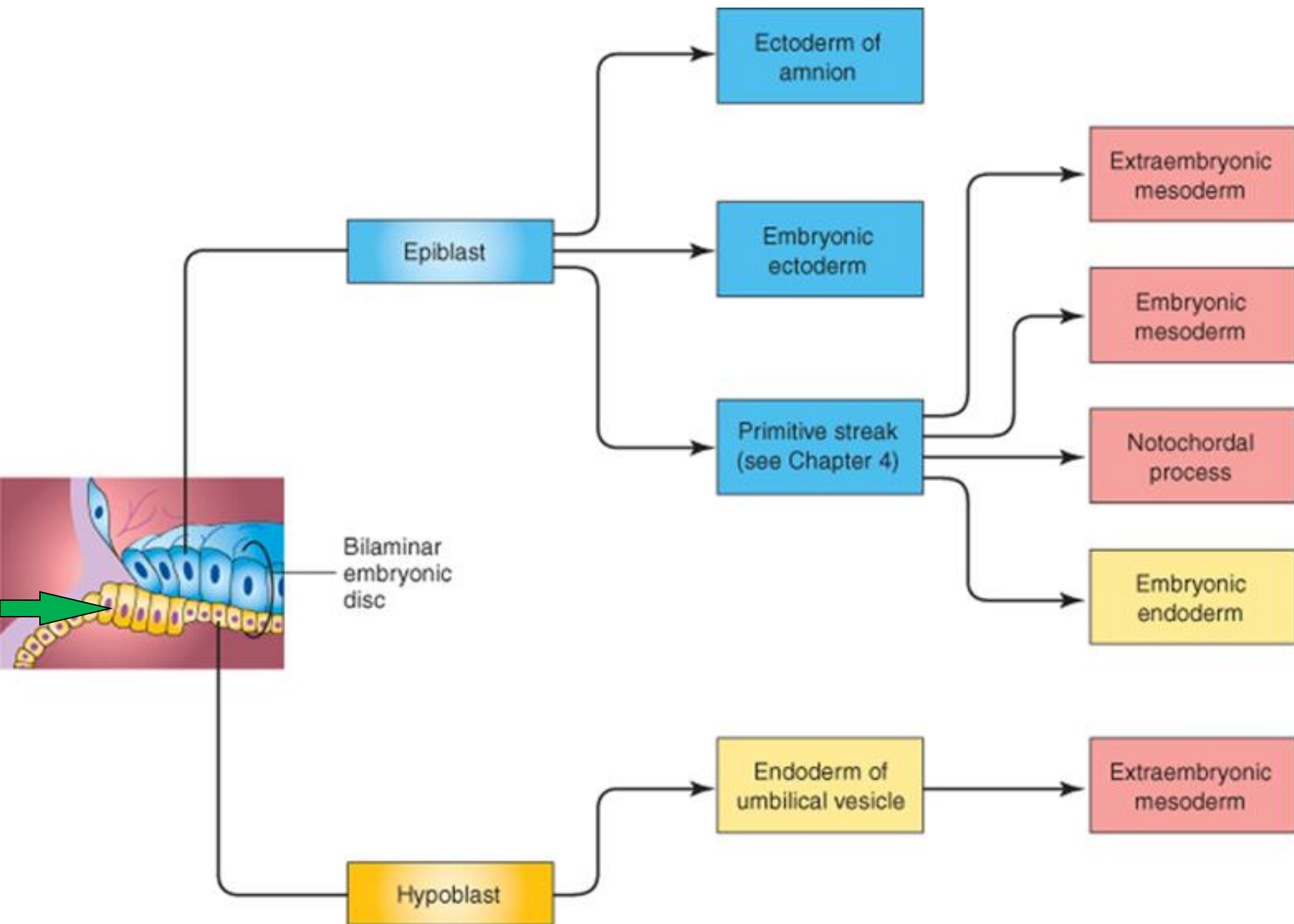
Drawings of sections through implanted human embryos, based mainly on Hertig and colleagues (1956).

Observe that

- (1) the **defect** in the endometrial epithelium has disappeared;
- (2) a small **secondary umbilical vesicle** has formed;
- (3) a large cavity, the **extraembryonic coelom**, now surrounds the umbilical vesicle and amnion, except where the amnion is attached to the chorion by the connecting stalk; and
- (4) the extraembryonic coelom splits the extraembryonic mesoderm into two layers: ***extraembryonic somatic mesoderm*** lining the trophoblast and covering the amnion and the ***extraembryonic splanchnic mesoderm*** around the umbilical vesicle.

A, At 13 days, illustrating the decrease in relative size of the primary umbilical vesicle and the early appearance of primary chorionic villi.

B, At 14 days, showing the newly formed secondary umbilical vesicle and the location of the prechordal plate in its roof.



Summary of the 2nd week

Developmental events	day/ days
The trophoblast differentiates into two layers	Day 7
The syncytiotrophoblast erodes endometrial tissues	Day 8
Blood-filled lacunae appear in the syncytiotrophoblast	Day 9
The blastocyst sinks beneath the endometrial epithelium, and the defect is filled by a closing plug	Day 10
Lacunar networks form by fusion of adjacent lacunae	Day 10 & 11
Establishing a uteroplacental circulation	Day 11 & 12
The defect in the endometrial epithelium is repaired	Day 12 & 13
Primary chorionic villi develop	Day 13 & 14

THANK YOU!